

ANALYSIS II

LECTURE NOTES

POSHEN CHANG

Contents

1	Lebesgue Measure	4
1.1	Preliminaries	4
1.2	Outer Measure	5
1.3	Measurable Sets and the Lebesgue Measure	8
1.3.1	Continuity of Lebesgue Measure	10
1.3.2	Approximation of Measurable Sets	10
1.3.3	Translation Invariance and Scaling	11
1.3.4	σ -Algebra and Borel Sets	11
1.3.5	Nonmeasurable Sets	12
1.4	The Brunn-Minkowski Inequality	13
1.5	Measurable Functions	14
1.5.1	Approximation of Measurable Functions	16
1.5.2	Littlewood's Three Principles	17
2	Integration Theory	19
2.1	The Lebesgue Integral	19
2.2	L^p -spaces	27
2.2.1	Invariance Properties	29
2.3	Fubini's Theorem	30
2.3.1	Convolution	35
3	Differentiation and Integration	37
3.1	Differentiation of the Integral	37
3.1.1	The hardy-Littlewood Maximal Function	37
3.2	Approximation to the Identity	41
3.3	Differentiability of functions	44
3.3.1	Absolutely Continuous Functions	48
3.3.2	Jump functions	50
3.4	Rectifiable curves	51
3.4.1	Isoperimetric Inequality	53
4	Hilbert Spaces	57
4.1	Orthogonal Decomposition	58
4.1.1	Self-Duality	59
4.1.2	Direct Sum Decomposition	60
4.1.3	Complete Orthonormal Sets	60
5	Fourier Analysis	64
5.1	Fourier Series	64
5.1.1	Fourier Series for L^2 -Functions	66
5.2	Fourier Transform	67
6	Abstract Measure	72
6.1	Abstract Measure Spaces	72
6.1.1	Outer Measures	72

6.1.2	Metric Outer Measures	74
6.1.3	Premeasures	76
6.2	Integration	78
6.2.1	Product Measures and Fubini's Theorem	79
6.2.2	Stieltjes Integrals	81
6.3	Absolute Continuity of Measures	83
6.3.1	Signed Measures	83
6.3.2	Mutual Singularity and Absolute Continuity	84
7	Fractal Geometry	88
7.1	Self-similarity	88
7.1.1	Coding Theorem	89

Lebesgue Measure

CHAPTER 1

This chapter is devoted to the construction of Lebesgue measure in $\mathbb{R}^d$ and the study of the resulting class of measurable functions.

1.1. Preliminaries

The goal of this section is to construct a measure on $\mathbb{R}$, which assigns to each subset E of $\mathbb{R}$ a non-negative number $m(E)$, such that

- (i) $m(E) = b - a$ if E is an interval $[a, b]$;
- (ii) m is countably additive, i.e.,

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} m(E_n)$$

for any sequence of disjoint sets $E_1, E_2, \dots$;

- (i) m is translation invariant, i.e., $m(E + x) = m(E)$ for any $x \in \mathbb{R}$.

A (closed) rectangle in $\mathbb{R}^d$ is a set of the form

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_d, b_d].$$

The volume of R is defined to be

$$|R| = (b_1 - a_1)(b_2 - a_2) \cdots (b_d - a_d).$$

A cube is a rectangle with $b_1 - a_1 = b_2 - a_2 = \cdots = b_d - a_d$. A union of rectangles is said to be almost disjoint if the interiors of the rectangles are disjoint.

LEMMA 1.1. *If a rectangle is the union of finitely many other rectangles $R_1, R_2, \dots, R_N$ that are almost disjoint, then*

$$|R| = |R_1| + |R_2| + \cdots + |R_N|.$$

Proof. Consider the grid formed by extending the sides of the rectangles. The resulting grid divides the original rectangle into finitely many almost disjoint rectangles $R_1', R_2', \dots, R_{M'}$ and a partition $J_1, J_2, \dots, J_N$ of integers from 1 to M , such that

$$R = \bigcup_{j=1}^M R_{j'} \quad \text{and} \quad R_k = \bigcup_{j \in J_k} R_{j'} \quad \text{for } k = 1, 2, \dots, N.$$

Since the grid partitions the sides of R , by simply expanding the products we have

$$|R| = \sum_{j=1}^M |R_{j'}|.$$

This also holds for each R_k , so

$$|R| = \sum_{j=1}^M |R_{j'}| = \sum_{k=1}^N \sum_{j \in J_k} |R_{j'}| = \sum_{k=1}^N |R_k|.$$

■

LEMMA 1.2. *If $R, R_1, R_2, \dots, R_N$ are rectangles and $R \subseteq \bigcup_{n=1}^N R_n$, then*

$$|R| \leq |R_1| + |R_2| + \dots + |R_N|.$$

Proof. We may employ the same grid as in the previous lemma, which partitions R into almost disjoint rectangles $R_{1'}, R_{2'}, \dots, R_{M'}$. R can be written as a almost disjoint union of a subset of these rectangles, and each $R_{j'}$ is contained in some R_n . Therefore,

$$|R| \leq \sum_{j=1}^M |R_{j'}| \leq \sum_{n=1}^N |R_n|.$$

■

If $\mathcal{U}$ is open in $\mathbb{R}$, then $\mathcal{U}$ can be written as a countable union of almost disjoint open intervals, say $\mathcal{U} = \bigcup_{n=1}^{\infty} I_n$. We define the measure of $\mathcal{U}$ to be

$$m(\mathcal{U}) = \sum_{n=1}^{\infty} |I_n|.$$

We wish to generalize this to higher dimensions.

PROPOSITION 1.3. *Every open subset $\mathcal{U} \subseteq \mathbb{R}^d$ can be written as a countable union of almost disjoint closed cubes.*

Proof. At a first step, consider the grid formed by taking all closed cubes of side length 1 with vertices in $\mathbb{Z}^d$. We either accept or reject cubes Q in the initial grid according to the following rule: if Q is entirely contained in $\mathcal{U}$ then we accept Q ; if Q intersects both $\mathcal{U}$ and $\mathcal{U}^c$ then we tentatively accept it; and if Q is entirely contained in $\mathcal{U}^c$ then we reject it.

As a second step, we bisect the tentatively accepted cubes into 2^d cubes with side length $\frac{1}{2}$. We then repeat our procedure, by accepting the smaller cubes if they are completely contained in $\mathcal{U}$, tentatively accepting them if they intersect both $\mathcal{U}$ and $\mathcal{U}^c$, and rejecting them if they are contained in $\mathcal{U}^c$.

This procedure is then repeated indefinitely, and (by construction) the resulting collection of all accepted cubes is countable and consists of almost disjoint cubes. To see why their union is all of $\mathcal{U}$, we note that given $x \in \mathcal{U}$ there exists a cube of side length 2^{-N} (obtained from successive bisections of the original grid) that contains x and that is entirely contained in $\mathcal{U}$. Either this cube has been accepted, or it is contained in a cube that has been previously accepted. This shows that the union of all cubes in $\mathcal{Q}$ covers $\mathcal{U}$. ■

We can try to define the measure of $\mathcal{U}$ to be the sum of the volumes of the cubes in the decomposition. The subtle issue is that the decomposition of $\mathcal{U}$ into cubes is not unique.

1.2. Outer Measure

DEFINITION 1.4. *Let $E \subseteq \mathbb{R}^d$. The outer measure of E is defined to be*

$$m_*(E) = \inf \left\{ \sum_{n=1}^{\infty} |Q_n| : E \subseteq \bigcup_{n=1}^{\infty} Q_n \right\},$$

where the infimum is taken over all countable collections of closed cubes $\{Q_n\}$ that cover E .

Example 1.1.

- Let Q be a closed cube. Since Q covers itself, we have $m_*(Q) \leq |Q|$. On the other hand, suppose $Q \subseteq \bigcup_{n=1}^{\infty} Q_n$. Let $\varepsilon > 0$. Choose an open cube $S_j \supseteq Q_j$ such that $|S_j| \leq (1 + \varepsilon)|Q_j|$. Then $Q \subseteq \bigcup_{n=1}^{\infty} S_n$. Since Q is compact, there exists a finite subcover $Q \subseteq \bigcup_{n=1}^N S_n$. By Lemma 1.2, we have

$$|Q| \leq \sum_{n=1}^N |S_n| \leq (1 + \varepsilon) \sum_{n=1}^N |Q_n| \leq (1 + \varepsilon) \sum_{n=1}^{\infty} |Q_n|.$$

Since ε is arbitrary, we conclude that $m_*(Q) \geq |Q|$. Therefore, $m_*(Q) = |Q|$.

- Let Q be an open cube. Since $\bar{Q}$ covers Q , we have $m_*(Q) \leq |\bar{Q}| = |Q|$. On the other hand, for a closed cube Q_0 contained in Q , we have $m_*(Q) \geq m_*(Q_0) = |Q_0|$. By letting $|Q_0| \rightarrow |Q|$, we conclude that $m_*(Q) \geq |Q|$. Therefore, $m_*(Q) = |Q|$.
- Let R be a closed rectangle. We can argue similarly as in (a) to show that $m_*(R) \geq |R|$. On the other hand, consider a grid formed by cubes of side length $\frac{1}{k}$. Let $\mathcal{Q}$ be the collection of cubes that are entirely contained in R , and let $\mathcal{Q}'$ be the collection of cubes that intersect both R and R^c . Note that

$$\sum_{Q \in \mathcal{Q}} |Q| \leq |R|$$

and $R \subseteq \bigcup_{Q \in \mathcal{Q}} Q \cup \bigcup_{Q \in \mathcal{Q}'} Q$. Since there are at most $O(k^{d-1})$ cubes in $\mathcal{Q}'$, we have $\sum_{Q \in \mathcal{Q}'} |Q| = O(\frac{1}{k})$. By letting $k \rightarrow \infty$, we conclude that $m_*(R) = |R|$.

- (i) $m_*(\mathbb{R}^d) = \infty$ since every cube is contained in $\mathbb{R}^d$, and these cubes can have arbitrarily large volume.
- (ii) The Cantor set $\mathcal{C}$ has zero outer measure. This follows from the construction of $\mathcal{C}$, where $\mathcal{C} \subseteq C_k$ for each k , and each C_k is a disjoint union of 2^k closed intervals of length 3^{-k} .

PROPOSITION 1.5. *For every $\varepsilon > 0$, there exists a covering $E \subseteq \bigcup_{n=1}^{\infty} Q_n$ by closed cubes such that*

$$\sum_{n=1}^{\infty} m_*(Q_n) \leq m_*(E) + \varepsilon.$$

Proof. Follows from the definition of $m_*(E)$ as an infimum and $|Q_n| = m_*(Q_n)$ for closed cubes Q_n . ■

PROPOSITION 1.6 (Monotonicity). *If $E \subseteq F$, then $m_*(E) \leq m_*(F)$.*

Proof. Follows immediately from the definition of m_* as an infimum over a set of coverings. ■

PROPOSITION 1.7 (Countable Subadditivity). *If $E = \bigcup_{n=1}^{\infty} E_n$, then $m_*(E) \leq \sum_{n=1}^{\infty} m_*(E_n)$.*

Proof. First assume that $m_*(E_n) < \infty$ for each n . Given $\varepsilon > 0$, for each n there exists a covering $E_n \subseteq \bigcup_{k=1}^{\infty} Q_{n,k}$ such that

$$\sum_{k=1}^{\infty} m_*(Q_{n,k}) \leq m_*(E_n) + \frac{\varepsilon}{2^n}.$$

Then $E \subseteq \bigcup_{n=1}^{\infty} \bigcup_{k=1}^{\infty} Q_{n,k}$ and

$$\sum_{n=1}^{\infty} \sum_{k=1}^{\infty} m_*(Q_{n,k}) \leq \sum_{n=1}^{\infty} \left(m_*(E_n) + \frac{\varepsilon}{2^n} \right) = \sum_{n=1}^{\infty} m_*(E_n) + \varepsilon.$$

Since ε is arbitrary, we conclude that $m_*(E) \leq \sum_{n=1}^{\infty} m_*(E_n)$. ■

PROPOSITION 1.8 (Outer Regularity). *If $E \subseteq \mathbb{R}^d$, then*

$$m_*(E) = \inf\{m_*(\mathcal{U}) : \mathcal{U} \text{ is open and } E \subseteq \mathcal{U}\}.$$

Proof. Clearly $m_*(E) \leq \inf\{m_*(\mathcal{U})\}$. On the other hand, given $\varepsilon > 0$, there exists a covering $E \subseteq \bigcup_{n=1}^{\infty} Q_n$ by closed cubes such that

$$\sum_{n=1}^{\infty} m_*(Q_n) \leq m_*(E) + \frac{\varepsilon}{2}.$$

Let Q_j^0 denote an open cube containing Q_j such that $|Q_j^0| \leq |Q_j| + \frac{\varepsilon}{2^{j+1}}$. Then $\mathcal{U} = \bigcup_{n=1}^{\infty} Q_n^0$ is an open set containing E and by countable subadditivity we have

$$m_*(\mathcal{U}) \leq \sum_{n=1}^{\infty} m_*(Q_n^0) \leq \sum_{n=1}^{\infty} \left(m_*(Q_n) + \frac{\varepsilon}{2^{n+1}} \right) \leq m_*(E) + \varepsilon.$$

Since ε is arbitrary, we conclude that $m_*(E) \geq \inf\{m_*(\mathcal{U})\}$. ■

PROPOSITION 1.9. *If $E = E_1 \cup E_2$ and $d(E_1, E_2) > 0$, then $m_*(E) = m_*(E_1) + m_*(E_2)$.*

Proof. By subadditivity, we have $m_*(E) \leq m_*(E_1) + m_*(E_2)$. On the other hand, fix $\varepsilon > 0$. Choose $\delta > 0$ such that $d(E_1, E_2) > \delta$. Choose a cover $E \subseteq \bigcup_{n=1}^{\infty} Q_n$ by closed cubes with side length less than δ . Then each Q_n can intersect at most one of E_1 and E_2 . Let $\mathcal{Q}_1$ be the collection of cubes that intersect E_1 , and let $\mathcal{Q}_2$ be the collection of cubes that intersect E_2 . Then

$$m_*(E_1) + m_*(E_2) \leq \sum_{Q \in \mathcal{Q}_1} m_*(Q) + \sum_{Q \in \mathcal{Q}_2} m_*(Q) \leq \sum_{n=1}^{\infty} m_*(Q_n) \leq m_*(E) + \varepsilon.$$

Since ε is arbitrary, we conclude that $m_*(E) \geq m_*(E_1) + m_*(E_2)$. ■

PROPOSITION 1.10. *If $E = \bigcup_{n=1}^{\infty} Q_n$ is a countable union of almost disjoint closed cubes, then $m_*(E) = \sum_{n=1}^{\infty} |Q_n|$.*

Proof. By subadditivity, we have $m_*(E) \leq \sum_{n=1}^{\infty} m_*(Q_n) = \sum_{n=1}^{\infty} |Q_n|$. On the other hand, given $\varepsilon > 0$, let $Q_{j'}$ be a cube strictly contained in Q_j such that $|Q_j| \leq |Q_{j'}| + \frac{\varepsilon}{2^j}$. Then for every N , the cubes $Q_{1'}, Q_{2'}, \dots, Q_{N'}$ are disjoint and have positive distance from each other, so

$$m_*\left(\bigcup_{j=1}^N Q_{j'}\right) = \sum_{j=1}^N |Q_{j'}| \geq \sum_{j=1}^N \left(|Q_j| - \frac{\varepsilon}{2^j}\right).$$

Since $\bigcup_{j=1}^N Q_{j'} \subseteq E$, we have

$$m_*(E) \geq \sum_{j=1}^N \left(|Q_j| - \frac{\varepsilon}{2^j} \right) \geq \sum_{j=1}^N |Q_j| - \varepsilon.$$

By letting $N \rightarrow \infty$, we conclude that $m_*(E) \geq \sum_{n=1}^{\infty} |Q_n| - \varepsilon$. Since ε is arbitrary, we conclude that $m_*(E) \geq \sum_{n=1}^{\infty} |Q_n|$. ■

1.3. Measurable Sets and the Lebesgue Measure

DEFINITION 1.11. A subset $E \subseteq \mathbb{R}^d$ is said to be (Lebesgue) measurable if for every $\varepsilon > 0$, there exists an open set $\mathcal{U} \subseteq \mathbb{R}^d$ such that $E \subseteq \mathcal{U}$ and $m_*(\mathcal{U} \setminus E) \leq \varepsilon$. If E is measurable, we define the Lebesgue measure of E to be $m(E) = m_*(E)$.

PROPOSITION 1.12. Every open set is measurable.

Proof. Follows immediately from the definition of measurability, since we can take $\mathcal{U} = E$ and $m_*(\emptyset) = 0$. ■

PROPOSITION 1.13. If $m_*(E) = 0$, then E is measurable.

Proof. Since $m_*(E) = 0$, by outer regularity there exists an open set $\mathcal{U}$ containing E such that $m_*(\mathcal{U}) \leq \varepsilon$. Then $\mathcal{U} \setminus E \subseteq \mathcal{U}$, so $m_*(\mathcal{U} \setminus E) \leq m_*(\mathcal{U}) \leq \varepsilon$. Since ε is arbitrary, we conclude that E is measurable. ■

PROPOSITION 1.14. A countable union of measurable sets is measurable.

Proof. Using the $\frac{\varepsilon}{2^n}$ argument, we can find a sequence of open sets $\mathcal{U}_1, \mathcal{U}_2, \dots$ such that $E_n \subseteq \mathcal{U}_n$ and $m_*(\mathcal{U}_n \setminus E_n) \leq \frac{\varepsilon}{2^n}$. Then $\bigcup_{n=1}^{\infty} E_n \subseteq \bigcup_{n=1}^{\infty} \mathcal{U}_n$ and

$$m_* \left(\bigcup_{n=1}^{\infty} \mathcal{U}_n \setminus \bigcup_{n=1}^{\infty} E_n \right) \leq m_* \left(\bigcup_{n=1}^{\infty} (\mathcal{U}_n \setminus E_n) \right) \leq \sum_{n=1}^{\infty} m_*(\mathcal{U}_n \setminus E_n) \leq \varepsilon.$$

Since ε is arbitrary, we conclude that $\bigcup_{n=1}^{\infty} E_n$ is measurable. ■

PROPOSITION 1.15. Closed sets are measurable.

Proof. It suffices to show that compact sets are measurable since every closed set can be written as a countable union of compact sets (take the intersection of the closed set with a large closed cube).

Let $F \subseteq \mathbb{R}^d$ be compact and let $\varepsilon > 0$. We have $m_*(F) < \infty$ since F is bounded. By outer regularity, there exists an open set $\mathcal{U}$ containing F such that $m_*(\mathcal{U}) \leq m_*(F) + \varepsilon$. Since F is closed, $\mathcal{U} \setminus F$ is open, so $\mathcal{U} \setminus F$ can be written as a countable union of almost disjoint cubes

$$\mathcal{U} \setminus F = \bigcup_{n=1}^{\infty} Q_n.$$

For each N , the finite union $K = \bigcup_{n=1}^N Q_n$ is compact hence has positive distance from F . Therefore,

$$m_*(\mathcal{U}) \geq m_*(F) + m_*(K) = m_*(F) + \sum_{n=1}^N |Q_n|$$

$$\Rightarrow \sum_{n=1}^N |Q_n| \leq m_*(\mathcal{U}) - m_*(F) \leq \varepsilon.$$

Letting $N \rightarrow \infty$, we conclude that

$$m_*(\mathcal{U} \setminus F) = m_*\left(\bigcup_{n=1}^{\infty} Q_n\right) \leq \sum_{n=1}^{\infty} |Q_n| \leq \varepsilon.$$

Since ε is arbitrary, we conclude that F is measurable. ■

PROPOSITION 1.16. *The complement of a measurable set is measurable.*

Proof. Suppose that E is measurable. For positive integer n , there exists an open set $\mathcal{U}_n$ containing E such that $m_*(\mathcal{U}_n \setminus E) \leq \frac{1}{n}$. Then $\mathcal{U}_n^c$ is a closed set contained in E^c , so

$$S := \bigcup_{n=1}^{\infty} \mathcal{U}_n^c \subseteq E^c$$

is measurable. Also, $E^c \setminus S = \bigcap_{n=1}^{\infty} (\mathcal{U}_n \setminus E)$, so $m_*(E^c \setminus S) \leq m_*(\mathcal{U}_n \setminus E) \leq \frac{1}{n}$ for each n , thus $m_*(E^c \setminus S) = 0$. Since S is measurable and $E^c \setminus S$ has zero outer measure, we conclude that E^c is measurable. ■

PROPOSITION 1.17. *A countable intersection of measurable sets is measurable.*

Proof. Follows immediately from the previous proposition and De Morgan's laws. ■

The family of measurable sets is closed under countable unions, countable intersections and complements, and contains all open sets and closed sets, and hence G_δ sets and F_σ sets.

THEOREM 1.18. *If $E_1, E_2, \dots$ are pairwise disjoint measurable sets, then*

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} m(E_n).$$

Proof. Assume each E_n is bounded. Apply the definition of measurability to each E_n^c to find an open set $\mathcal{U}_n \supseteq E_n^c$ such that $m_*(\mathcal{U}_n \setminus E_n^c) \leq \frac{\varepsilon}{2^n}$. Then $F_n := \mathcal{U}_n^c$ is a closed set contained in E_n with $m_*(E_n \setminus F_n) \leq \frac{\varepsilon}{2^n}$. Note that $F_1, F_2, \dots$ are compact and pairwise disjoint, so we have

$$m\left(\bigcup_{n=1}^N F_n\right) = \sum_{n=1}^N m(F_n).$$

Since $\bigcup_{n=1}^N F_n \subseteq \bigcup_{n=1}^N E_n$, we have

$$m\left(\bigcup_{n=1}^N E_n\right) \geq m\left(\bigcup_{n=1}^N F_n\right) = \sum_{n=1}^N m(F_n) \geq \sum_{n=1}^N \left(m(E_n) - \frac{\varepsilon}{2^n}\right).$$

By letting $N \rightarrow \infty$, we conclude that

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) \geq \sum_{n=1}^{\infty} m(E_n) - \varepsilon.$$

Since ε is arbitrary, we conclude that $m\left(\bigcup_{n=1}^{\infty} E_n\right) \geq \sum_{n=1}^{\infty} m(E_n)$. The reverse inequality follows from countable subadditivity.

In the general case, we can write each E_n as a countable union of bounded measurable sets, and then apply the above argument to these bounded sets. ■

1.3.1. Continuity of Lebesgue Measure

DEFINITION 1.19. A sequence of sets $E_1, E_2, \dots$ increases to E if $E_1 \subseteq E_2 \subseteq \dots$ and $\bigcup_{n=1}^{\infty} E_n = E$, denoted by $E_n \uparrow E$. A sequence of sets $E_1, E_2, \dots$ decreases to E if $E_1 \supseteq E_2 \supseteq \dots$ and $\bigcap_{n=1}^{\infty} E_n = E$, denoted by $E_n \downarrow E$.

PROPOSITION 1.20. Suppose that $E_1, E_2, \dots$ are measurable sets.

- (i) If $E_n \uparrow E$, then $m(E_n) \rightarrow m(E)$.
- (ii) If $E_n \downarrow E$ and $m(E_1) < \infty$, then $m(E_n) \rightarrow m(E)$.

Proof.

- (i) Let $F_1 = E_1$ and $F_n = E_n \setminus E_{n-1}$ for $n \geq 2$. Then $F_1, F_2, \dots$ are pairwise disjoint measurable sets and $\bigcup_{n=1}^N F_n = E_N$ for each N . Therefore,

$$m(E) = m\left(\bigcup_{n=1}^{\infty} F_n\right) = \sum_{n=1}^{\infty} m(F_n) = \lim_{N \rightarrow \infty} \sum_{n=1}^N m(F_n) = \lim_{N \rightarrow \infty} m(E_N).$$

- (ii) Let $F_k = E_k \setminus E_{k+1}$ for each k . Then $F_1, F_2, \dots$ are pairwise disjoint measurable sets and $E_1 = E \cup \bigcup_{k=1}^{\infty} F_k$. Since $m(E_k) < \infty$ for each k , we have

$$m(E_k) = m(F_k) + m(E_{k+1}) \Rightarrow m(F_k) = m(E_k) - m(E_{k+1}).$$

Therefore,

$$\begin{aligned} m(E_1) &= m(E) + \sum_{k=1}^{\infty} m(F_k) = m(E) + m(E_1) - \lim_{N \rightarrow \infty} m(E_{N+1}) \\ &\Rightarrow \lim_{N \rightarrow \infty} m(E_N) = m(E). \end{aligned}$$

■

1.3.2. Approximation of Measurable Sets

PROPOSITION 1.21. Suppose that $E \subseteq \mathbb{R}^d$ is measurable. Let $\varepsilon > 0$.

- (i) There exists a closed set $F \subseteq E$ such that $m(E \setminus F) \leq \varepsilon$.
- (ii) If $m(E) < \infty$, then there exists a compact set $K \subseteq E$ such that $m(E \setminus K) \leq \varepsilon$.
- (iii) If $m(E) < \infty$, then there exists a finite union F of closed cubes such that $m(E \triangle F) \leq \varepsilon$.

Proof.

- (i) Follows from outer regularity and the fact that the complement of measurable sets are measurable.
- (ii) By (a), there exists a closed set $F \subseteq E$ such that $m(E \setminus F) \leq \frac{\varepsilon}{2}$. Define $K_n = F \cap [-n, n]^d$ for each n . Then $K_n \uparrow F$, so $E \setminus K_n \downarrow E \setminus F$. By continuity of measure, we have $m(E \setminus K_n) \rightarrow m(E \setminus F) \leq \frac{\varepsilon}{2}$. Therefore, there exists N such that $m(E \setminus K_N) \leq \varepsilon$.
- (iii) By definition of outer measure, there exists a covering $E \subseteq \bigcup_{n=1}^{\infty} Q_n$ by closed cubes such that $\sum_{n=1}^{\infty} |Q_n| \leq m(E) + \frac{\varepsilon}{2}$. Since $m(E) < \infty$, the series is conver-

gent, so there exists N such that $\sum_{n=N+1}^{\infty} |Q_n| \leq \frac{\varepsilon}{2}$. Let $F = \bigcup_{n=1}^N Q_n$. Then F is a finite union of closed cubes and

$$\begin{aligned}
 m(E \triangle F) &\leq m(E \setminus F) + m(F \setminus E) \\
 &\leq m\left(E \setminus \bigcup_{n=1}^N Q_n\right) + m\left(\bigcup_{n=1}^N Q_n \setminus E\right) \\
 &\leq m\left(\bigcup_{n=N+1}^{\infty} Q_n\right) + m\left(\bigcup_{n=1}^{\infty} Q_n \setminus E\right) \\
 &\leq \sum_{n=N+1}^{\infty} |Q_n| + \sum_{n=1}^{\infty} |Q_n| - m(E) \\
 &\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.
 \end{aligned}$$

■

1.3.3. Translation Invariance and Scaling

PROPOSITION 1.22. *If $E \subseteq \mathbb{R}^d$ is measurable, then so is $E + h = \{x + h : x \in E\}$. Moreover, $m(E + h) = m(E)$.*

Proof. This holds when E is a closed cube. From the definition of outer measure, we have $m_*(E + h) = m_*(E)$. It remains to show that $E + h$ is measurable. Given $\varepsilon > 0$, there exists an open set $\mathcal{U}$ containing E such that $m(\mathcal{U} \setminus E) \leq \varepsilon$. Then $\mathcal{U} + h$ is an open set containing $E + h$, and

$$m_*((\mathcal{U} + h) \setminus (E + h)) = m_*(\mathcal{U} \setminus E) \leq \varepsilon,$$

so $E + h$ is measurable and $m(E + h) = m_*(E + h) = m_*(E) = m(E)$. ■

Remark. Similarly, one can prove that $m(aE) = |a|^d m(E)$ for any nonzero real number a .

1.3.4. σ -Algebra and Borel Sets

DEFINITION 1.23. *A σ -algebra of sets is a collection of subsets of a set Ω that consists of $\emptyset$, and is closed under countable unions and complements.*

Example 1.2.

- All measurable sets in $\mathbb{R}^d$ form a σ -algebra.
- The power set of Ω is a σ -algebra.
- $\{\emptyset, \Omega\}$ is a σ -algebra.

Note that the intersection of any collection of σ -algebras is still a σ -algebra. Therefore, given a family $\mathcal{F}$ of subsets of Ω , there exists a smallest σ -algebra containing $\mathcal{F}$, which is the intersection of all σ -algebras containing $\mathcal{F}$.

The Borel σ -algebra on $\mathbb{R}^d$, denoted by $\mathcal{B}_{\mathbb{R}^d}$, is the smallest σ -algebra containing all open sets in $\mathbb{R}^d$. The elements of $\mathcal{B}_{\mathbb{R}^d}$ are called Borel sets.

We can easily see that $\mathcal{B}_{\mathbb{R}^d}$ is contained in the family of measurable sets. In fact, the containment is strict. However, a measurable set can be approximated by Borel sets in the following sense.

PROPOSITION 1.24. *The following are equivalent for a subset $E \subseteq \mathbb{R}^d$:*

- (i) E is measurable.
- (ii) $m(G \setminus E) = 0$ for some G_δ set $G \supseteq E$.
- (iii) $m(E \setminus F) = 0$ for some F_σ set $F \subseteq E$.

Proof. (i) $\Rightarrow$ (ii): Suppose that E is measurable. For each positive integer n , there exists an open set $\mathcal{U}_n$ containing E such that $m_*(\mathcal{U}_n \setminus E) \leq \frac{1}{n}$. Then $G := \bigcap_{n=1}^{\infty} \mathcal{U}_n$ is a G_δ set containing E , and

$$m(G \setminus E) \leq m(\mathcal{U}_n \setminus E) \leq \frac{1}{n}$$

for each n , so $m(G \setminus E) = 0$.

(ii) $\Rightarrow$ (iii): Follows from the fact that the complement of a G_δ set is an F_σ set, and measurable sets are closed under complements.

(iii) $\Rightarrow$ (i): Obvious since E is now the union of a measurable set F and a set of measure zero. ■

The Lebesgue measurable sets can be seen as the completion of the Borel σ -algebra with respect to the Lebesgue measure.

1.3.5. Nonmeasurable Sets

Define the equivalence relation $\sim$ on $[0, 1]$ by $x \sim y$ if $x - y \in \mathbb{Q}$. Write

$$[0, 1] = \bigsqcup_{\alpha \in A} E_\alpha,$$

where E_α are the equivalence classes. Using the axiom of choice, we can choose a representative x_α from each E_α , and let $N = \{x_\alpha : \alpha \in A\}$. Then N is a nonmeasurable set.

THEOREM 1.25. *N is not measurable.*

Proof. Suppose that N is measurable. Let (r_k) be the enumeration of rational numbers in $[-1, 1]$. Define

$$N_k = N + r_k.$$

If $N_k \cap N_l \neq \emptyset$ for some $k \neq l$, then there exist $x_\alpha, x_\beta \in N$ such that $x_\alpha + r_k = x_\beta + r_l$, so $x_\alpha - x_\beta = r_l - r_k \in \mathbb{Q}$. This contradicts the definition of N . Therefore, the sets N_k are pairwise disjoint.

Let $x \in [0, 1]$. Then $x \sim x_\alpha$ for some α , so $x - x_\alpha \in \mathbb{Q} \cap [-1, 1]$. Therefore, $x \in N_k$ for some k , so

$$[0, 1] \subseteq \bigcup_{k=1}^{\infty} N_k \subseteq [-1, 2].$$

Since N_k are pairwise disjoint and have the same measure as N , we have

$$1 \leq \sum_{k=1}^{\infty} m(N) \leq 3.$$

This is a contradiction since $m(N)$ must be zero or positive. Therefore, N is not measurable. ■

1.4. The Brunn-Minkowski Inequality

For two measurable sets $A, B \subseteq \mathbb{R}^d$, we define their (Minkowski) sum to be

$$A + B = \{a + b : a \in A, b \in B\}.$$

In general, it is impossible to upper bound $m(A + B)$ in terms of $m(A)$ and $m(B)$, since there exists sets A, B of measure zero such that $A + B$ has positive measure.

THEOREM 1.26 (Brunn-Minkowski Inequality). *Suppose $A, B \subseteq \mathbb{R}^d$ are measurable sets, and $A + B$ is also measurable. Then*

$$m(A + B)^{\frac{1}{d}} \geq m(A)^{\frac{1}{d}} + m(B)^{\frac{1}{d}}.$$

Proof. We first assume that A, B are rectangles with side lengths (a_i) and (b_i) respectively. Then $A + B$ is a rectangle with side lengths $(a_i + b_i)$. Note that

$$\begin{aligned} \left(\prod_{i=1}^d \frac{a_i}{a_i} + b_i \right)^{\frac{1}{d}} + \left(\prod_{i=1}^d \frac{b_i}{a_i} + b_i \right)^{\frac{1}{d}} &\leq \frac{1}{d} \sum_{i=1}^d \frac{a_i}{a_i} + b_i + \frac{1}{d} \sum_{i=1}^d \frac{b_i}{a_i} + b_i = 1, \\ \Rightarrow m(A + B)^{\frac{1}{d}} &= \prod_{i=1}^d (a_i + b_i)^{\frac{1}{d}} \geq \prod_{i=1}^d a_i^{\frac{1}{d}} + \prod_{i=1}^d b_i^{\frac{1}{d}} = m(A)^{\frac{1}{d}} + m(B)^{\frac{1}{d}}. \end{aligned}$$

Now suppose that A, B are unions of finitely almost disjoint rectangles. We proceed by induction on the number of rectangles. Choose a pair of disjoint rectangles $R_1, R_2 \subseteq A$, and we may WLOG assume that after some translation, R_1 and R_2 are separated by a coordinate hyperplane $x_j = 0$. Let $A_- = A \cap \{x_j \leq 0\}$ and $A_+ = A \cap \{x_j \geq 0\}$, then both A_- and A_+ have fewer rectangles than A . We then translate B so that $B_- = B \cap \{x_j \leq 0\}$ and $B_+ = B \cap \{x_j \geq 0\}$ satisfy

$$\frac{m(B_+)}{m(B)} = \frac{m(A_+)}{m(A)}.$$

Note that A_+ has strictly fewer rectangles than A , and B_+ has at most as many rectangles as B , and same goes for A_- and B_- . We have

$$A + B \supset (A_+ + B_+) \cup (A_- + B_-).$$

By induction hypothesis, we have

$$\begin{aligned} m(A + B) &\geq m(A_+ + B_+) + m(A_- + B_-) \\ &\geq \left(m(A_+)^{\frac{1}{d}} + m(B_+)^{\frac{1}{d}} \right)^d + \left(m(A_-)^{\frac{1}{d}} + m(B_-)^{\frac{1}{d}} \right)^d \\ &\geq m(A_+) \left(1 + \left(\frac{m(B_+)}{m(A_+)} \right)^{\frac{1}{d}} \right)^d + m(A_-) \left(1 + \left(\frac{m(B_-)}{m(A_-)} \right)^{\frac{1}{d}} \right)^d \\ &\geq m(A) \left(1 + \left(\frac{m(B)}{m(A)} \right)^{\frac{1}{d}} \right)^d = \left(m(A)^{\frac{1}{d}} + m(B)^{\frac{1}{d}} \right)^d. \end{aligned}$$

Next, suppose that A, B are open sets with finite measures. For any $\varepsilon > 0$, we can find finite unions of almost disjoint cubes $A_\varepsilon \subseteq A$ and $B_\varepsilon \subseteq B$ such that $m(A) \leq m(A_\varepsilon) + \varepsilon$ and $m(B) \leq m(B_\varepsilon) + \varepsilon$. Then $A_\varepsilon + B_\varepsilon \subseteq A + B$, so

$$m(A + B) \geq m(A_\varepsilon + B_\varepsilon)$$

$$\begin{aligned}
&\geq \left(m(A_\varepsilon)^{\frac{1}{d}} + m(B_\varepsilon)^{\frac{1}{d}} \right)^d \\
&\geq \left((m(A) - \varepsilon)^{\frac{1}{d}} + (m(B) - \varepsilon)^{\frac{1}{d}} \right)^d.
\end{aligned}$$

By letting $\varepsilon \rightarrow 0$, we conclude that $m(A + B) \geq \left(m(A)^{\frac{1}{d}} + m(B)^{\frac{1}{d}} \right)^d$.

Suppose A, B are compact sets. If A, B are compact, then $A + B$ is also compact, in particular measurable. Define $A_\varepsilon = \{x \in \mathbb{R}^d : d(x, A) \leq \varepsilon\}$ and $B_\varepsilon = \{x \in \mathbb{R}^d : d(x, B) \leq \varepsilon\}$. Then $A_\varepsilon \downarrow A$ and $B_\varepsilon \downarrow B$, so $A_\varepsilon + B_\varepsilon \downarrow A + B$. By continuity of measure, we have

$$m(A + B) = \lim_{\varepsilon \rightarrow 0} m(A_\varepsilon + B_\varepsilon) \geq \lim_{\varepsilon \rightarrow 0} \left(m(A_\varepsilon)^{\frac{1}{d}} + m(B_\varepsilon)^{\frac{1}{d}} \right)^d = \left(m(A)^{\frac{1}{d}} + m(B)^{\frac{1}{d}} \right)^d.$$

Finally, for general measurable sets A, B , if $m(A) = \infty$ or $m(B) = \infty$, then $m(A + B) = \infty$, so the inequality is trivial. Otherwise, we can find compact sets $K_A \subseteq A$ and $K_B \subseteq B$ such that $m(A \setminus K_A) \leq \varepsilon$ and $m(B \setminus K_B) \leq \varepsilon$. Then $K_A + K_B \subseteq A + B$, so

$$\begin{aligned}
m(A + B) &\geq m(K_A + K_B) \\
&\geq \left(m(K_A)^{\frac{1}{d}} + m(K_B)^{\frac{1}{d}} \right)^d \\
&\geq \left((m(A) - \varepsilon)^{\frac{1}{d}} + (m(B) - \varepsilon)^{\frac{1}{d}} \right)^d.
\end{aligned}$$

By letting $\varepsilon \rightarrow 0$, we conclude that $m(A + B) \geq \left(m(A)^{\frac{1}{d}} + m(B)^{\frac{1}{d}} \right)^d$. ■

Remark. In general, A, B being measurable does not imply that $A + B$ is measurable. For example, let N be the nonmeasurable set constructed above, and let $A = \{0\} \times N$ and $B = N \times \{0\}$, then $A + B = N \times N$ is not measurable.

1.5. Measurable Functions

Consider functions from $\mathbb{R}^d$ to extended real numbers $[-\infty, \infty] := \mathbb{R} \cup \{\pm\infty\}$.

DEFINITION 1.27. A function $f : E \rightarrow [-\infty, \infty]$ is measurable if for every $a \in \mathbb{R}$, the set

$$f^{-1}([-\infty, a)) = \{x \in E : f(x) < a\}$$

is measurable.

Roughly speaking, measurable functions are those that we would like to integrate. The simplest function to integrate is the characteristic function χ_E of a measurable set E .

PROPOSITION 1.28. f is measurable if and only if

$$f^{-1}((-\infty, a]) = \{x \in E : f(x) \leq a\}$$

is measurable for every $a \in \mathbb{R}$.

Proof. If $f^{-1}([-\infty, a))$ is measurable for every a , then $f^{-1}((-\infty, a]) = \bigcap_{n=1}^{\infty} f^{-1}([-\infty, a + \frac{1}{n}))$ is measurable for every a . Conversely, if $f^{-1}((-\infty, a])$ is measurable for every a , then $f^{-1}([-\infty, a)) = \bigcup_{n=1}^{\infty} f^{-1}((-\infty, a - \frac{1}{n}])$ is measurable for every a . ■

In the same way, we can show that f is measurable if and only if $f^{-1}((a, b))$ is measurable for every $a < b$.

PROPOSITION 1.29. *A finite-valued function f is measurable if and only if $f^{-1}(\mathcal{O})$ is measurable for every open set $\mathcal{O} \subseteq \mathbb{R}$, and if and only if $f^{-1}(\mathcal{F})$ is measurable for every closed set $\mathcal{F} \subseteq \mathbb{R}$.*

PROPOSITION 1.30. *If f is continuous on $\mathbb{R}^d$, then f is measurable.*

Proof. Since f is continuous, it is finite-valued. By the previous proposition it is measurable. ■

PROPOSITION 1.31. *If f is measurable and Φ is continuous, then $\Phi \circ f$ is measurable.*

Proof. Since Φ is continuous, $\Phi^{-1}((-\infty, a))$ is open, hence

$$(\Phi \circ f)^{-1}((-\infty, a)) = f^{-1}(\Phi^{-1}((-\infty, a)))$$

is measurable for every a , so $\Phi \circ f$ is measurable. ■

PROPOSITION 1.32. *If (f_n) is a sequence of measurable functions, then*

$$\sup_n f_n, \quad \inf_n f_n, \quad \limsup_{n \rightarrow \infty} f_n, \quad \liminf_{n \rightarrow \infty} f_n$$

are all measurable.

Proof.

$$\left\{ \sup_n f_n > a \right\} = \bigcup_{n=1}^{\infty} \{f_n > a\}, \quad \left\{ \inf_n f_n < a \right\} = \bigcup_{n=1}^{\infty} \{f_n < a\}.$$

Therefore, $\sup_n f_n$ and $\inf_n f_n$ are measurable. Since

$$\limsup_{n \rightarrow \infty} f_n = \inf_N \sup_{n \geq N} f_n, \quad \liminf_{n \rightarrow \infty} f_n = \sup_N \inf_{n \geq N} f_n,$$

$\limsup_{n \rightarrow \infty} f_n$ and $\liminf_{n \rightarrow \infty} f_n$ are measurable as well. ■

CLAIM 1.33. *If (f_n) is a sequence of measurable functions and $\lim_{n \rightarrow \infty} f_n = f$ pointwise, then f is measurable.*

PROPOSITION 1.34. *If f is measurable, then so is f^k for any positive integer k .*

Proof. If k is odd, then $\{f^k > a\} = \{f > a^{\frac{1}{k}}\}$. If k is even, then $\{f^k > a\} = \{f > a^{\frac{1}{k}}\} \cup \{f < -a^{\frac{1}{k}}\}$. In either case, f^k is measurable. ■

PROPOSITION 1.35. *If f, g are finite-valued measurable functions, then $f + g$ and fg are measurable.*

Proof. Note that

$$\{f + g > a\} = \bigcup_{q \in \mathbb{Q}} \{f > q\} \cap \{g > a - q\},$$

hence $f + g$ is measurable. Write

$$fg = \frac{1}{4}((f + g)^2 - (f - g)^2),$$

hence fg is measurable. ■

DEFINITION 1.36. *Two functions f, g defined on a measurable set E are equal almost everywhere, denoted by $f = g$ a.e., if the set $\{x \in E : f(x) \neq g(x)\}$ has measure zero.*

PROPOSITION 1.37. *If f is measurable and $g = f$ a.e., then g is measurable.*

Proof. $\{g < a\}$ and $\{f < a\}$ differ by a subset of $\{x \in E : f(x) \neq g(x)\}$, which has measure zero, so g is measurable. ■

1.5.1. Approximation of Measurable Functions

DEFINITION 1.38. *A step function is a finite linear combination of characteristic functions of rectangles, i.e. a function of the form*

$$f = \sum_{k=1}^N a_k \chi_{R_k},$$

where a_k are real numbers and R_k are rectangles.

DEFINITION 1.39. *A simple function is a finite linear combination of characteristic functions of measurable sets with finite measures, i.e. a function of the form*

$$f = \sum_{k=1}^N a_k \chi_{E_k},$$

where a_k are real numbers and E_k are measurable sets with finite measures.

THEOREM 1.40. *Suppose that f is a nonnegative measurable function. Then there exists an increasing sequence of nonnegative simple functions (φ_k) such that $\varphi_k \rightarrow f$ pointwise.*

Proof. For $N \geq 1$, let Q_N be the cube centered at the origin with side length N . Define the truncated function

$$F_N(x) = \begin{cases} f(x) & x \in Q_N \text{ and } f(x) \leq N \\ N & x \in Q_N \text{ and } f(x) > N \\ 0 & x \notin Q_N \end{cases}$$

Then $F_N \rightarrow f$ pointwise as $N \rightarrow \infty$. Also, each F_N is measurable and bounded. For fixed $N, M \geq 1$, define

$$E_{l,M} = \left\{ x \in Q_N : \frac{l}{M} < F_N(x) \leq \frac{l+1}{M} \right\}$$

for $l = 0, 1, \dots, MN - 1$. Then $E_{l,M}$ are measurable sets with finite measures. Define

$$F_{N,M} = \sum_{l=0}^{MN-1} \frac{l}{M} \chi_{(E_{l,M})}.$$

Then $F_{N,M}$ are simple functions and

$$0 \leq F_N - F_{N,M} \leq \frac{1}{M}.$$

Choose $N = M = 2^k$ and set $\varphi_k = F_{2^k, 2^k}$ for each k . Then (φ_k) is an increasing sequence of nonnegative simple functions such that $\varphi_k \rightarrow f$ pointwise as $k \rightarrow \infty$.

Note that the result holds for extended-value nonnegative measurable functions, if we allow the limit to be ∞ . ■

THEOREM 1.41. *Suppose that f is a measurable function. Then there exists a sequence of simple functions (φ_k) such that*

$$|\varphi_{k(x)}| \leq |\varphi_{k+1}(x)|, \quad \lim_{k \rightarrow \infty} \varphi_{k(x)} = f(x)$$

for every x .

Proof. Write $f = f_+ - f_-$, where $f_+ = \max(f, 0)$ and $f_- = \max(-f, 0)$. Then f_+ and f_- are nonnegative measurable functions. By the previous theorem, there exist increasing sequences of nonnegative simple functions (φ_k^+) and (φ_k^-) which increase to f_+ and f_- pointwise respectively. Define $\varphi_k = \varphi_k^+ - \varphi_k^-$, then $\varphi_k \rightarrow f$ pointwise as $k \rightarrow \infty$. Moreover, we have

$$|\varphi_k| = \varphi_k^+ + \varphi_k^- \leq \varphi_{k+1}^+ + \varphi_{k+1}^- = |\varphi_{k+1}|.$$

■

THEOREM 1.42. *Suppose that f is a measurable function. Then there exists a sequence of step functions (ψ_k) such that $\psi_k \rightarrow f$ a.e. as $k \rightarrow \infty$.*

Proof. By the previous theorem, it suffices to show the result for χ_E , where E is a measurable set with finite measure. Let $\varepsilon > 0$. Recall that there exists a finite union $\bigcup_{n=1}^N Q_n$ of cubes such that

$$m\left(E \triangle \bigcup_{n=1}^N Q_n\right) \leq \varepsilon.$$

Extend the sides these cubes and obtain a grid, then we can write $\bigcup_{n=1}^N Q_n$ as a finite union of almost disjoint rectangles $R_1, \dots, R_M$. By taking rectangles R_j slightly smaller than R_j , we can ensure that $R_1, \dots, R_M$ are pairwise disjoint rectangles such that

$$m\left(E \triangle \bigcup_{j=1}^M R_j\right) \leq 2\varepsilon.$$

Hence $f(x) = \sum_{j=1}^M \chi_{R_j}(x)$ for all x except for a set of measure at most 2ε . Consequently, for each k , we can find a step function ψ_k such that

$$m(\underbrace{\{x : \psi_k(x) \neq f(x)\}}_{=: E_k}) \leq 2^{-k}.$$

Note that f converges to f on the complement of $\bigcap_{j=1}^{\infty} \bigcup_{k=j}^{\infty} E_k$, which has measure zero. Therefore, $\psi_k \rightarrow f$ a.e. as $k \rightarrow \infty$. ■

1.5.2. Littlewood's Three Principles

- (i) Every measurable set in $\mathbb{R}$ is nearly a finite union of disjoint intervals.
- (ii) Every measurable function on $\mathbb{R}$ is nearly continuous.
- (iii) Every convergent sequence of measurable functions is nearly uniformly convergent.

THEOREM 1.43 (Egorov). *Suppose that (f_k) is a sequence of measurable functions on a measurable set E with finite measure, and $f_k \rightarrow f$ almost everywhere on E . Then for every $\varepsilon > 0$, there exists a closed set $A_\varepsilon \subseteq E$ such that $m(E \setminus A_\varepsilon) \leq \varepsilon$ and $f_k \rightarrow f$ uniformly on A_ε .*

Proof. We may assume WLOG that $f_k \rightarrow f$ everywhere on E . For $n, k \geq 1$, define

$$E_k^n = \left\{ x \in E : |f_{j(x)} - f(x)| < \frac{1}{n} \text{ for all } j \geq k \right\}.$$

Then $E_k^n \uparrow E$ as $k \rightarrow \infty$, so by continuity of measure, we have $m(E \setminus E_k^n) \rightarrow 0$ as $k \rightarrow \infty$. Therefore, there exists k_n such that $m(E \setminus E_{k_n}^n) \leq 2^{-n}$ for each n . Choose N large enough such that $\sum_{n=N}^{\infty} 2^{-n} \leq \frac{\varepsilon}{2}$. Let $A_{\varepsilon'} = \bigcap_{n=N}^{\infty} E_{k_n}^n$, then $m(E \setminus A_{\varepsilon'}) \leq \frac{\varepsilon}{2}$.

Let $\delta > 0$. Choose $n \geq N$ such that $\frac{1}{n} < \delta$. Then for every $x \in A_{\varepsilon'}$ and $j \geq k_n$, we have $|f_{j(x)} - f(x)| < \frac{1}{n} < \delta$. Therefore, $f_j \rightrightarrows f$ uniformly on $A_{\varepsilon'}$. Since $A_{\varepsilon'}$ is measurable, there exists a closed set $A_{\varepsilon} \subseteq A_{\varepsilon'}$ such that $m(A_{\varepsilon'} \setminus A_{\varepsilon}) \leq \frac{\varepsilon}{2}$. Then

$$m(E \setminus A_{\varepsilon}) \leq m(E \setminus A_{\varepsilon'}) + m(A_{\varepsilon'} \setminus A_{\varepsilon}) \leq \varepsilon.$$

Moreover, $f_j \rightrightarrows f$ uniformly on A_{ε} since $A_{\varepsilon} \subseteq A_{\varepsilon'}$. ■

THEOREM 1.44 (Lusin). *Suppose that f is a finite-valued measurable function on a measurable set E with finite measure. Then for every $\varepsilon > 0$, there exists a closed set $F_{\varepsilon} \subseteq E$ such that $m(E \setminus F_{\varepsilon}) \leq \varepsilon$ and $f|_{F_{\varepsilon}}$ is continuous.*

Proof. By Theorem 1.42, there exists a sequence of step functions (ψ_k) such that $\psi_k \rightarrow f$ a.e. as $k \rightarrow \infty$. Find measurable sets E_k such that $m(E_k) \leq 2^{-k}$ and ψ_k is continuous outside E_k . By Egorov's theorem, there exists a closed set $A_{\varepsilon} \subseteq E$ such that $m(E \setminus A_{\varepsilon}) \leq \frac{\varepsilon}{3}$ and $\psi_k \rightrightarrows f$ uniformly on A_{ε} . Pick N large enough such that $\sum_{k=N}^{\infty} 2^{-k} \leq \frac{\varepsilon}{3}$, and let

$$F_{\varepsilon'} = A_{\varepsilon} \setminus \bigcup_{k=N}^{\infty} E_k.$$

Then for all $n \geq N$, ψ_n is continuous on $F_{\varepsilon'}$, so f is the uniform limit of continuous functions on $F_{\varepsilon'}$, hence continuous on $F_{\varepsilon'}$. Since $F_{\varepsilon'}$ is measurable, there exists a closed set $F_{\varepsilon} \subseteq F_{\varepsilon'}$ such that $m(F_{\varepsilon'} \setminus F_{\varepsilon}) \leq \frac{\varepsilon}{3}$. Then

$$m(E \setminus F_{\varepsilon}) \leq m(E \setminus A_{\varepsilon}) + m(A_{\varepsilon} \setminus F_{\varepsilon'}) + m(F_{\varepsilon'} \setminus F_{\varepsilon}) \leq \varepsilon.$$

Moreover, $f|_{F_{\varepsilon}}$ is continuous since $F_{\varepsilon} \subseteq F_{\varepsilon'}$. ■

Integration Theory

CHAPTER 2

In this chapter, we will define the Lebesgue integral for integrable functions in a progressive manner. Starting from simple functions, bounded measurable functions on sets of finite measure, nonnegative measurable functions, and finally general integrable functions.

2.1. The Lebesgue Integral

A simple function is a finite sum

$$\varphi = \sum_{k=1}^N a_k \chi_{E_k},$$

where a_k are real numbers and E_k are measurable sets with finite measures. There is usually more than one way to write a simple function as a linear combination of characteristic functions. A decomposition of φ is canonical if all E_k are pairwise disjoint and $a_k \neq 0$ are distinct for each k . Every simple function has a unique canonical decomposition.

If φ is a simple function with canonical decomposition $\varphi = \sum_{k=1}^N a_k \chi_{E_k}$, then we define the Lebesgue integral of φ to be

$$\int_{\mathbb{R}^d} \varphi(x) dx = \sum_{k=1}^N a_k m(E_k).$$

If $E \subseteq \mathbb{R}^d$ is a measurable set, then we define

$$\int_E \varphi(x) dx = \int_{\mathbb{R}^d} \varphi(x) \chi_{E(x)} dx.$$

PROPOSITION 2.1. *If $\varphi = \sum_{k=1}^M b_k \chi_{F_k}$ is any other decomposition of φ , then*

$$\int \varphi = \sum_{k=1}^M b_k m(F_k).$$

Proof. Assume that F_k are pairwise disjoint, then for each nonzero value b in $\{b_k\}$, define $F_{b'} = \bigcup_{b_k=b} F_k$. Then clearly $\varphi = \sum_b b \chi_{F_{b'}}$ is the canonical decomposition of φ , and the finite sum equals $\int \varphi$.

In general, we can refine $\bigcup_{k=1}^M F_k$ into a finite union of pairwise disjoint sets $F_{1'}, \dots, F_{N'}$ such that $\bigcup_{k=1}^M F_k = \bigcup_{n=1}^{N'} F_{n'}$, and $F_k = \bigcup_{n: F_{n'} \subseteq F_k} F_{n'}$ for each k . For each n , let $b_{n'} = \sum_{k: F_{n'} \subseteq F_k} b_k$, then clearly $\varphi = \sum_{n=1}^{N'} b_{n'} \chi_{F_{n'}}$. This reduces to the previous case, so the finite sum equals $\int \varphi$. ■

PROPOSITION 2.2. *If φ, ψ are simple functions and $a, b \in \mathbb{R}$, then*

$$\int (a\varphi + b\psi) = a \int \varphi + b \int \psi.$$

Proof. Follows from the previous proposition and the linearity of finite sums. ■

PROPOSITION 2.3. *if E, F are disjoint subsets of $\mathbb{R}^d$ with finite measures, then*

$$\int_{E \cup F} \varphi = \int_E \varphi + \int_F \varphi.$$

Proof. Follows from the previous proposition and the fact that $\chi_{E \cup F} = \chi_E + \chi_F$. ■

PROPOSITION 2.4. *If $\varphi \leq \psi$ are simple functions, then $\int \varphi \leq \int \psi$.*

Proof. It suffices to show that if $\eta \geq 0$ is a simple function, then $\int \eta \geq 0$. This is clear since the canonical decomposition of η has nonnegative coefficients, so $\int \eta$ is a sum of nonnegative terms. ■

PROPOSITION 2.5. *If φ is a simple function, then so is $|\varphi|$, and*

$$\int |\varphi| \geq \left| \int \varphi \right|.$$

Proof. Write $|\varphi| = \sum_{k=1}^N |a_k| \chi_{E_k}$, then

$$\int |\varphi| = \sum_{k=1}^N |a_k| m(E_k) \geq \left| \sum_{k=1}^N a_k m(E_k) \right| = \left| \int \varphi \right|.$$

■

PROPOSITION 2.6. *If two simple functions φ, ψ are equal a.e., then $\int \varphi = \int \psi$.*

Proof. Let $\eta = |\varphi - \psi|$, then η is nonzero only on a set of measure zero, so $\int \eta = 0$. ■

The support of a measurable function f is defined as

$$\text{supp } f = \{x : f(x) \neq 0\}.$$

We say that f is supported on a set E if $\text{supp } f \subseteq E$. If f is measurable, then $\text{supp } f$ is measurable. We'll consider bounded measurable functions with supports of finite measures.

Recall that given a measurable function f with $|f| \leq M$ and supported on E , there exists a sequence of simple functions (φ_k) such that $|\varphi_k| \leq |\varphi_{k+1}|$ and $\varphi_k \rightarrow f$ pointwise as $k \rightarrow \infty$. If f is bounded and supported on a set of finite measure, then we can choose φ_k to be bounded and supported on a set of finite measure as well.

LEMMA 2.7. *Let f be as above, and assume that $m(E) < \infty$. If (φ_n) is any sequence of simple functions with $|\varphi_n| \leq M$ and $\text{supp } \varphi_n \subseteq E$ such that $\varphi_n \rightarrow f$ a.e. as $n \rightarrow \infty$, then*

- (i) $\int \varphi_n$ converges to a limit as $n \rightarrow \infty$.
- (ii) If $f = 0$ a.e., then $\int \varphi_n \rightarrow 0$ as $n \rightarrow \infty$.

Proof. If $\varphi_k \rightrightarrows f$ uniformly on E , then both (i) and (ii) hold.

In general, by Egorov's theorem, for every $\varepsilon > 0$, there exists a closed set $A_\varepsilon \subseteq E$ such that $m(E \setminus A_\varepsilon) \leq \varepsilon$ and $\varphi_n \rightrightarrows f$ uniformly on A_ε . Then for all $m, n \geq 1$, we have

$$\begin{aligned} \left| \int \varphi_n - \int \varphi_m \right| &= \left| \int (\varphi_n - \varphi_m) \right| \\ &\leq \int_E |\varphi_n - \varphi_m| \\ &\leq \int_{A_\varepsilon} |\varphi_n - \varphi_m| + \int_{E \setminus A_\varepsilon} |\varphi_n - \varphi_m| \\ &\leq m(A_\varepsilon) \sup_{x \in A_\varepsilon} |\varphi_n(x) - \varphi_m(x)| + 2Mm(E \setminus A_\varepsilon) \\ &\leq m(E) \sup_{x \in A_\varepsilon} |\varphi_n(x) - \varphi_m(x)| + 2M\varepsilon. \end{aligned}$$

By uniform convergence of φ_n , the first term goes to zero as $n, m \rightarrow \infty$, so $\int \varphi_n$ is a Cauchy sequence, hence converges to a limit.

Part (ii) follows from the same argument. ■

Define

$$\int f = \lim_{n \rightarrow \infty} \int \varphi_n,$$

where (φ_n) is any sequence of simple functions with $|\varphi_n| \leq M$ and $\text{supp } \varphi_n \subseteq \text{supp } f$ such that $\varphi_n \rightarrow f$ a.e. as $n \rightarrow \infty$. The previous lemma guarantees that the limit exists and is independent of the choice of (φ_n) , so $\int f$ is well-defined.

Since the integral is defined as a limit of integrals of simple functions, it inherits properties such as linearity and monotonicity from the integral of simple functions.

THEOREM 2.8 (Bounded Convergence Theorem). *Let (f_n) be a sequence of measurable functions such that $|f_n| \leq M$ and $\text{supp } f_n \subseteq E$ for some $M > 0$ and some measurable set E with finite measure. If $f_n \rightarrow f$ a.e. as $n \rightarrow \infty$, then f is measurable and $\int f_n \rightarrow \int f$ as $n \rightarrow \infty$.*

Proof. Clearly, f is measurable, $|f| \leq M$ and $\text{supp } f \subseteq E$. Let $\varepsilon > 0$. By Egorov's theorem, there exists a closed set $A_\varepsilon \subseteq E$ such that $m(E \setminus A_\varepsilon) \leq \varepsilon$ and $f_n \rightrightarrows f$ uniformly on A_ε . Then for all $n \geq 1$, we have

$$\begin{aligned} \left| \int f_n - \int f \right| &= \left| \int (f_n - f) \right| \\ &\leq \int_E |f_n - f| \\ &\leq \int_{A_\varepsilon} |f_n - f| + \int_{E \setminus A_\varepsilon} |f_n - f| \\ &\leq m(A_\varepsilon) \sup_{x \in A_\varepsilon} |f_n(x) - f(x)| + 2Mm(E \setminus A_\varepsilon) \\ &\leq m(E) \sup_{x \in A_\varepsilon} |f_n(x) - f(x)| + 2M\varepsilon. \end{aligned}$$

By uniform convergence of f_n , the first term goes to zero as $n \rightarrow \infty$, so $\int f_n \rightarrow \int f$ as $n \rightarrow \infty$. ■

PROPOSITION 2.9. *If $f \geq 0$ is bounded and supported on E with $m(E) < \infty$, and if $\int f = 0$, then $f = 0$ a.e.*

Proof. For $k \geq 1$, define

$$E_k = \left\{ x : f(x) \geq \frac{1}{k} \right\}.$$

Then E_k are measurable sets with finite measures and $f \geq (\frac{1}{k})\chi_{E_k}$, so by monotonicity,

$$\frac{1}{k}m(E_k) \leq \int f = 0 \Rightarrow m(E_k) = 0.$$

Therefore, $\{x : f(x) > 0\} = \bigcup_{k=1}^{\infty} E_k$ has measure zero, so $f = 0$ a.e. ■

In fact, we can show that Lebesgue integral is a generalization of Riemann integral.

Let (φ_n) and (ψ_n) be two sequences of step functions such that (φ_n) is increasing, (ψ_n) is decreasing,

$$\varphi_n \leq f \leq \psi_n$$

for every n , and

$$\lim_{n \rightarrow \infty} \int^{\mathcal{R}} \varphi_n = \lim_{n \rightarrow \infty} \int^{\mathcal{R}} \psi_n.$$

On the other hand,

$$\int^{\mathcal{R}} \varphi_n = \int^{\mathcal{L}} \varphi_n, \quad \int^{\mathcal{R}} \psi_n = \int^{\mathcal{L}} \psi_n$$

since they are step functions. By monotonicity,

$$\varphi := \lim_{n \rightarrow \infty} \varphi_n, \quad \psi := \lim_{n \rightarrow \infty} \psi_n$$

exists pointwise and are bounded measurable functions. Also, $\varphi \leq f \leq \psi$ and $\int^{\mathcal{L}} \varphi = \int^{\mathcal{L}} \psi$. By bounded convergence theorem,

$$\begin{aligned} \int^{\mathcal{R}} \varphi &= \lim_{n \rightarrow \infty} \int^{\mathcal{R}} \varphi_n = \lim_{n \rightarrow \infty} \int^{\mathcal{L}} \varphi_n = \int^{\mathcal{L}} \varphi, \\ \int^{\mathcal{R}} \psi &= \lim_{n \rightarrow \infty} \int^{\mathcal{R}} \psi_n = \lim_{n \rightarrow \infty} \int^{\mathcal{L}} \psi_n = \int^{\mathcal{L}} \psi. \end{aligned}$$

Since $\psi \geq \varphi$ and $\int^{\mathcal{L}} \psi = \int^{\mathcal{L}} \varphi$, we have $\psi = \varphi$ a.e. by Proposition 2.9, so $f = \varphi = \psi$ a.e. and

$$\int^{\mathcal{L}} f = \int^{\mathcal{L}} \varphi = \lim_{n \rightarrow \infty} \int^{\mathcal{L}} \varphi_n = \int^{\mathcal{R}} f.$$

So the Lebesgue integral of f equals the Riemann integral of f if f is Riemann integrable.

We now allow the functions to be unbounded and take on ∞ as values. Define the Lebesgue integral of such a function by

$$\int f = \sup_g \int g,$$

where the supremum is taken over all bounded measurable functions g with supports of finite measures such that $0 \leq g \leq f$. When the integral of f is finite, we say that f is Lebesgue integrable. Again, if $E \subseteq \mathbb{R}^d$ is a measurable set, then we define

$$\int_E f = \int f \chi_E.$$

PROPOSITION 2.10. *If $f, g \geq 0$ are measurable and $a, b \geq 0$, then*

$$\int (af + bg) = a \int f + b \int g.$$

Proof. First consider the case that $a = b = 1$. If φ, ψ are nonnegative bounded measurable functions with supports of finite measures such that $\varphi \leq f$ and $\psi \leq g$, then $\varphi + \psi \leq f + g$, so

$$\int f + \int g \leq \int (f + g).$$

On the other hand, suppose that η is a nonnegative bounded measurable function with support of finite measure such that $\eta \leq f + g$. Define

$$\varphi = \min\{f, \eta\}, \quad \psi = \eta - \varphi.$$

Then $\varphi \leq f$ and

$$\psi = \eta - \min\{f, \eta\} = \max\{\eta - f, 0\} \leq g.$$

Both φ and ψ are bounded measurable functions with supports of finite measures, so

$$\int (f + g) \leq \int f + \int g.$$

The general case follows from the case $a = b = 1$ by homogeneity. ■

PROPOSITION 2.11. *If E, F are disjoint measurable sets, and $f \geq 0$, then*

$$\int_{E \cup F} f = \int_E f + \int_F f.$$

Proof. Follows from the previous proposition and the fact that $\chi_{E \cup F} = \chi_E + \chi_F$. ■

PROPOSITION 2.12. *If $0 \leq f \leq g$ are measurable, then $\int f \leq \int g$.*

Proof. If φ is a nonnegative bounded measurable function with support of finite measure such that $\varphi \leq f$, then $\varphi \leq g$ as well, so

$$\int \varphi \leq \int g.$$

■

PROPOSITION 2.13. *If g is integrable and $0 \leq f \leq g$, then f is integrable as well.*

Proof. Since the subset of nonnegative functions that are bounded measurable and supported on sets of finite measures and $\leq f$ is a subset of that of g , we have $\int f \leq \int g < \infty$, so f is integrable. ■

PROPOSITION 2.14. *If f is integrable, then $f(x) < \infty$ a.e.*

Proof. Let

$$E_k := \{x : f(x) \geq k\}$$

and $E_\infty := \{x : f(x) = \infty\}$. Then

$$\int f \geq \int_{E_k} f \geq km(E_k),$$

hence $m(E_k) \rightarrow 0$ as $k \rightarrow \infty$. Since $E_k \downarrow E_\infty$ as $k \rightarrow \infty$, we have $m(E_\infty) = 0$, so $f(x) < \infty$ a.e. ■

PROPOSITION 2.15. *If $\int f = 0$, then $f = 0$ a.e.*

Proof. Similar to the proof of Proposition 2.9. ■

THEOREM 2.16 (Fatou's Lemma). *Suppose that (f_n) is a sequence of nonnegative measurable functions. Then*

$$\int \liminf_{n \rightarrow \infty} f_n \leq \liminf_{n \rightarrow \infty} \int f_n.$$

Proof. Let g be a bounded measurable function with support of finite measure such that $0 \leq g \leq \liminf_{n \rightarrow \infty} f_n$. For $n \geq 1$, define

$$g_{n(x)} := \min \left\{ g(x), \inf_{k \geq n} f_{k(x)} \right\}.$$

Then g_n is a bounded measurable function with support of finite measure and $g_n \rightarrow g$ pointwise. By the bounded convergence theorem, we have

$$\int g = \lim_{n \rightarrow \infty} \int g_n.$$

For each n , we have $g_n \leq f_k$ for all $k \geq n$, so $\int g_n \leq \int f_n$, implying that

$$\int g \leq \liminf_{n \rightarrow \infty} \int f_n.$$

Taking supremum over all such g , we obtain

$$\int \liminf_{n \rightarrow \infty} f_n \leq \liminf_{n \rightarrow \infty} \int f_n.$$

■

CLAIM 2.17. *Suppose that f is nonnegative and measurable, and (f_n) is a sequence of nonnegative measurable functions such that $f_n \leq f$ and $f_n \rightarrow f$ a.e. as $n \rightarrow \infty$. Then*

$$\lim_{n \rightarrow \infty} \int f_n = \int \lim_{n \rightarrow \infty} f_n = \int f.$$

Proof. Since $f_n \leq f$, we have $\int f_n \leq \int f$ for each n , so $\limsup_{n \rightarrow \infty} \int f_n \leq \int f$. On the other hand, by Fatou's lemma, we have

$$\int f = \int \liminf_{n \rightarrow \infty} f_n \leq \liminf_{n \rightarrow \infty} \int f_n.$$

Therefore, $\lim_{n \rightarrow \infty} \int f_n = \int f$. ■

THEOREM 2.18 (Monotone Convergence Theorem). *Suppose that (f_n) is an increasing sequence of nonnegative measurable functions such that $f_n \uparrow f$ a.e. as $n \rightarrow \infty$. Then*

$$\lim_{n \rightarrow \infty} \int f_n = \int f.$$

Proof. Follows from the previous corollary. ■

CLAIM 2.19. *Consider a series of nonnegative measurable functions $\sum_{k=1}^{\infty} a_k$. Then*

$$\int \sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} \int a_k.$$

If the right hand side is finite, then $\sum_{k=1}^{\infty} a_{k(x)}$ converges for a.e. x .

Proof. Follows from the monotone convergence theorem by considering the sequence of partial sums. ■

Consider

$$f(x) = \begin{cases} \frac{1}{|x|^{d+1}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

We'll show that f is integrable outside any ball $\{x : |x| < \varepsilon\}$. Moreover, $\exists C > 0$ such that

$$\int_{|x| \geq \varepsilon} f(x) dx \leq C\varepsilon^{-1}.$$

Define

$$A_k = \{x : 2^k \varepsilon \leq |x| < 2^{k+1} \varepsilon\},$$

$$a_{k(x)} = \frac{1}{(2^k \varepsilon)^{d+1}} \chi_{A_k}(x), \quad g(x) = \sum_{k=0}^{\infty} a_{k(x)}.$$

Then

$$f(x) \chi_{\{|x| \geq \varepsilon\}}(x) \leq g(x)$$

$$\Rightarrow \int_{|x| \geq \varepsilon} f(x) dx \leq \int g(x) dx = \sum_{k=0}^{\infty} \frac{m(A_k)}{(2^k \varepsilon)^{d+1}}.$$

Let $A = \{x : 1 \leq |x| < 2\}$, then $A_k = 2^k \varepsilon A$, so $m(A_k) = (2^k \varepsilon)^d m(A)$. Therefore,

$$\int_{|x| \geq \varepsilon} f(x) dx \leq \sum_{k=0}^{\infty} \frac{(2^k \varepsilon)^d m(A)}{(2^k \varepsilon)^{d+1}} = m(A) \varepsilon^{-1} \sum_{k=0}^{\infty} 2^{-k} = 2m(A) \varepsilon^{-1}.$$

For general measurable f on $\mathbb{R}^d$, we say that f is Lebesgue integrable if $|f|$ is integrable. In this case, we define

$$\int f = \int f^+ - \int f^-,$$

where

$$f^+(x) = \max\{f(x), 0\}, \quad f^-(x) = \max\{-f(x), 0\}.$$

Since $f^+ \leq |f|$ and $f^- \leq |f|$, both $\int f^+$ and $\int f^-$ are finite, so $\int f$ is well-defined.

PROPOSITION 2.20. *The integral does not depend on the decomposition, i.e. if $f = f_1 - f_2$ is another decomposition of f with $f_1, f_2 \geq 0$ and integrable, then $\int f = \int f_1 - \int f_2$.*

Proof. Since $f_1 - f_2 = f^+ - f^-$, we have $f_1 + f^- = f^+ + f_2$, hence

$$\begin{aligned} \int f_1 + \int f^- &= \int f^+ + \int f_2 \\ \Rightarrow \int f_1 - \int f_2 &= \int f^+ - \int f^- = \int f. \end{aligned}$$

■

Note that the integrability of a function and the value of its integral are unchanged if we modify f on a set of measure zero, so sometimes we allow our function to be undefined on sets of measure zero. When we add two integrable functions f and g , we might encounter $\infty - \infty$, but this only happens on a set of measure zero, so it still makes sense to write $f + g$. When we are talking about a function f , in many situations we are actually talking about the collection of all functions that are equal to f a.e.

PROPOSITION 2.21. *Suppose that f is integrable and $\varepsilon > 0$. Then there exists a set B of finite measure such that*

$$\int_{B^c} |f| < \varepsilon.$$

Also, $\exists \delta > 0$ such that whenever $m(E) < \delta$, we have $\int_E |f| < \varepsilon$.

Proof. Replacing f by $|f|$, we may assume that $f \geq 0$. Define $f_N = f\chi_{B(0,N)}$, then $f_N \geq 0$ is measurable, $f_N \leq f_{N+1}$ and $\lim_{N \rightarrow \infty} f_N = f$ pointwise. By the monotone convergence theorem, we have

$$\lim_{N \rightarrow \infty} \int f_N = \int f < \infty,$$

so there exists N such that $\int f_N > \int f - \varepsilon$. Let $B = B(0, N)$, then

$$\int_{B^c} f = \int f - \int f_N < \varepsilon.$$

For the second part, define

$$E_N = \{x : f(x) \leq N\}, \quad f_N = f\chi_{E_N}.$$

Again, $f_N \geq 0$ is measurable, $f_N \leq f_{N+1}$ and $\lim_{N \rightarrow \infty} f_N = f$ a.e. (except at the points where $f(x) = \infty$). By the monotone convergence theorem, we have

$$\lim_{N \rightarrow \infty} \int f_N = \int f < \infty,$$

so there exists N such that $\int (f - f_N) < \frac{\varepsilon}{2}$. Let $\delta = \frac{\varepsilon}{2}N$, then for every measurable set E with $m(E) < \delta$, we have

$$\int_E f = \int_E f_N + \int_E (f - f_N) \leq N \cdot m(E) + \int (f - f_N) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

■

THEOREM 2.22 (Lebesgue Dominated Convergence Theorem). *Suppose that (f_n) is a sequence of measurable functions such that $f_n \rightarrow f$ a.e. as $n \rightarrow \infty$. If $|f_n(x)| \leq g(x)$ for some integrable function g , then*

$$\lim_{n \rightarrow \infty} \int |f_n - f| = 0,$$

hence $\int f_n \rightarrow \int f$ as $n \rightarrow \infty$.

Proof. For each $N \geq 0$, define

$$E_N := \{x : |x| \leq N, g(x) \leq N\}.$$

Let $\varepsilon > 0$. Similar to the previous proposition, we can find N such that $\int_{E_N^c} g < \frac{\varepsilon}{4}$. Note that $|f_n \chi_{E_N}| \leq g \chi_{E_N} \leq N$ and $f_n \chi_{E_N} \rightarrow f \chi_{E_N}$ a.e., so by the bounded convergence theorem, we have

$$\int_{E_N} |f_n - f| < \frac{\varepsilon}{2}$$

for all sufficiently large n . Therefore,

$$\int |f_n - f| = \int_{E_N} |f_n - f| + \int_{E_N^c} |f_n - f| \leq \int_{E_N} |f_n - f| + 2 \int_{E_N^c} g < \varepsilon$$

for all sufficiently large n , so $\int |f_n - f| \rightarrow 0$ as $n \rightarrow \infty$. The convergence of $\int f_n$ follows from the triangle inequality:

$$\left| \int f_n - \int f \right| = \left| \int (f_n - f) \right| \leq \int |f_n - f| \rightarrow 0.$$

■

Let $f = u + iv$ be a complex-valued function, where u, v are real-valued functions. We say that f is measurable if both u and v are measurable. We say that f is Lebesgue integrable if

$$|f(x)| = \sqrt{u(x)^2 + v(x)^2}$$

is Lebesgue integrable. Note that

$$|u(x)|, |v(x)| \leq |f(x)| \leq |u(x)| + |v(x)|,$$

so f is integrable iff both u and v are integrable. In this case, we define

$$\int f = \int u + i \int v.$$

Define $\int_E f = \int f \chi_E$ for every measurable set E . It's easy to see that

$$\{f : E \rightarrow \mathbb{C} \mid f \text{ is integrable}\}$$

forms a vector space over $\mathbb{C}$.

2.2. L^p -spaces

For any measurable function f on $\mathbb{R}^d$ and $p \in [1, \infty)$, we define the L^p -norm of f by

$$|f|_p = \left(\int |f|^p \right)^{\frac{1}{p}}.$$

provided that $|f|^p$ is integrable.

DEFINITION 2.23. Let $f : \mathbb{R}^d \rightarrow [0, \infty]$ be measurable. Set

$$A = \{\alpha \in \mathbb{R} : m(\{x : f(x) \in (\alpha, \infty]\}) = 0\}.$$

Define $s = \inf A$. s is called the essential supremum of f , denoted by $|f|_\infty$. In general, if $f : \mathbb{R}^d \rightarrow \mathbb{C}$ is measurable, we define $|f|_\infty = \|f\|_\infty$.

It's easy to check that for all $p \in [1, \infty]$, we have $|f|_p \geq 0$, $|af|_p = |a| |f|_p$ for all $a \in \mathbb{C}$, and $|f + g|_p \leq |f|_p + |g|_p$. However, $|f|_p = 0$ only implies that $f = 0$ a.e. By considering equivalence classes of functions that are equal a.e., we can turn

$$\{f : \mathbb{R}^d \rightarrow \mathbb{C} \mid f \text{ is measurable, } |f|_p < \infty\}$$

into a normed vector space, denoted by $L^p(\mathbb{R}^d)$.

PROPOSITION 2.24. *If $\frac{1}{p} + \frac{1}{q} = 1$ with $p, q \in [1, \infty]$, then for every $f \in L^p(\mathbb{R}^d)$ and $g \in L^q(\mathbb{R}^d)$, we have $fg \in L^1(\mathbb{R}^d)$ and*

$$|fg|_1 \leq |f|_p |g|_q.$$

THEOREM 2.25. *$L^p(\mathbb{R}^d)$ is a Banach space for every $p \in [1, \infty]$.*

Proof. Let (f_n) be a Cauchy sequence in $L^p(\mathbb{R}^d)$. Choose a subsequence (f_{n_k}) such that

$$|f_{n_k} - f_{n_{k+1}}|_p < 2^{-k}$$

for every $k \geq 1$. Consider

$$\begin{aligned} f(x) &= f_{n_1}(x) + \sum_{k=1}^{\infty} (f_{n_{k+1}}(x) - f_{n_k}(x)), \\ g(x) &= |f_{n_1}(x)| + \sum_{k=1}^{\infty} |f_{n_{k+1}}(x) - f_{n_k}(x)|. \end{aligned}$$

Note that

$$|g|_p \leq |f_{n_1}|_p + \sum_{k=1}^{\infty} |f_{n_{k+1}} - f_{n_k}|_p \leq |f_{n_1}|_p + 1 < \infty,$$

so g^p is integrable, hence $g(x) < \infty$ a.e. Since $|f(x)| \leq g(x)$, the series defining $f(x)$ converges absolutely for a.e. x , so f is finite a.e. The partial sums of the series defining f are exactly f_{n_k} , so $f_{n_k} \rightarrow f$ a.e. as $k \rightarrow \infty$. Since $|f_{n_k} - f| \leq g$ and g is integrable, by the dominated convergence theorem, we have $|f_{n_k} - f|_p \rightarrow 0$ as $k \rightarrow \infty$. Finally, since (f_n) is a Cauchy sequence, we have $|f_n - f|_p \rightarrow 0$ as $n \rightarrow \infty$, so $f_n \rightarrow f$ in $L^p(\mathbb{R}^d)$ as $n \rightarrow \infty$.

If $p = \infty$, let

$$A_k := \{x \in \mathbb{R}^d : |f_{k(x)}| > |f_k|_{\infty}\},$$

$$B_{m,n} := \{x \in \mathbb{R}^d : |f_{m(x)} - f_{n(x)}| > |f_m - f_n|_{\infty}\}.$$

Write $E = \bigcup_{k=1}^{\infty} A_k \cup \bigcup_{m,n=1}^{\infty} B_{m,n}$, then $m(E) = 0$. On E^c , the sequence (f_n) converges uniformly to a bounded function f . Here, we use the fact that the space of bounded functions is complete under the supremum norm. Define $f(x) = 0$ for $x \in E$, then $f \in L^{\infty}(\mathbb{R}^d)$ and $|f_n - f|_{\infty} \rightarrow 0$ as $n \rightarrow \infty$. ■

CLAIM 2.26. *If (f_n) converges to f in $L^p(\mathbb{R}^d)$ for some $p \in [1, \infty]$, then there exists a subsequence (f_{n_k}) such that $f_{n_k} \rightarrow f$ a.e. as $k \rightarrow \infty$.*

Proof. Follows from the proof of the previous theorem by considering a subsequence of (f_n) that satisfies the condition in the proof. ■

THEOREM 2.27. *Suppose that $p \in [1, \infty)$. The following families are dense in $L^p(\mathbb{R}^d)$:*

- (i) *The simple functions;*
- (ii) *The step functions;*
- (iii) *The continuous functions with compact supports.*

Proof. Let $f \in L^p(\mathbb{R}^d)$. We may assume that f is nonnegative, since we may approximate its real and imaginary parts, and their positive and negative parts separately. Recall that there exists (φ_k) nonnegative simple functions such that $\varphi_k \uparrow f$ pointwise. By the monotone convergence theorem, we have $|\varphi_k - f|_p \rightarrow 0$ as $k \rightarrow \infty$, so (a) holds.

For part (b), it suffices to approximate χ_E by step functions where $m(E) < \infty$. Recall that for every $\varepsilon > 0$, there exists almost disjoint closed rectangles (R_j) such that

$$m\left(E \triangle \bigcup_{j=1}^M R_j\right) < \varepsilon^p.$$

Then χ_E and $\sum_{j=1}^M \chi_{R_j}$ differ only on a set of measure less than ε^p , so $|\chi_E - \sum_{j=1}^M \chi_{R_j}|_p < \varepsilon$.

For part (c), it suffices to approximate χ_R by continuous functions with compact supports where R is a rectangle. Consider $d = 1$, then $R = [a, b]$ for some $a < b$. Define

$$f_{\varepsilon}(x) = \begin{cases} 0 & \text{if } x \leq a - \varepsilon \\ \frac{x - (a - \varepsilon)}{\varepsilon} & \text{if } a - \varepsilon < x < a \\ 1 & \text{if } a \leq x \leq b \\ \frac{(b + \varepsilon) - x}{\varepsilon} & \text{if } b < x < b + \varepsilon \\ 0 & \text{if } x \geq b + \varepsilon \end{cases}$$

Then f_{ε} is continuous with compact support and $|f_{\varepsilon} - \chi_R|_p < 2\varepsilon$. For general d , we can approximate χ_R by the product of d such functions, which is also continuous with compact support. ■

Remark. The completeness and denseness results hold for $L^p(E)$ where $E \subseteq \mathbb{R}^d$ is a measurable set and $m(E) > 0$.

For all $p \in [1, \infty)$, $|\cdot|_p$ is a norm on $C_c(\mathbb{R}^d)$, the space of continuous functions with compact supports. Also, the essential supremum is the same as the actual supremum for functions in $C_c(\mathbb{R}^d)$. Since $C_c(\mathbb{R}^d)$ is dense in $L^p(\mathbb{R}^d)$, we can identify $L^p(\mathbb{R}^d)$ with the completion of $C_c(\mathbb{R}^d)$ under the L^p -norm.

Therefore, Lebesgue integration is the “correct” generalization of Riemann integration in the sense that it allows us to complete the space of continuous functions with compact supports under the L^p -norm.

$L^1(\mathbb{R}^d)$ is the completion of the set of integrable functions under the L^1 -norm. For $p = \infty$, the L^∞ -completion of $C_c(\mathbb{R}^d)$ is not $L^\infty(\mathbb{R}^d)$, but $C_0(\mathbb{R}^d)$, the space of continuous functions that vanish at infinity.

2.2.1. Invariance Properties

DEFINITION 2.28. Let f be a measurable function on $\mathbb{R}^d$. The translation of f by a vector $h \in \mathbb{R}^d$ is the function f_h defined by

$$f_{h(x)} = f(x - h).$$

PROPOSITION 2.29. *If f is integrable, then f_h is integrable and $\int f_h = \int f$ for every $h \in \mathbb{R}^d$.*

Proof. We first prove this for $f = \chi_E$ where E is a measurable set with finite measure. By translation invariance of Lebesgue measure, we have $m(E + h) = m(E)$, so $\int \chi_E = \int \chi_{E+h}$. By linearity, the result holds for simple functions.

If f is nonnegative and integrable, then there exists a sequence of nonnegative simple functions (φ_n) such that $\varphi_n \uparrow f$ pointwise. Then $((\varphi_n)_h) \uparrow f_h$ pointwise as well, so by the monotone convergence theorem, we have

$$\int f_h = \lim_{n \rightarrow \infty} \int (\varphi_n)_h = \lim_{n \rightarrow \infty} \int \varphi_n = \int f.$$

Finally, if f is complex-valued and integrable, then $\int |f_h| = \int |f| \rightsquigarrow f$ is integrable. By linearity again $\int f_h = \int f$. ■

Similarly, we can show that the Lebesgue integral is invariant under dilations: for all $a \in \mathbb{R} \setminus \{0\}$, we have $|a|^d \int f(ax) = \int f$.

A consequence of the invariance properties is the commutativity of convolution: if f, g are measurable and $f(x-y)g(y)$ is integrable in y , then by replacing y by $x-y$, we have

$$\int f(x-y)g(y)dy = \int f(y)g(x-y)dy.$$

By dilation invariance, for all $\varepsilon > 0$,

$$\begin{aligned} \int_{|x| \geq \varepsilon} \frac{1}{|x|} dx &= \varepsilon^{-a+d} \int_{|x| \geq 1} \frac{1}{|x|} dx, < \infty \text{ if } a > d, \\ \int_{|x| \leq \varepsilon} \frac{1}{|x|} dx &= \varepsilon^{-a+d} \int_{|x| \leq 1} \frac{1}{|x|} dx < \infty \text{ if } a < d. \end{aligned}$$

PROPOSITION 2.30. *Let $f \in L^1(\mathbb{R}^d)$, then $|f_h - f|_1 \rightarrow 0$ as $h \rightarrow 0$.*

Proof. Let $f \in L^1(\mathbb{R}^d)$ and $\varepsilon > 0$. Since $C_c(\mathbb{R}^d)$ is dense in $L^1(\mathbb{R}^d)$, there exists $g \in C_c(\mathbb{R}^d)$ such that $|f - g|_1 < \frac{\varepsilon}{3}$, then $|f_h - g_h|_1 < \frac{\varepsilon}{3}$ for every h . Since g is uniformly continuous, there exists $\delta > 0$ such that $|g_h - g|_1 < \frac{\varepsilon}{3}$ for every $|h| < \delta$. Therefore,

$$|f_h - f|_1 \leq |f_h - g_h|_1 + |g_h - g|_1 + |g - f|_1 < \varepsilon$$

for every $|h| < \delta$, so $|f_h - f|_1 \rightarrow 0$ as $h \rightarrow 0$. ■

2.3. Fubini's Theorem

Let f be a function on $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2} = \mathbb{R}^d$, then for fixed y , define $f^{y(x)} = f(x, y)$, and for fixed x , define $f_{x(y)} = f(x, y)$. For $E \subseteq \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$, define the slices by

$$E^y = \{x \in \mathbb{R}^{d_1} : (x, y) \in E\}, \quad E_x = \{y \in \mathbb{R}^{d_2} : (x, y) \in E\}.$$

THEOREM 2.31 (Fubini's Theorem). *Suppose that $f(x, y)$ is integrable on $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$. Then for a.e. $y \in \mathbb{R}^{d_2}$,*

- (i) *The slice f^y is integrable on $\mathbb{R}^{d_1}$;*
- (ii) *The function defined by $y \mapsto \int_{\mathbb{R}^{d_1}} f^{y(x)} dx$ is integrable on $\mathbb{R}^{d_2}$;*

(iii)

$$\int_{\mathbb{R}^{d_2}} F(y) dy = \int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} f(x, y) dx \right) dy = \int_{\mathbb{R}^d} f.$$

Proof. We assume that f is real-valued, since the complex-valued case can be reduced to the real-valued case by considering the real and imaginary parts separately.

Let $\mathcal{F}$ be the set of intergrable functions that satisfy all 3 conclusions in the theorem. It suffices to show that $L^1(\mathbb{R}^d) \subseteq \mathcal{F}$.

First, we show that $\mathcal{F}$ is a vector space. Let $f, g \in \mathcal{F}$ and $a, b \in \mathbb{R}$. Since union of two null sets is still a null set, the conclusions holds for $af + bg$ by linearity of the integral.

CLAIM 2.32. *If $(f_n) \subseteq \mathcal{F}$ and $f_n \uparrow f$ or $f_n \downarrow f$ pointwise where f is integrable, then $f \in \mathcal{F}$.*

Proof. We only prove the case $f_n \uparrow f$. Replace f_n by $f_n - f_1$ so that the sequence is nonnegative. By the monotone convergence theorem, $\lim_{n \rightarrow \infty} \int f_n = \int f$. By assumption, for each n , there exists a null set E_n such that f_n^y is integrable for every $y \in E_n^c$. Let $E = \bigcup_{n=1}^{\infty} E_n$, then $m(E) = 0$. If $y \notin E$, then f_n^y is integrable for every n . Note that

$$\underbrace{\int_{\mathbb{R}^{d_1}} f_n^{y(x)} dx}_{=: g_n(y)} \uparrow \underbrace{\int_{\mathbb{R}^{d_1}} f^{y(x)} dx}_{=: g(y)}$$

as $n \rightarrow \infty$ for all $y \notin E$, so by the monotone convergence theorem again, we have

$$\int_{\mathbb{R}^{d_2}} g(y) dy = \lim_{n \rightarrow \infty} \int_{\mathbb{R}^{d_2}} g_n(y) dy = \lim_{n \rightarrow \infty} \int f_n = \int f.$$

Since f is integrable, g is finite a.e., hence f^y is integrable for a.e. y and

$$\int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} f(x, y) dx \right) dy = \int f,$$

thus $f \in \mathcal{F}$.

CLAIM 2.33. *If E is a G_δ set such that $m(E) < \infty$, then $\chi_E \in \mathcal{F}$.*

Proof. If E is a bounded open cube, then write $E = I \times J$, where I, J are open cubes in $\mathbb{R}^{d_1}, \mathbb{R}^{d_2}$ respectively. Then for each y , the function $\chi_{E(x, y)}$ is measurable in x and integrable with

$$g(y) := \int_{\mathbb{R}^{d_1}} \chi_{E(x, y)} dx = \begin{cases} m(I) & \text{if } y \in J \\ 0 & \text{if } y \notin J \end{cases}$$

Therefore, $g = |I| \chi_J$ is integrable, and

$$\int_{\mathbb{R}^{d_2}} g(y) dy = m(I)m(J) = m(E) = \int \chi_E,$$

so $\chi_E \in \mathcal{F}$.

If E is a subset of the boundary of some closed cubes, then since $m(E) = 0$, $\int_{\mathbb{R}^d} \chi_E = 0$. It's also easy to see that for almost every y , the slice E^y has measure zero in $\mathbb{R}$, hence $\int_{\mathbb{R}^{d_1}} \chi_{E(x, y)} dx = 0$ for a.e. y . Therefore, $\int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} \chi_{E(x, y)} dx \right) dy = 0$, so $\chi_E \in \mathcal{F}$.

If E is a finite union of almost disjoint closed cubes, then χ_E can be written as a finite linear combination of the characteristic functions of open cubes and subsets of boundaries of closed cubes, so $\chi_E \in \mathcal{F}$.

If E is open and $m(E) < \infty$, then $E = \bigcup_{j=1}^{\infty} Q_j$, where (Q_j) are almost disjoint closed cubes. Define $f_k = \sum_{j=1}^k \chi_{Q_j}$, then $f_k \uparrow \chi_E$ pointwise, hence by the previous claim, $\chi_E \in \mathcal{F}$.

If E is a G_δ set such that $m(E) < \infty$, then exist open sets (U_k) such that $E = \bigcap_{k=1}^{\infty} U_k$. Since $m(E) < \infty$, there exists an open set U_0 of finite measure such that $E \subseteq U_0$. Define $V_k = \bigcap_{j=0}^k U_j$, then $V_k \downarrow E \rightsquigarrow \chi_{V_k} \downarrow \chi_E$ and each V_k has finite measure. By the previous paragraph, $\chi_{V_k} \in \mathcal{F}$ for every k , so by the previous claim again, $\chi_E \in \mathcal{F}$.

CLAIM 2.34. *If E has measure zero, then $\chi_E \in \mathcal{F}$.*

Proof. Since $m(E) = 0$, there exists a G_δ set G such that $E \subseteq G$ and $m(G) = 0$. By the previous claim, $\chi_G \in \mathcal{F}$ and

$$\int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} \chi_{G(x,y)} dx \right) dy = \int \chi_G = 0,$$

so $\int_{\mathbb{R}^{d_1}} \chi_{G(x,y)} dx = 0$ for a.e. y , i.e. the slice G^y has measure zero for a.e. y . Since $E^y \subseteq G^y$, E^y also has measure zero for a.e. y , so

$$\int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} \chi_{E(x,y)} dx \right) dy = 0 = \int \chi_E,$$

hence $\chi_E \in \mathcal{F}$.

CLAIM 2.35. *If E has finite measure, then $\chi_E \in \mathcal{F}$.*

Proof. Exists a G_δ set G of finite measure such that $E \subseteq G$ and $m(G \setminus E) = 0$, so $\chi_E = \chi_G - \chi_{G \setminus E} \in \mathcal{F}$.

By Claim 4 and linearity, simple functions with finite measure supports are in $\mathcal{F}$. By the monotone convergence theorem and Claim 1, every nonnegative integrable function is in $\mathcal{F}$. Finally, by linearity again, every integrable function is in $\mathcal{F}$. ■

THEOREM 2.36 (Tonelli). *Suppose that f is nonnegative and measurable on $\mathbb{R}^d = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$. Then for a.e. $y \in \mathbb{R}^{d_2}$,*

- (i) *The slice f^y is measurable on $\mathbb{R}^{d_1}$;*
- (ii) *The function defined by $y \mapsto \int_{\mathbb{R}^{d_1}} f^{y(x)} dx$ is measurable on $\mathbb{R}^{d_2}$;*
- (iii)

$$\int_{\mathbb{R}^{d_2}} F(y) dy = \int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} f(x, y) dx \right) dy = \int_{\mathbb{R}^d} f$$

where the integrals can be infinite.

Proof. Consider the truncations

$$f_{k(x,y)} = \begin{cases} f(x, y) & \text{if } |(x, y)| < k \text{ and } |f(x, y)| < k \\ 0 & \text{otherwise} \end{cases}$$

then each f_k is integrable, so by Fubini's theorem, for each k , there exists a measure zero set $E_k \subseteq \mathbb{R}^{d_2}$ such that the slice $(f_k)^y$ is measurable for every $y \in E_k^c$. Let $E = \bigcup_{k=1}^{\infty} E_k$, then $m(E) = 0$. Since $(f_k)^y \uparrow f^y$, f^y is measurable for every $y \notin E$, proving part (i). By the monotone convergence theorem, for every $y \notin E$, we have

$$\int_{\mathbb{R}^{d_1}} f^{y(x)} dx = \lim_{k \rightarrow \infty} \int_{\mathbb{R}^{d_1}} (f_k)^{y(x)} dx,$$

so by Fubini's theorem again, the function F defined by $y \mapsto \int_{\mathbb{R}^{d_1}} f^{y(x)} dx$ is measurable, proving part (ii). Finally, by the monotone convergence theorem once more, we have

$$\int_{\mathbb{R}^{d_2}} F(y) dy = \lim_{k \rightarrow \infty} \int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} (f_k)^{y(x)} dx \right) dy = \lim_{k \rightarrow \infty} \int f_k = \int f,$$

proving part (iii). ■

Given a measurable function f on $\mathbb{R}^d$ and we want to compute $\int f$. We can first apply Tonelli's theorem to $|f|$, then compute or estimate the iterated integrals of $|f|$ to check whether f is integrable. If f is integrable, then by Tonelli $\int |f| < \infty$, hence f is integrable, thus we can apply Fubini's theorem to complete the calculation of $\int f$.

CLAIM 2.37. *Let $E \subseteq \mathbb{R}^d = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ be a measurable set. Then for a.e. $y \in \mathbb{R}^{d_2}$, the slice E^y is measurable in $\mathbb{R}^{d_1}$. Also, $m(E^y)$ is a measurable function of y and*

$$m(E) = \int_{\mathbb{R}^{d_2}} m(E^y) dy.$$

Proof. Follows by applying Tonelli's theorem to χ_E . ■

Remark. The converse of the previous corollary is not true: let $N \subseteq \mathbb{R}$ be a nonmeasurable set and consider $E = [0, 1] \times N$, then

$$E^y = \begin{cases} [0, 1] & \text{if } y \in N \\ \emptyset & \text{if } y \notin N \end{cases}$$

so E^y is measurable for every y . But if E were measurable, then by Tonelli's theorem E_x would be measurable for a.e. x , which is not the case since $E_x = N$ for every $x \in [0, 1]$.

PROPOSITION 2.38. *If $E = E_1 \times E_2$ is measurable and $m_*(E_2) > 0$, then E_1 is measurable. Note that E_2 can be nonmeasurable.*

Proof. By Claim 2.37, for a.e. $y \in \mathbb{R}^{d_2}$,

$$(\chi_E)^{y(x)} = \chi_{E_1}(x) \cdot \chi_{E_2}(y)$$

is measurable in x . If exists $y \in E_2$ such that $(\chi_E)^y$ is measurable, then χ_{E_1} is measurable, so E_1 is measurable. Otherwise, for a.e. $y \in E_2$, $(\chi_E)^y$ is not measurable, which implies that E_2 is contained in a null set, contradicting the assumption that $m_*(E_2) > 0$. ■

LEMMA 2.39. *If $E_1 \subseteq \mathbb{R}^{d_1}$ and $E_2 \subseteq \mathbb{R}^{d_2}$, then*

$$m_*(E_1 \times E_2) \leq m_*(E_1)m_*(E_2),$$

with the convention that $0 \cdot \infty = 0$ and $\infty \cdot \infty = \infty$.

Proof. Let $\varepsilon > 0$. By definition of outer measure, there exists countable collections of closed cubes (Q_j^1) and (Q_k^2) such that $E_1 \subseteq \bigcup_{j=1}^{\infty} Q_j^1$, $E_2 \subseteq \bigcup_{k=1}^{\infty} Q_k^2$, and

$$\sum_{j=1}^{\infty} m(Q_j^1) \leq m_*(E_1) + \varepsilon, \quad \sum_{k=1}^{\infty} m(Q_k^2) \leq m_*(E_2) + \varepsilon.$$

Then $E_1 \times E_2 \subseteq \bigcup_{j,k=1}^{\infty} Q_j^1 \times Q_k^2$, so

$$\begin{aligned} m_*(E_1 \times E_2) &\leq \sum_{j,k=1}^{\infty} m(Q_j^1) m(Q_k^2) \\ &= \left(\sum_{j=1}^{\infty} m(Q_j^1) \right) \left(\sum_{k=1}^{\infty} m(Q_k^2) \right) \\ &\leq (m_*(E_1) + \varepsilon)(m_*(E_2) + \varepsilon). \end{aligned}$$

If both E_1 and E_2 have positive outer measures, then by letting $\varepsilon \rightarrow 0$, we have $m_*(E_1 \times E_2) \leq m_*(E_1)m_*(E_2)$.

Otherwise, suppose that $m_*(E_1) = 0$, then define $E_2^j = E_2 \cap B(0, j)$ for each $j \geq 1$. Since E_2^j is now bounded, by the above argument $m_*(E_1 \times E_2^j) \leq m_*(E_1)m_*(E_2^j) = 0$ for every j . Since $E_1 \times E_2^j \uparrow E_1 \times E_2$, by the continuity of measure, we have $m_*(E_1 \times E_2) = \lim_{j \rightarrow \infty} m_*(E_1 \times E_2^j) = 0$. ■

PROPOSITION 2.40. *Suppose that E_1, E_2 are measurable, then $E = E_1 \times E_2$ is measurable and $m(E) = m(E_1)m(E_2)$.*

Proof. We can find G_δ sets G_1, G_2 such that $E_j \subseteq G_j$ and $m(G_j \setminus E_j) = 0$ for $j = 1, 2$. Then $G = G_1 \times G_2$ is measurable. Note that

$$(G_1 \times G_2) \setminus (E_1 \times E_2) \subseteq ((G_1 \setminus E_1) \times G_2) \cup (E_1 \times (G_2 \setminus E_2)).$$

By the previous lemma,

$$\begin{aligned} m_*((G_1 \setminus E_1) \times G_2) &\leq m_*(G_1 \setminus E_1)m_*(G_2) = 0, \\ m_*(E_1 \times (G_2 \setminus E_2)) &\leq m_*(E_1)m_*(G_2 \setminus E_2) = 0, \end{aligned}$$

so $m((G_1 \times G_2) \setminus (E_1 \times E_2)) = 0$, hence E is measurable. By Claim 2.37,

$$m(E) = \int_{\mathbb{R}^{d_2}} m(E^y) dy = m(E_1)m(E_2).$$

■

CLAIM 2.41. *Suppose that f is measurable on $\mathbb{R}^{d_1}$, then the function defined by $g(x, y) = f(x)$ is measurable on $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$.*

Proof. We may assume that f is real-valued. If $a \in \mathbb{R}$, then $\{x : f(x) < a\}$ is measurable, so $\{(x, y) : g(x, y) < a\} = \{x : f(x) < a\} \times \mathbb{R}^{d_2}$ is measurable as well, hence g is measurable. ■

CLAIM 2.42. *Suppose that f is nonnegative on $\mathbb{R}^d$. Let*

$$\mathcal{A} = \{(x, y) \in \mathbb{R}^d \times \mathbb{R} : 0 \leq y \leq f(x)\}.$$

Then f is measurable if and only if $\mathcal{A}$ is measurable, and in this case, $\int f = m(\mathcal{A})$.

Proof. If f is measurable, consider $F(x, y) = y - f(x)$, which by the previous corollary is the difference of two measurable functions hence measurable, hence $\mathcal{A} = \{y \geq 0\} \cap$

$\{F \leq 0\}$ is measurable. Conversely, if $\mathcal{A}$ is measurable, then for each $a \in \mathbb{R}$, the slice $A_x = [0, f(x)]$ is a closed interval, hence $m(A_x) = f(x)$. By Tonelli's theorem, $x \mapsto m(A_x)$ is measurable, so f is measurable. Finally, by Tonelli's theorem again, we have

$$\int_{\mathbb{R}^d} f = \int_{\mathbb{R}^d} m(A_x) dx = m(\mathcal{A}).$$

■

PROPOSITION 2.43. *If f is measurable on $\mathbb{R}^d$, then $g(x, y) = f(x - y)$ is measurable on $\mathbb{R}^d \times \mathbb{R}^d$.*

Proof. We need to show that for each $a \in \mathbb{R}$, the set $E = \{(x, y) : f(x - y) < a\}$ is measurable.

CLAIM 2.44. *If $G \subseteq \mathbb{R}^d$ is measurable, then $G' = \{(x, y) : x - y \in G\}$ is measurable.*

Proof. If $\mathcal{U}$ is open, then $\mathcal{U}'$ is also open. Taking countable intersection, if $\mathcal{U}$ is G_δ , then $\mathcal{U}'$ is also G_δ . Define $B_k = \{(x, y) : |y| \leq k\}$, then

$$\chi_{\mathcal{U}' \cap B_k}(x, y) = \chi_{\mathcal{U}}(x - y) \cdot \chi_{B(0, k)}(y),$$

hence by Tonelli's theorem,

$$m(\mathcal{U}' \cap B_k) = \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} \chi_{\mathcal{U}}(x - y) \cdot \chi_{B(0, k)}(y) dx \right) dy = m(\mathcal{U})m(B(0, k)).$$

If $m(G) = 0$, then there exists a sequence of open sets $(\mathcal{U}_j)$ such that $G \subseteq \mathcal{U}_j$ and $m(\mathcal{U}_j) \rightarrow 0$ as $j \rightarrow \infty$, so $m(\mathcal{U}_{j'} \cap B_k) \rightarrow 0$ as well, hence $m(G' \cap B_k) = 0$ for every k . Since $G' \cap B_k \uparrow G'$, by the continuity of measure, we have $m(G') = \lim_{k \rightarrow \infty} m(G' \cap B_k) = 0$. Since every measurable set can be written as the difference of a G_δ set and a null set, the result follows. ■

By the claim, let $F = \{x : f(x) < a\}$, then $E = F'$, so E is measurable, hence g is measurable. ■

2.3.1. Convolution

Recall that the convolution of two functions f, g defined by

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x - y)g(y)dy,$$

which makes sense only when $f(x - y)g(y)$ is integrable in y .

PROPOSITION 2.45. *If $f, g \in L^1(\mathbb{R}^d)$, then $f * g \in L^1(\mathbb{R}^d)$. Moreover, $|f * g|_1 \leq |f|_1 |g|_1$.*

Proof. Define $h(x, y) = f(x - y)g(y)$, then h is measurable on $\mathbb{R}^d \times \mathbb{R}^d$. By Tonelli's theorem,

$$\begin{aligned} \int_{\mathbb{R}^d \times \mathbb{R}^d} |h(x, y)| dx dy &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |f(x - y)g(y)| dx \right) dy \\ &= \int_{\mathbb{R}^d} |g(y)| \left(\int_{\mathbb{R}^d} |f(x - y)| dx \right) dy = |f|_1 |g|_1 < \infty, \end{aligned}$$

so h is integrable. By Fubini's theorem, $\int_{\mathbb{R}^d} |f(x-y)g(y)| dy < \infty$ for a.e. x , so $f * g$ is well-defined a.e. and

$$\begin{aligned} |f * g|_1 &= \int_{\mathbb{R}^d} \left| \int_{\mathbb{R}^d} f(x-y)g(y)dy \right| dx \\ &\leq \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |f(x-y)g(y)| dy dx = |f|_1 |g|_1. \end{aligned}$$

■

PROPOSITION 2.46. *Suppose that $f, g \in L^1(\mathbb{R})$ such that g is differentiable and g' is uniformly bounded. Then $f * g$ is differentiable and $(f * g)' = f * g'$.*

Proof. For $h \neq 0$, we have

$$\frac{(f * g)(x+h) - (f * g)(x)}{h} = \int_{\mathbb{R}} f(y) \frac{g(x+h-y) - g(x-y)}{h} dy.$$

Let $M > 0$ be a bound for $|g'|$. By the mean value theorem, there exists ξ between $x+h-y$ and $x-y$ such that

$$\begin{aligned} \left| \frac{g(x+h-y) - g(x-y)}{h} \right| &= |g'(\xi)| \leq M \\ \Rightarrow \left| f(y) \frac{g(x+h-y) - g(x-y)}{h} \right| &\leq M |f(y)|. \end{aligned}$$

By the dominated convergence theorem,

$$(f * g)'(x) = \lim_{h \rightarrow 0} \frac{(f * g)(x+h) - (f * g)(x)}{h} = \int_{\mathbb{R}} f(y) g'(x-y) dy = (f * g')(x).$$

■

Differentiation and Integration

CHAPTER 3

In this chapter, we will study the relationship between differentiation and integration. We will see that under some mild assumptions, differentiation and integration are inverse operations of each other.

3.1. Differentiation of the Integral

If f is integrable on $[a, b]$, we define

$$F(x) = \int_a^x f(y) dy$$

Consider the derivative

$$\lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h}.$$

For simplicity, assume that $h > 0$. The difference quotient can be written as

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) dt = \frac{1}{|I|} \int_I f(y) dy,$$

where $I = [x, x+h]$. As $|I| \rightarrow 0$, we expect that the average tends to $f(x)$, but this is not always the case. We generalize this to higher dimensions. Suppose f is integrable on $\mathbb{R}^d$, then we want to know that whether

$$\lim_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \frac{1}{m(B)} \int_B f(y) dy = f(x)$$

for a.e. x , where B ranges over all open balls containing x .

3.1.1. The Hardy-Littlewood Maximal Function

DEFINITION 3.1. If $f \in L^1(\mathbb{R}^d)$, define the (Hardy-Littlewood) maximal function of f by

$$f^*(x) = \sup_{B \ni x} \frac{1}{m(B)} \int_B |f(y)| dy,$$

where B ranges over all open balls containing x .

PROPOSITION 3.2. f^* is measurable.

Proof. Let $\alpha > 0$ and let $E_\alpha = \{x : f^*(x) > \alpha\}$. Let $y \in E_\alpha$, then there exists an open ball B containing y such that

$$\frac{1}{m(B)} \int_B |f| > \alpha.$$

Since B is open, there exists $\varepsilon > 0$ such that $B(y, \varepsilon) \subseteq B$, so for every $x \in B(y, \varepsilon)$, we have

$$\frac{1}{m(B)} \int_B |f| > \alpha,$$

hence $B(y, \varepsilon) \subseteq E_\alpha$. Therefore, E_α is open, so f^* is measurable. ■

LEMMA 3.3 (Vitali Covering Lemma). *Suppose that $\mathcal{B} = \{B_1, \dots, B_N\}$ is a finite collection of open balls in $\mathbb{R}^d$. Then there exists a disjoint subcollection $\{B_{i_1}, \dots, B_{i_k}\}$ of $\mathcal{B}$ such that*

$$m\left(\bigcup_{l=1}^N B_l\right) \leq 3^d \sum_{j=1}^k m(B_{i_j}).$$

Proof. We adopt the notation that $cB(x, r) = B(x, cr)$ throughout this proof.

Suppose B and B' are two open balls that intersect with the radius of B' not larger than that of B , then clearly $B' \subseteq 3B$.

We construct the desired subcollection by induction. Let B_{i_1} be a ball in $\mathcal{B}$ with the largest radius, then delete all the balls that intersect with B_{i_1} from $\mathcal{B}$. Let B_{i_2} be a ball in the remaining collection with the largest radius, then delete all the balls that intersect with B_{i_2} from the remaining collection. We repeat this process until there is no ball left. Since the collection is finite, the process will terminate after finitely many steps, and we end up with a disjoint subcollection $\{B_{i_1}, \dots, B_{i_k}\}$. By the above observation, every ball in $\mathcal{B}$ is contained in $3B_{i_j}$ for some j , so

$$m\left(\bigcup_{l=1}^N B_l\right) \leq m\left(\bigcup_{j=1}^k 3B_{i_j}\right) \leq 3^d \sum_{j=1}^k m(B_{i_j}).$$

■

THEOREM 3.4. *Suppose f is integrable on $\mathbb{R}^d$, then*

$$m(\{x : f^*(x) > \alpha\}) \leq \frac{3^d}{\alpha} \|f\|_1,$$

for every $\alpha > 0$.

Proof. Write $E_\alpha = \{x : f^*(x) > \alpha\}$. If $x \in E_\alpha$, then there exists an open ball B_x containing x such that

$$\frac{1}{m(B_x)} \int_{B_x} |f(y)| dy > \alpha \Leftrightarrow m(B_x) < \frac{1}{\alpha} \int_{B_x} |f(y)| dy.$$

Let $K \subseteq E_\alpha$ be a compact set, then $\{B_x : x \in E_\alpha\}$ is an open cover of K , so there exists a finite subcover, say $\{B_{x_1}, \dots, B_{x_N}\}$. By the Vitali covering lemma, there exists a disjoint subcollection $\{B_{x_{i_1}}, \dots, B_{x_{i_k}}\}$ such that

$$\begin{aligned} m(K) &\leq m\left(\bigcup_{l=1}^N B_{x_l}\right) \leq 3^d \sum_{j=1}^k m(B_{x_{i_j}}) \\ &< \frac{3^d}{\alpha} \sum_{j=1}^k \int_{B_{x_{i_j}}} |f(y)| dy = \frac{3^d}{\alpha} \int_{\bigcup_{j=1}^k B_{x_{i_j}}} |f(y)| dy \leq \frac{3^d}{\alpha} \|f\|_1. \end{aligned}$$

By inner regularity of Lebesgue measure, we have

$$m(E_\alpha) = \sup\{m(K) : K \subseteq E_\alpha \text{ is compact}\} \leq \frac{3^d}{\alpha} \|f\|_1.$$

THEOREM 3.5 (Lebesgue Differentiation Theorem). *If $f \in L^1(\mathbb{R}^d)$, then for a.e. $x \in \mathbb{R}^d$,*

$$\lim_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \frac{1}{m(B)} \int_B f(y) dy = f(x).$$

Proof. For $\alpha > 0$, define

$$E_\alpha = \left\{ x : \limsup_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \left| \frac{1}{m(B)} \int_B f(y) dy - f(x) \right| > \alpha \right\}.$$

We claim that $m(E_\alpha) = 0$ for all $\alpha > 0$.

Fix $\alpha, \varepsilon > 0$. Recall that there exists $g \in C_c(\mathbb{R}^d)$ such that $\|f - g\|_1 < \varepsilon$. By continuity,

$$\lim_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \frac{1}{m(B)} \int_B g(y) dy = g(x)$$

for every x . Write

$$\begin{aligned} & \frac{1}{m(B)} \int_B f(y) dy - f(x) \\ &= \frac{1}{m(B)} \int_B (f(y) - g(y)) dy + \frac{1}{m(B)} \int_B g(y) dy - g(x) + g(x) - f(x) \\ &\Rightarrow \limsup_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \left| \frac{1}{m(B)} \int_B f(y) dy - f(x) \right| \leq (f - g)^*(x) + |g(x) - f(x)|. \end{aligned}$$

Define

$$F_\alpha = \left\{ x : (f - g)^*(x) > \frac{\alpha}{2} \right\}, \quad G_\alpha = \left\{ x : |g(x) - f(x)| > \frac{\alpha}{2} \right\},$$

then $E_\alpha \subseteq F_\alpha \cup G_\alpha$. Note that

$$m(G_\alpha) \leq \frac{2}{\alpha} \|f - g\|_1 \leq \frac{2}{\alpha} \varepsilon$$

by Markov's inequality, and

$$m(F_\alpha) \leq \frac{3^d}{2} \|f - g\|_1 \leq 2 \cdot \frac{3^d}{\alpha} \varepsilon,$$

by the previous theorem, so $m(E_\alpha) \leq m(F_\alpha) + m(G_\alpha) \leq \frac{2(3^d+1)}{\alpha} \varepsilon$. Since ε is arbitrary, we have $m(E_\alpha) = 0$ for every $\alpha > 0$, hence $\bigcup_{n=1}^{\infty} E_{\frac{1}{n}}$ has measure zero, i.e. for a.e. x ,

$$\limsup_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \left| \frac{1}{m(B)} \int_B f(y) dy - f(x) \right| = 0,$$

which is equivalent to the desired conclusion. ■

CLAIM 3.6. $f^*(x) \geq |f(x)|$ for a.e. x .

Proof. By applying the Lebesgue differentiation theorem to $|f|$, we have

$$|f(x)| = \lim_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \frac{1}{m(B)} \int_B |f(y)| dy \leq f^*(x)$$

for a.e. x . ■

The class $L^1(\mathbb{R}^d)$ is the globally integrable functions, but since differentiation is a local property, we expect that the Lebesgue differentiation theorem should hold for functions that are locally integrable.

DEFINITION 3.7. A measurable function f on $\mathbb{R}^d$ is locally integrable if for every ball B , the function $f \cdot \chi_B$ is integrable. The space of all locally integrable functions on $\mathbb{R}^d$ is denoted by $L^1_{\text{loc}}(\mathbb{R}^d)$.

Example 3.1. $f(x) = x$ is locally integrable on $\mathbb{R}$, but not integrable.

Clearly, the Lebesgue differentiation theorem also holds for locally integrable functions, since the average of f over a ball B is the same as the average of $f \cdot \chi_B$ over B .

DEFINITION 3.8. If $E \subseteq \mathbb{R}^d$ is measurable and $x \in \mathbb{R}^d$, we say that x is a density point of E if

$$\lim_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \frac{m(E \cap B)}{m(B)} = 1.$$

CLAIM 3.9. If $E \subseteq \mathbb{R}^d$ is measurable, then for a.e. $x \in E$, x is a density point of E , and for a.e. $x \in E^c$, x is not a density point of E .

Proof. Apply the Lebesgue differentiation theorem to χ_E . ■

DEFINITION 3.10. Let $f \in L^1_{\text{loc}}(\mathbb{R}^d)$. The Lebesgue set of f is the set of all points $x \in \mathbb{R}^d$ such that $|f(x)| < \infty$ and

$$\lim_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \frac{1}{m(B)} \int_B |f(y) - f(x)| dy = 0.$$

If f is continuous at x , then x belongs to the Lebesgue set of f . If x belongs to the Lebesgue set of f , then by the triangle inequality, we have

$$\lim_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \int_B f(y) dy = f(x).$$

CLAIM 3.11. If $f \in L^1_{\text{loc}}(\mathbb{R}^d)$, then for a.e. x , x belongs to the Lebesgue set of f .

Proof. For each $r \in \mathbb{Q}$, by the Lebesgue differentiation theorem, exists a set E_r of measure zero such that for every $x \notin E_r$,

$$\lim_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \frac{1}{m(B)} \int_B |f(y) - r| dy = |f(x) - r|.$$

Let $E = \bigcup_{r \in \mathbb{Q}} E_r$, then $m(E) = 0$. Suppose that $x \notin E$ and $|f(x)| < \infty$. Let $\varepsilon > 0$ and choose $r \in \mathbb{Q}$ such that $|f(x) - r| < \frac{\varepsilon}{2}$, then

$$\begin{aligned} \frac{1}{m(B)} \int_B |f(y) - f(x)| dy &\leq \frac{1}{m(B)} \int_B |f(y) - r| dy + |r - f(x)| \\ \Rightarrow \limsup_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \frac{1}{m(B)} \int_B |f(y) - f(x)| dy &\leq 2 |f(x) - r| < \varepsilon. \end{aligned}$$

Since ε is arbitrary, we have

$$\lim_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \frac{1}{m(B)} \int_B |f(y) - f(x)| dy = 0,$$

so x belongs to the Lebesgue set of f . ■

Remark. Recall that elements of $L^1(\mathbb{R}^d)$ are equivalence classes of functions that may differ on a measure zero set, but the set

$$\left\{ x \in \mathbb{R}^d : \text{the limit } \lim_{\substack{m(B) \rightarrow 0 \\ B \ni x}} \frac{1}{m(B)} \int_B f(y) dy \text{ exists} \right\}$$

is independent of the choice of representatives. Nevertheless, the Lebesgue set of f may depend on the choice of representatives.

3.2. Approximation to the Identity

DEFINITION 3.12. A family of functions $(K_\delta)_{\delta>0}$ on $\mathbb{R}^d$ is an approximation to the identity if they are integrable and exists a constant $A > 0$ such that

- (i) $\int_{\mathbb{R}^d} K_{\delta(x)} dx = 1$ for every $\delta > 0$;
- (ii) $|K_{\delta(x)}| \leq A\delta^{-d}$ for all $\delta > 0$ and $x \in \mathbb{R}^d$;
- (iii) $|K_{\delta(x)}| \leq A\delta |x|^{-d-1}$ for all $\delta > 0$ and $x \in \mathbb{R}^d$.

PROPOSITION 3.13. If $(K_\delta)_{\delta>0}$ is an approximation to the identity, then exists a constant $C > 0$ such that

$$\int_{\mathbb{R}^d} |K_{\delta(x)}| dx \leq C$$

for every $\delta > 0$.

Proof. Recall that

$$\int_{|x| \geq \varepsilon} \frac{dx}{|x|^{d+1}} \leq \frac{C'}{\varepsilon}$$

for some constant $C' > 0$. By the properties of K_δ , we have

$$\begin{aligned} \int_{\mathbb{R}^d} |K_{\delta(x)}| dx &= \int_{|x| < \delta} |K_{\delta(x)}| dx + \int_{|x| \geq \delta} |K_{\delta(x)}| dx \\ &\leq A \int_{|x| < \delta} \delta^{-d} dx + A \int_{|x| \geq \delta} \delta |x|^{-d-1} dx \\ &\leq Am(B(0, 1)) + AC' = C. \end{aligned}$$

■

PROPOSITION 3.14. *If $(K_\delta)_{\delta>0}$ is an approximation to the identity, then for every $\eta > 0$,*

$$\lim_{\delta \rightarrow 0} \int_{|x| \geq \eta} |K_{\delta(x)}| dx = 0.$$

Proof. By the properties of K_δ , we have

$$\int_{|x| \geq \eta} |K_{\delta(x)}| dx \leq A \int_{|x| \geq \eta} \delta |x|^{-d-1} dx \leq AC' \frac{\delta}{\eta} \rightarrow 0$$

as $\delta \rightarrow 0$. ■

LEMMA 3.15. *Suppose that $f \in L^1(\mathbb{R}^d)$ and x belongs to the Lebesgue set of f . For $r > 0$, define*

$$A(r) = \frac{1}{r^d} \int_{|y| \leq r} |f(x-y) - f(x)| dy.$$

Then A is continuous on $(0, \infty)$, $\lim_{r \rightarrow 0} A(r) = 0$ and exists $M > 0$ such that $A(r) \leq M$ for every $r > 0$.

Proof. Recall that if $f \in L^1(\mathbb{R}^d)$ and $\varepsilon > 0$, then exists $\delta > 0$ such that for every measurable set E with $m(E) < \delta$, we have $\int_E |f| < \varepsilon$. Clearly $f(x-y) - f(x)$ is integrable in y , so by absolute continuity of the integral, A is continuous on $(0, \infty)$.

Since x belongs to the Lebesgue set of f , $A(r) \rightarrow 0$ as $r \rightarrow 0$ by definition. Since A is continuous, it implies that A is bounded on $(0, 1]$. On the other hand, for $r > 1$, we have

$$\begin{aligned} A(r) &\leq \frac{1}{r^d} \int_{|y| \leq r} |f(x-y)| dy + \frac{1}{r^d} \int_{|y| \leq r} |f(x)| dy \\ &\leq \frac{1}{r^d} \|f\|_1 + m(B(0, 1)) |f(x)| < \infty. \end{aligned}$$

■

THEOREM 3.16. *If $(K_\delta)_{\delta>0}$ is an approximation to the identity and $f \in L^1(\mathbb{R}^d)$, then*

$$\lim_{\delta \rightarrow 0} (f * K_\delta)(x) = f(x)$$

for all x in the Lebesgue set of f . In particular, the limit holds for a.e. x .

Proof. Write

$$\begin{aligned} (f * K_\delta)(x) - f(x) &= \int_{\mathbb{R}^d} f(x-y) K_{\delta(y)} dy - f(x) \\ &= \int_{\mathbb{R}^d} (f(x-y) - f(x)) K_{\delta(y)} dy, \\ \Rightarrow |(f * K_\delta)(x) - f(x)| &\leq \int_{\mathbb{R}^d} |f(x-y) - f(x)| |K_{\delta(y)}| dy \\ &= \int_{|y| < \delta} |f(x-y) - f(x)| |K_{\delta(y)}| dy \end{aligned}$$

$$+ \sum_{k=0}^{\infty} \int_{2^k \delta \leq |y| < 2^{k+1} \delta} |f(x-y) - f(x)| |K_{\delta(y)}| dy.$$

By the previous lemma, we have

$$\int_{|y| < \delta} |f(x-y) - f(x)| |K_{\delta(y)}| dy \leq \frac{C}{\delta^d} \int_{|y| < \delta} |f(x-y) - f(x)| dy = CA(\delta).$$

For $k \geq 0$, we have

$$\begin{aligned} \int_{2^k \delta \leq |y| < 2^{k+1} \delta} |f(x-y) - f(x)| |K_{\delta(y)}| dy &\leq \frac{C\delta}{(2^k \delta)^{d+1}} \int_{|y| < 2^{k+1} \delta} |f(x-y) - f(x)| dy \\ &\leq \frac{C'}{2^k} A(2^{k+1} \delta). \end{aligned}$$

Therefore,

$$|(f * K_{\delta})(x) - f(x)| \leq CA(\delta) + C' \sum_{k=0}^{\infty} \frac{1}{2^k} A(2^{k+1} \delta).$$

Let $\varepsilon > 0$. Choose N large such that $\sum_{k=N}^{\infty} \frac{1}{2^k} < \varepsilon$, then choose δ small enough such that $A(2^{k+1} \delta) < \frac{\varepsilon}{N}$ for every $0 \leq k < N$. Since A is bounded, we have

$$\begin{aligned} |(f * K_{\delta})(x) - f(x)| &\leq C\varepsilon + C' \sum_{k=0}^{N-1} \frac{1}{2^k} \cdot \frac{\varepsilon}{N} + C' \sum_{k=N}^{\infty} \frac{1}{2^k} \cdot M \\ &< (C + C' + C'M)\varepsilon, \end{aligned}$$

so $\lim_{\delta \rightarrow 0} (f * K_{\delta})(x) = f(x)$. ■

THEOREM 3.17. Suppose that $f \in L^1(\mathbb{R}^d)$ and $(K_{\delta})_{\delta>0}$ is an approximation to the identity, then $f * K_{\delta} \in L^1(\mathbb{R}^d)$ and $|f * K_{\delta} - f|_1 \rightarrow 0$ as $\delta \rightarrow 0$.

Proof. Write $f_{h(x)} = f(x+h)$, then

$$\begin{aligned} |(f * K_{\delta})(x) - f(x)| &\leq \int_{\mathbb{R}^d} |f(x-y) - f(x)| |K_{\delta(y)}| dy \\ &= \int_{\mathbb{R}^d} |f_{-y}(x) - f(x)| |K_{\delta(y)}| dy. \end{aligned}$$

By Tonelli's theorem,

$$\begin{aligned} |f * K_{\delta} - f|_1 &\leq \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |f_{-y}(x) - f(x)| |K_{\delta(y)}| dy dx \\ &= \int_{\mathbb{R}^d} |f_{-y} - f|_1 |K_{\delta(y)}| dy. \end{aligned}$$

Recall that if $f \in L^1(\mathbb{R}^d)$, then $|f_h - f|_1 \rightarrow 0$ as $h \rightarrow 0$. Let $\varepsilon > 0$, then exists $\eta > 0$ such that $|f_h - f|_1 < \varepsilon$ for every $|h| < \eta$. By the properties of K_{δ} , we have

$$\begin{aligned} |f * K_{\delta} - f|_1 &\leq \int_{|y| < \eta} |f_{-y} - f|_1 |K_{\delta(y)}| dy + \int_{|y| \geq \eta} |f_{-y} - f|_1 |K_{\delta(y)}| dy \\ &\leq C\varepsilon + 2 \|f\|_1 \int_{|y| \geq \eta} |K_{\delta(y)}| dy \rightarrow C\varepsilon \end{aligned}$$

as $\delta \rightarrow 0$, so $|f * K_{\delta} - f|_1 \rightarrow 0$ as $\delta \rightarrow 0$. ■

3.3. Differentiability of functions

We want to find a class of functions that satisfies the fundamental theorem of calculus:

$$F(b) - F(a) = \int_a^b F'(x)dx.$$

DEFINITION 3.18. Let $F : [a, b] \rightarrow \mathbb{C}$ be a function, and let $a = t_0 < t_1 < \dots < t_N = b$ be a partition of $[a, b]$. The variation of F on this partition is

$$\sum_{j=1}^N |F(t_j) - F(t_{j-1})|.$$

F is of bounded variation if the variations of F on all partitions are uniformly bounded, i.e.

$$\sup \sum_{j=1}^N |F(t_j) - F(t_{j-1})| < \infty.$$

If $\mathcal{P}$ is a partition of $[a, b]$ and $\mathcal{P}'$ is a refinement of $\mathcal{P}$, then the variation of F on $\mathcal{P}'$ is at least that of F on $\mathcal{P}$.

DEFINITION 3.19. The total variation of F on $[a, x]$ (where $a \leq x \leq b$) is defined by

$$T_{F(a,x)} = \sup \sum_{j=1}^N |F(t_j) - F(t_{j-1})|,$$

where the supremum is taken over all partitions of $[a, x]$. If F is real-valued, the positive variation of F on $[a, x]$ is

$$P_{F(a,x)} = \sup \sum_{(+)} (F(t_j) - F(t_{j-1})),$$

where the supremum is taken over all partitions of $[a, x]$ and the sum is taken over all j such that $F(t_j) - F(t_{j-1}) \geq 0$. Similarly, the negative variation of F on $[a, x]$ is

$$N_{F(a,x)} = \sup \sum_{(-)} (F(t_{j-1}) - F(t_j)),$$

where the supremum is taken over all partitions of $[a, x]$ and the sum is taken over all j such that $F(t_j) - F(t_{j-1}) \leq 0$.

LEMMA 3.20. Suppose that F is real-valued and of bounded variation on $[a, b]$, then for all $x \in [a, b]$,

$$F(x) - F(a) = P_{F(a,x)} - N_{F(a,x)}, \quad T_{F(a,x)} = P_{F(a,x)} + N_{F(a,x)}.$$

Proof. Let $\varepsilon > 0$, then exists a partition $\mathcal{P}$ of $[a, x]$ such that

$$|P_{F(a,x)} - \sum_{(+)} (F(t_j) - F(t_{j-1}))| < \frac{\varepsilon}{2}, \quad |N_{F(a,x)} - \sum_{(-)} (F(t_{j-1}) - F(t_j))| < \frac{\varepsilon}{2}.$$

Therefore,

$$F(x) - F(a) = \sum_{j=1}^N (F(t_j) - F(t_{j-1})) = \sum_{(+)} (F(t_j) - F(t_{j-1})) - \sum_{(-)} (F(t_{j-1}) - F(t_j))$$

$$\Rightarrow |F(x) - F(a) - (P_{F(a,x)} - N_{F(a,x)})| < \varepsilon.$$

For the second identity, we have

$$\sum_{j=1}^N |F(t_j) - F(t_{j-1})| = \sum_{(+)} (F(t_j) - F(t_{j-1})) + \sum_{(-)} (F(t_{j-1}) - F(t_j))$$

for every partition $\mathcal{P}$ of $[a, x]$, so

$$T_{F(a,x)} \leq P_{F(a,x)} + N_{F(a,x)}.$$

On the other hand, let $\varepsilon > 0$, then exists a partition $\mathcal{P}$ of $[a, x]$ such that

$$|T_{F(a,x)} - \sum_{j=1}^N |F(t_j) - F(t_{j-1})|| < \varepsilon.$$

Since

$$\sum_{j=1}^N |F(t_j) - F(t_{j-1})| = P_{F(a,x)} + N_{F(a,x)},$$

we have $T_{F(a,x)} \leq P_{F(a,x)} + N_{F(a,x)} + \varepsilon$. Since ε is arbitrary, we have $T_{F(a,x)} \leq P_{F(a,x)} + N_{F(a,x)}$, hence $T_{F(a,x)} = P_{F(a,x)} + N_{F(a,x)}$. ■

CLAIM 3.21. *A real-valued function F on $[a, b]$ is of bounded variation if and only if F is a difference of two increasing bounded functions.*

Proof. Since increasing functions are of bounded variation, the difference of two increasing bounded functions is also of bounded variation. Conversely, if F is of bounded variation, then we can set $F_1(x) = P_{F(a,x)} + F(a)$ and $F_2(x) = N_{F(a,x)}$, then F_1 and F_2 are increasing bounded functions and $F = F_1 - F_2$. ■

As a consequence, any complex-valued function of bounded variation can be written as a linear combination of four increasing bounded functions.

LEMMA 3.22 (Rising Sun Lemma). *Suppose that G is real-valued and continuous on $\mathbb{R}$. Let*

$$E = \{x : G(x+h) > G(x) \text{ for some } h > 0\}.$$

If $E \neq \emptyset$, then it is open, and we can write $E = \bigcup_{n=1}^{\infty} (a_n, b_n)$, where (a_n, b_n) are disjoint open intervals. Moreover, if (a_k, b_k) is a finite interval, then $G(a_k) = G(b_k)$.

Proof. Since G is continuous, we can easily see that E is open. If (a_k, b_k) is a finite interval, then $a_k \notin E$ hence we cannot have $G(b_k) > G(a_k)$. Suppose that $G(a_k) > G(b_k)$. By continuity, there exists $c \in (a_k, b_k)$ such that

$$G(c) = \frac{G(a_k) + G(b_k)}{2}.$$

We may choose c to be the rightmost such point, then $c \in E$, so there exists some $d > c$ such that $G(d) > G(c)$. Since $b_k \notin E$, we have $G(x) \leq G(b_k)$ for every $x \geq b_k$, so

$$G(d) > G(c) = \frac{G(a_k) + G(b_k)}{2} > G(b_k) \rightsquigarrow d < b_k.$$

By continuity, there exists $c' \in (d, b_k)$ such that $G(c') = G(c) = \frac{G(a_k) + G(b_k)}{2}$, which contradicts the choice of c . ■

Remark. If $G : [a, b] \rightarrow \mathbb{R}$ is continuous and

$$E = \{x \in (a, b) : G(x+h) > G(x) \text{ for some } h > 0\},$$

then the above lemma still holds, with the exception that if $a_k = a$, then we can only conclude that $G(a_k) \leq G(b_k)$.

THEOREM 3.23. *If F is of bounded variation, then F is differentiable a.e.*

Proof. It suffices to show that an increasing bounded function is differentiable a.e., and since increasing functions only have countably many discontinuities, we can assume that F is continuous.

Define the difference quotient of F at x by

$$\Delta_h(F)(x) = \frac{F(x+h) - F(x)}{h}.$$

The Dini derivatives of F at x are defined by

$$\begin{aligned} D^+(F)(x) &= \limsup_{h \rightarrow 0^+} \Delta_h(F)(x), & D_+(F)(x) &= \liminf_{h \rightarrow 0^+} \Delta_h(F)(x), \\ D^-(F)(x) &= \limsup_{h \rightarrow 0^-} \Delta_h(F)(x), & D_-(F)(x) &= \liminf_{h \rightarrow 0^-} \Delta_h(F)(x). \end{aligned}$$

Our goal is to show that the Dini derivatives are finite and equal for a.e. x . Clearly $D_+(F)(x) \leq D^+(F)(x)$ and $D_-(F)(x) \leq D^-(F)(x)$ for every x . It suffices to show that

- (i) $D^+(F)(x) < \infty$ for a.e. x ;
- (ii) $D^+(F)(x) \leq D_-(F)(x)$ for a.e. x .

Indeed, by applying (ii) to $-F(-x)$, we get $D^-(F)(x) \leq D_+(F)(x)$ for a.e. x , hence

$$D_+(F)(x) \leq D^+(F)(x) \leq D_-(F)(x) \leq D^-(F)(x) \leq D_+(F)(x) < \infty$$

for a.e. x , so F is differentiable a.e.

For (i), let

$$E_\gamma = \{x \in [a, b] : D^+(F)(x) > \gamma\}$$

for $\gamma > 0$. Let $G(x) = F(x) - \gamma x$, then by Lemma 3.22, we have

$$\begin{aligned} E &:= \{x \in (a, b) : G(x+h) > G(x) \text{ for some } h > 0\} \\ &= \left\{x \in (a, b) : \frac{F(x+h) - F(x)}{h} > \gamma \text{ for some } h > 0\right\} \\ &\Rightarrow E_\gamma \setminus \{a\} \subseteq E = \bigcup_{n=1}^{\infty} (a_n, b_n). \end{aligned}$$

We have $G(a_n) \leq G(b_n) \rightsquigarrow F(b_n) - F(a_n) \geq \gamma(b_n - a_n)$ for every n , so

$$\begin{aligned} m(E_\gamma) &\leq \sum_{n=1}^{\infty} (b_n - a_n) \leq \frac{1}{\gamma} \sum_{n=1}^{\infty} (F(b_n) - F(a_n)) \\ &\leq \frac{1}{\gamma} (F(b) - F(a)). \end{aligned}$$

As $\gamma \rightarrow \infty$, we have $m(E_\gamma) \rightarrow 0$, so $D^+(F)(x) < \infty$ for a.e. x .

For (ii), fix rational numbers $r < R$ and let

$$H = \{x \in (a, b) : D_-(F)(x) < r < R < D^+(F)(x)\}.$$

If we can show that $m(H) = 0$, then taking union over all rational numbers $r < R$ gives

$$\{x \in (a, b) : D_-(F)(x) < D^+(F)(x)\} = \bigcup_{r < R} H$$

has measure zero, hence $D^+(F)(x) \leq D_-(F)(x)$ for a.e. x .

Assume $m(H) > 0$ and $\frac{R}{r} > 1$. By outer regularity, there exists an open set $\mathcal{U}$ such that $H \subseteq \mathcal{U} \subseteq (a, b)$ such that $m(\mathcal{U}) < m(H) \cdot \frac{R}{r}$. Write $\mathcal{U} = \bigcup_{n=1}^{\infty} I_n$, where I_n are disjoint open intervals. We may assume that $I_n \cap H \neq \emptyset$ for every n , otherwise we can just remove those intervals. Fix n and define

$$E_n = \{x \in -I_n : G(x+h) > G(x) \text{ for some } h > 0\}.$$

Note that if $x \in (-I_n) \cap (-F)$, then

$$\frac{F(-x+h) - F(-x)}{h} < r \text{ for some } h > 0 \Rightarrow G(x+h) > G(x) \Rightarrow x \in E_n,$$

so $(-I_n) \cap (-F) \subseteq E_n$, in particular E_n is nonempty. By Lemma 3.22, we can write $-E_n = \bigcup_{k=1}^{\infty} (a_k, b_k)$ so that

$$G(b_k) \leq G(a_k) \rightsquigarrow F(b_k) - F(a_k) \leq r(b_k - a_k)$$

We then apply Lemma 3.22 to each interval (a_k, b_k) again to $G(x) = F(x) - Rx$, we then obtain a family of disjoint open intervals $\mathcal{O}_n = \bigcup_{k,j} (a_{k,j}, b_{k,j})$ such that

$$G(a_{k,j}) \leq G(b_{k,j}) \rightsquigarrow F(b_{k,j}) - F(a_{k,j}) \geq R(b_{k,j} - a_{k,j})$$

for every j . Using the fact that F is increasing, we have

$$\begin{aligned} m(\mathcal{O}_n) &= \sum_{k,j} (b_{k,j} - a_{k,j}) \leq \frac{1}{R} \sum_{k,j} (F(b_{k,j}) - F(a_{k,j})) \\ &\leq \frac{1}{R} \sum_k (F(b_k) - F(a_k)) \leq \frac{r}{R} \sum_k (b_k - a_k) \leq \frac{r}{R} m(I_n). \end{aligned}$$

Note that $\mathcal{O}_n \supset H \cap I_n$ since $D^+(F)(x) > R$ and $D_-(F)(x) < r$ for every $x \in H \cap I_n$. Therefore,

$$m(H) = \sum_n m(H \cap I_n) \leq \sum_n m(\mathcal{O}_n) \leq \frac{r}{R} \sum_n m(I_n) = \frac{r}{R} m(\mathcal{U}) < m(H),$$

which is a contradiction. Hence $m(H) = 0$. ■

CLAIM 3.24. *If F is increasing and continuous on $[a, b]$, then F' exists a.e. Moreover, F' is measurable, nonnegative and*

$$\int_a^b F'(x) dx \leq F(b) - F(a).$$

In particular, if F is bounded on $\mathbb{R}$, then $F' \in L^1(\mathbb{R})$.

Proof. Since $F'(x) = \lim_{n \rightarrow \infty} (f(x + \frac{1}{n}) - f(x))n$ for every x such that $F'(x)$ exists, F' is measurable. By Fatou's lemma, we have

$$\begin{aligned} \int_a^b F'(x) dx &\leq \liminf_{n \rightarrow \infty} \int_a^b \left(F\left(x + \frac{1}{n}\right) - F(x) \right) n dx \\ &= \liminf_{n \rightarrow \infty} \left(n \int_a^b F\left(x + \frac{1}{n}\right) dx - n \int_a^b F(x) dx \right) \\ &= \liminf_{n \rightarrow \infty} \left(n \int_{a+\frac{1}{n}}^{b+\frac{1}{n}} F(x) dx - n \int_a^b F(x) dx \right) \end{aligned}$$

$$= \liminf_{n \rightarrow \infty} \left(n \int_b^{b+\frac{1}{n}} F(x) dx - n \int_a^{a+\frac{1}{n}} F(x) dx \right) = F(b) - F(a).$$

■

In general, the inequality $\int_a^b F'(x) dx \leq F(b) - F(a)$ can be strict. For example, if F is the Cantor function, then $F' = 0$ a.e. but $F(1) - F(0) = 1$.

3.3.1. Absolutely Continuous Functions

DEFINITION 3.25. A function $F : [a, b] \rightarrow \mathbb{R}$ is absolutely continuous if for every $\varepsilon > 0$, exists $\delta > 0$ such that

$$\sum_{j=1}^N (b_j - a_j) < \delta \Rightarrow \sum_{j=1}^N |F(b_j) - F(a_j)| < \varepsilon$$

for every finite collection of disjoint intervals $\{(a_j, b_j)\}_{j=1}^N$ in $[a, b]$.

Clearly, absolute continuity implies uniform continuity and bounded variation.

PROPOSITION 3.26. If $F(x) = \int_a^x f(t) dt$ for some $f \in L^1([a, b])$, then F is absolutely continuous.

Proof. Follows from the absolute continuity of the integral. ■

Absolute continuity is a necessary condition for the fundamental theorem of calculus to hold. In fact, it is also sufficient.

DEFINITION 3.27. A collection $\mathcal{B}$ is said to be a Vitali covering of a set E if for every $x \in E$ and $\eta > 0$, there exists some $B \in \mathcal{B}$ such that $x \in B$ and $m(B) < \eta$.

LEMMA 3.28 (Actual Vitali Covering Lemma). Suppose that E is a set of finite measure and $\mathcal{B}$ is a Vitali covering of E , then for any $\delta > 0$, exists a finite collection of disjoint sets $B_1, \dots, B_N$ in $\mathcal{B}$ such that

$$m(E) \leq \sum_{i=1}^N m(B_i) + \delta.$$

Proof. Assume that $m(E) > \delta$. Take a compact $E' \subseteq E$ such that $m(E') \geq \delta$. We can cover E' by finitely many balls from $\mathcal{B}$. From the baby Vitali covering lemma, we can find a disjoint subcollection of these balls $\{B_1, \dots, B_N\}$ such that

$$\delta \leq m(E') \leq 3^d \sum_{i=1}^N m(B_i).$$

If $m(E) \leq \sum_{i=1}^N m(B_i) + \delta$, we are done. Otherwise, consider $E_2 = E \setminus \bigcup_{i=1}^N \overline{B_i}$, then $m(E_2) > \delta$. Also, the balls in $\mathcal{B}$ that are disjoint from $\bigcup_{i=1}^N \overline{B_i}$ form a Vitali covering of E_2 , so we can repeat the argument to find a finite collection of disjoint balls $\{B_{N+1}, \dots, B_{N_2}\}$ such that

$$\delta \leq m(E_2) \leq 3^d \sum_{i=N+1}^{N_2} m(B_i),$$

then $\sum_{i=1}^{N_2} m(B_i) \geq 2 \cdot 3^{-d} \delta$. If this process continues indefinitely, the sum of the measures of the disjoint balls we have found will exceed $m(E) < \infty$, which is a contradiction. Hence the process must terminate after finitely many steps, so we can find a finite collection of disjoint balls $\{B_1, \dots, B_N\}$ such that $m(E) \leq \sum_{i=1}^N m(B_i) + \delta$. ■

THEOREM 3.29. *If F is absolutely continuous on $[a, b]$, then F' exists a.e. Moreover, if $F' = 0$ a.e., then F is constant.*

Proof. Since F is absolutely continuous, it is of bounded variation, so F' exists a.e. For the second part, it suffices to show that $F(b) - F(a) = 0$ since the same argument can be applied to any subinterval of $[a, b]$. Let

$$E = \{x \in (a, b) : F'(x) = 0\}.$$

By assumption, $m(E) = b - a$. For $x \in E$, we have

$$\lim_{h \rightarrow 0} \left| \frac{F(x+h) - F(x)}{h} \right| = 0.$$

For each $\eta > 0$, exists an open interval (depending on η, x, ε) $I = (a_x, b_x) \subseteq [a, b]$ containing x such that

$$|F(b_x) - F(a_x)| \leq \varepsilon(b_x - a_x).$$

The collection of all such intervals forms a Vitali covering of E , so by the Vitali covering lemma, for all $\delta > 0$, exists a finite collection of disjoint intervals $\{I_1, \dots, I_N\}$ where $I_i = (a_i, b_i)$ such that

$$m(E) \leq \sum_{i=1}^N (b_i - a_i) + \delta.$$

By the inequality above, the complement $[a, b] \setminus \bigcup_{i=1}^N I_i$ can be written as a union of finitely many closed intervals $\bigcup_{j=1}^M [\alpha_j, \beta_j]$ such that the total length does not exceed δ . By absolute continuity of F , we can choose δ small enough such that

$$\sum_{j=1}^M |F(\beta_j) - F(\alpha_j)| \leq \varepsilon.$$

Also, we have

$$\sum_{i=1}^N |F(b_i) - F(a_i)| \leq \varepsilon \sum_{i=1}^N (b_i - a_i) \leq \varepsilon(b - a).$$

Therefore,

$$\begin{aligned} F(b) - F(a) &= \sum_{i=1}^N (F(b_i) - F(a_i)) + \sum_{j=1}^M (F(\beta_j) - F(\alpha_j)) \\ &\leq \varepsilon(b - a) + \varepsilon = (b - a + 1)\varepsilon. \end{aligned}$$

Taking $\varepsilon \rightarrow 0$ gives $F(b) - F(a) = 0$, so F is constant. ■

CLAIM 3.30. *Suppose that F is absolutely continuous on $[a, b]$, then F' exists a.e. and is integrable. Moreover, we have*

$$F(x) = F(a) + \int_a^x F'(t) dt$$

for every $x \in [a, b]$. Conversely, if $f \in L^1([a, b])$, then $F(x) = \int_a^x f(t)dt$ is absolutely continuous on $[a, b]$ and $F' = f$ a.e.

Proof. From Claim 3.24, we have

$$\int_a^x F'(t)dt \leq F(x) - F(a) < \infty,$$

so F' is integrable. Set $G(x) = \int_a^x F'(t)dt$, then G is absolutely continuous and so is the difference $G - F$. By Lebesgue differentiation theorem, we have $G' = F'$ a.e., so $(G - F)' = 0$ a.e. By the previous theorem, $G - F$ is constant, thus

$$F(x) = F(a) + \int_a^x F'(t)dt$$

for every $x \in [a, b]$. The converse direction follows from the first part and the absolute continuity of the integral. ■

3.3.2. Jump functions

We know that a bounded increasing function F on $[a, b]$ has countably many discontinuities: since F must be bounded, say $|F| \leq 2M$, then there are at most $2\frac{M}{\varepsilon}$ points where F has a jump of size at least ε , hence the total number of discontinuities must be countable. Let (x_n) be the sequence of discontinuities of F . Define

$$f(x_n^+) = \lim_{x \rightarrow x_n^+} F(x), \quad f(x_n^-) = \lim_{x \rightarrow x_n^-} F(x), \quad \alpha_n = f(x_n^+) - f(x_n^-).$$

Since F is increasing, exists $\theta_n \in [0, 1]$ such that

$$F(x_n) = F(x_n^-) + \theta_n \alpha_n.$$

Define

$$j_{n(x)} = \begin{cases} 0 & x < x_n \\ \theta_n & x = x_n \\ 1 & x > x_n \end{cases}$$

and define the jump function of F by

$$J_{F(x)} = \sum_{n=1}^{\infty} \alpha_n j_{n(x)}.$$

Note that $\sum_{n=1}^{\infty} \alpha_n \leq F(b) - F(a) < \infty$, so the series defining J_F converges absolutely and uniformly on $[a, b]$.

LEMMA 3.31. *Suppose that F is increasing and bounded on $[a, b]$, then J_F is discontinuous precisely at (x_n) . Moreover, the difference $F - J_F$ is continuous and increasing on $[a, b]$.*

Proof. If $x \neq x_n$ for every n , then j_n is continuous at x for every n , and since the series converges uniformly, J_F is continuous at x . If $x = x_N$ for some N , then we write

$$J(x) = \sum_{n=1}^N \alpha_n j_{n(x)} + \sum_{n=N+1}^{\infty} \alpha_n j_{n(x)}.$$

By the same argument as above, the second series is continuous at x_N , while the first series has a jump of α_N at x_N , so J_F has a jump of α_N at x_N . By definition of J_F , $F - J_F$ is continuous. For $x < y$, we have

$$J(y) - J(x) \leq \sum_{x < x_n \leq y} \alpha_n \leq F(y) - F(x),$$

so $F - J_F$ is increasing. ■

THEOREM 3.32. *If J is the jump function considered above, then J' exists a.e. and $J' = 0$ a.e.*

Proof. Given any $\varepsilon > 0$, let E be the set of those x where

$$\limsup_{h \rightarrow 0} \left| \frac{J(x+h) - J(x)}{h} \right| > \varepsilon$$

is measurable. Suppose $\delta = m(E)$. We want to show that $\delta = 0$. Since the series $\sum \alpha_n$ converges, for any $\eta > 0$, exists N such that $\sum_{n > N} \alpha_n < \eta$. Write

$$J_0(x) = \sum_{n > N} \alpha_n j_{n(x)},$$

then we have $J_0(b) - J_0(a) < \eta$. Let E_0 be the set of those x where

$$\limsup_{h \rightarrow 0} \left| \frac{J_0(x+h) - J_0(x)}{h} \right| > \varepsilon,$$

then $E_0 \subseteq E$ and E_0 differs from E by at most a finite set, namely $\{x_1, \dots, x_N\}$, so $m(E_0) = m(E) = \delta$. By inner regularity, there exists a compact set $K \subseteq E_0$ such that $m(K) > \frac{\delta}{2}$. For each $x \in K$, there exists an interval $(a_x, b_x) \ni x$ such that

$$J_0(b_x) - J_0(a_x) > \varepsilon(b_x - a_x).$$

By compactness, exists a finite collection of disjoint intervals that covers K , and by Vitali covering lemma, we can find a finite collection of disjoint intervals $\{(a_{i_k}, b_{i_k})\}_{k=1}^N$ such that

$$\sum_{k=1}^N (b_{i_k} - a_{i_k}) > \frac{1}{3} m(K).$$

Therefore,

$$J_0(b) - J_0(a) \geq \sum_{k=1}^N (J_0(b_{i_k}) - J_0(a_{i_k})) > \varepsilon \sum_{k=1}^N (b_{i_k} - a_{i_k}) > \frac{\varepsilon}{3} m(K) > \frac{\varepsilon}{6} \delta.$$

Since η is arbitrary, we can choose $\eta < \frac{\varepsilon}{6} \delta$, which is a contradiction. Hence $\delta = 0$, so $J' = 0$ a.e. ■

3.4. Rectifiable curves

DEFINITION 3.33. *Let γ be a parametrized curve in the plane given by $z(t) = (x(t), y(t))$ for $t \in [a, b]$, where x, y are continuous real-valued functions. γ is rectifiable if there exists $M < \infty$ such that for all partitions $a = t_0 < t_1 < \dots < t_N = b$ of $[a, b]$, we have*

$$\sum_{j=1}^N |z(t_j) - z(t_{j-1})| \leq M.$$

The length of γ is defined by

$$L(\gamma) = \sup \sum_{j=1}^N |z(t_j) - z(t_{j-1})|,$$

where the supremum is taken over all partitions of $[a, b]$.

PROPOSITION 3.34. γ is rectifiable if and only if x and y are of bounded variation.

Proof. Write $z(t) = x(t) + iy(t)$, then

$$z(t_i) - z(t_{i-1}) = (x(t_i) - x(t_{i-1})) + i(y(t_i) - y(t_{i-1})).$$

The conclusion follows from

$$|a + bi| \leq |a| + |b| \leq 2 |a + bi|$$

for every $a, b \in \mathbb{R}$. ■

We'd like to know when the length of a rectifiable curve can be computed by integration, i.e. when we have

$$L(\gamma) = \int_a^b \sqrt{|x'(t)|^2 + |y'(t)|^2} dt.$$

However, this is not always the case. For example, if $x(t) = y(t) = F(t)$ where F is the Cantor function, then γ traces out the line segment from $(0, 0)$ to $(1, 1)$, so $L(\gamma) = \sqrt{2}$, but $z'(t) = 0$ a.e., so $\int_0^1 |z'(t)| dt = 0$.

PROPOSITION 3.35. If $F : [a, b] \rightarrow \mathbb{C}$ is absolutely continuous, then

$$T_{F(a,b)} = \int_a^b |F'(t)| dt,$$

where $T_{F(a,b)}$ is the total variation of F on $[a, b]$.

Proof. By absolute continuity, for any partition $a = t_0 < t_1 < \dots < t_N = b$ of $[a, b]$, we have

$$\begin{aligned} \sum_{j=1}^N |F(t_j) - F(t_{j-1})| &= \sum_{j=1}^N \left| \int_{t_{j-1}}^{t_j} F'(t) dt \right| \\ &\leq \sum_{j=1}^N \int_{t_{j-1}}^{t_j} |F'(t)| dt = \int_a^b |F'(t)| dt, \end{aligned}$$

so $T_{F(a,b)} \leq \int_a^b |F'(t)| dt$. On the other hand, by absolute continuity, F' is integrable, so there exists a step function g on $[a, b]$ such that $|F' - g|_1 < \frac{\varepsilon}{2}$. Let $h = F' - g$, then $|h|_1 = \int_a^b |h(t)| dt < \frac{\varepsilon}{2}$. Set $G(x) = \int_a^x g(t) dt$ and $H(x) = \int_a^x h(t) dt$, then

$$F(x) - F(a) = G(x) + H(x) \Rightarrow T_{F(a,b)} \geq T_{G(a,b)} - T_{H(a,b)}.$$

By the first part, we have $T_{H(a,b)} \leq \int_a^b |h(t)| dt < \frac{\varepsilon}{2}$, hence

$$T_{F(a,b)} \geq T_{G(a,b)} - \frac{\varepsilon}{2}.$$

Partition the interval $[a, b]$ so that g is constant on each subinterval, then we have

$$T_{G(a,b)} \geq \sum_{j=1}^N |G(t_j) - G(t_{j-1})| = \sum_{j=1}^N \left| \int_{t_{j-1}}^{t_j} g(t) dt \right|$$

$$= \sum_{j=1}^N \int_{t_{j-1}}^{t_j} |g(t)| dt = \int_a^b |g(t)| dt.$$

Since $|F' - g|_1 < \frac{\varepsilon}{2}$, we have $\int_a^b |g(t)| dt > \int_a^b |F'(t)| dt - \frac{\varepsilon}{2}$, so

$$T_{F(a,b)} \geq \int_a^b |F'(t)| dt - \varepsilon.$$

Since ε is arbitrary, we have $T_{F(a,b)} \geq \int_a^b |F'(t)| dt$, hence $T_{F(a,b)} = \int_a^b |F'(t)| dt$. ■

THEOREM 3.36. *If x, y are absolutely continuous on $[a, b]$, then γ is rectifiable and*

$$L(\gamma) = \int_a^b \sqrt{|x'(t)|^2 + |y'(t)|^2} dt.$$

Proof. Since x and y are absolutely continuous, they are of bounded variation, so γ is rectifiable. The length formula follows from the previous proposition. ■

A rectifiable curve has a natural parametrization, called the arc-length parametrization. Suppose that γ is parametrized by $t \mapsto z(t)$. Write $s(t)$ for the length of the segment of γ that arises as the image of $z([a, t])$. Then s is continuous, nondecreasing and it maps $[a, b]$ to $[0, L(\gamma)]$. The arc-length parametrization of γ is given by $\tilde{z}(s)$, where $\tilde{z}(s) = z(t)$ for $s = s(t)$. If $s(t_1) = s(t_2)$ for some $t_1 < t_2$, then $z(t)$ does not vary on $[t_1, t_2]$, so $z(t_1) = z(t_2)$. Therefore, $\tilde{z}$ is well-defined. Moreover, $|\tilde{z}(s_1) - \tilde{z}(s_2)| \leq |s_1 - s_2|$ for every s_1, s_2 , so $\tilde{z}$ is Lipschitz continuous with constant 1, hence $|\tilde{z}'(s)| \leq 1$. By definition of total variation,

$$L(\gamma) = T_{\tilde{z}}(0, L(\gamma)) = \int_0^{L(\gamma)} |\tilde{z}'(s)| ds,$$

so $|\tilde{z}'(s)| = 1$ for a.e. s . In particular, writing $\tilde{z}(s) = (\tilde{x}(s), \tilde{y}(s))$, we have

$$L = \int_0^L \sqrt{|\tilde{x}'(s)|^2 + |\tilde{y}'(s)|^2} ds.$$

3.4.1. Isoperimetric Inequality

DEFINITION 3.37. *A curve parametrized by $z(t) \mapsto (x(t), y(t))$ is simple if $t \mapsto z(t)$ is injective. A curve is simple and closed if $t \mapsto z(t)$ is injective on $[a, b)$ and $z(a) = z(b)$. A curve is quasi-simple if $t \mapsto z(t)$ is injective for t in the complement of finitely many points in $[a, b]$.*

For a compact set $K \subset \mathbb{R}^2$ and $\delta > 0$, define

$$K^\delta = \{x \in \mathbb{R}^2 : d(x, K) < \delta\}.$$

We say that K has (one-dimensional) Minkowski content if the limit

$$\mathcal{M}(K) = \lim_{\delta \rightarrow 0} \frac{m(K^\delta)}{2\delta}$$

exists and is finite.

LEMMA 3.38. *If $\gamma = \{z(t) : t \in [a, b]\}$ is a curve, and if $D := |z(b) - z(a)|$ is the distance between the endpoints of γ , then $m(\gamma^\delta) \geq 2\delta D$.*

Proof. By translation and rotation, we may assume that $z(a) = (A, 0)$ and $z(b) = (B, 0)$ for some $A < B$, then $D = B - A$. By continuity of $t \mapsto x(t)$, for all $x \in [A, B]$, there exists $\bar{t}$ such that $x(\bar{t}) = x$. Define $\bar{Q} = (x(\bar{t}), y(\bar{t}))$, then $\bar{Q} \in \gamma$ and γ^δ contains a segment of the vertical line of length 2δ centered at $\bar{Q}$. In other words, $m_1((\gamma^\delta)_x) \geq 2\delta$ for every $x \in [A, B]$. By Fubini's theorem, we have

$$m(\gamma^\delta) = \int_{\mathbb{R}} m_1((\gamma^\delta)_x) dx \geq \int_A^B 2\delta dx = 2\delta D.$$

■

Consider

$$\overline{\mathcal{M}}(K) = \limsup_{\delta \rightarrow 0} \frac{m(K^\delta)}{2\delta}, \quad \underline{\mathcal{M}}(K) = \liminf_{\delta \rightarrow 0} \frac{m(K^\delta)}{2\delta}.$$

PROPOSITION 3.39. *Suppose that $\gamma = \{z(t) : t \in [a, b]\}$ is quasi-simple. If $\underline{\mathcal{M}}(\gamma) < \infty$, then γ is rectifiable and $L(\gamma) \leq \underline{\mathcal{M}}(\gamma)$.*

Proof. Let P be any partition $a = t_0 < t_1 < \dots < t_N = b$ of $[a, b]$. Define

$$L_P = \sum_{j=1}^N |z(t_j) - z(t_{j-1})|,$$

then L_P is the length of the polygonal curve obtained by connecting $z(t_0), z(t_1), \dots, z(t_N)$ in order. Let $\varepsilon > 0$. By continuity of z , there exists subintervals $I_j := [a_j, b_j] \subset (t_{j-1}, t_j)$ such that

$$\sum_{j=1}^N |z(b_j) - z(a_j)| \geq L_P - \varepsilon.$$

Since γ is quasi-simple, we can choose these subintervals so that their image under z are disjoint. Write $\gamma_j = \{z(t) : t \in I_j\}$. Also,

$$\gamma \supseteq \bigcup_{j=1}^N \gamma_j \rightsquigarrow \gamma^\delta \supseteq \bigcup_{j=1}^N \gamma_j^\delta.$$

If δ is sufficiently small, then γ_j^δ are disjoint for distinct j , so

$$m(\gamma^\delta) \geq m\left(\bigcup_{j=1}^N \gamma_j^\delta\right) = \sum_{j=1}^N m(\gamma_j^\delta) \geq 2\delta \sum_{j=1}^N |z(b_j) - z(a_j)| \geq 2\delta(L_P - \varepsilon).$$

Since ε is arbitrary, we have $m(\gamma^\delta) \geq 2\delta L_P$ for every partition P . Taking supremum over all partitions gives $m(\gamma^\delta) \geq 2\delta L(\gamma)$, so $\underline{\mathcal{M}}(\gamma) \geq L(\gamma)$. ■

PROPOSITION 3.40. *Suppose that $\gamma = \{z(t) : t \in [a, b]\}$ is rectifiable, then $\overline{\mathcal{M}}(\gamma) \leq L(\gamma)$.*

Proof. Since both quantities are independent of the parametrization, we may assume that γ is parametrized by arc-length. Write the arc-length parametrization of γ as $z(s) = (x(s), y(s))$ for $s \in [0, L]$, then z is absolutely continuous and $|z'(s)| = 1$ for a.e. s . For each $n \in \mathbb{N}$, define

$$F_{n(s)} = \sup_{0 < |h| < \frac{1}{n}} \left| \frac{z(s+h) - z(s)}{h} - z'(s) \right|.$$

where we set $z(s) = z(0)$ for $s < 0$ and $z(s) = z(L)$ for $s > L$. Since z is continuous, the supremum can be replaced by supremum over countably many h , so F_n is measurable. Moreover, $F_{n(s)} \rightarrow 0$ as $n \rightarrow \infty$ for a.e. s . Fix $\varepsilon \in (0, 1)$. By the Egorov theorem, there exists a measurable set $E_\varepsilon \subseteq [0, L]$ with $m(E_\varepsilon) < \varepsilon$ such that the convergence $F_{n(s)} \rightarrow 0$ is uniform on $[0, L] \setminus E_\varepsilon$. Choose $r_\varepsilon = \frac{1}{n}$ for sufficiently large n such that

$$\sup_{0 < |h| < r_\varepsilon} \left| \frac{z(s+h) - z(s)}{h} - z'(s) \right| < \varepsilon$$

for every $s \in [0, L] \setminus E_\varepsilon$. By enlarging E_ε if necessary, we can assume that $|z'(s)| = 1$ for every $s \in [0, L] \setminus E_\varepsilon$. Fix $\rho \in (0, r_\varepsilon)$. Partition $[0, L]$ into subintervals of length ρ , then there are $N \leq \frac{L}{\rho} + 1$ subintervals, say $I_1, \dots, I_N$. We say that I_j is good if $I_j \not\subseteq E_\varepsilon$ and bad if $I_j \subseteq E_\varepsilon$. The total length of bad intervals is at most $m(E_\varepsilon) < \varepsilon$. Define $\gamma_j = \{z(s) : s \in I_j\}$, then $m(\gamma^\delta) \leq \sum_{j=1}^N m(\gamma_j^\delta)$.

First consider that I_j is good, then there exists $s_0 \in I_j \setminus E_\varepsilon$ such that

$$\sup_{0 < |h| < r_\varepsilon} \left| \frac{z(s_0+h) - z(s_0)}{h} - z'(s_0) \right| < \varepsilon, \quad |z'(s_0)| = 1.$$

By translation and rotation, we may assume that $z(s_0) = (0, 0)$ and $z'(s_0) = (1, 0)$. Note that as h varies over $[a_j - s_0, b_j - s_0]$, $h + s_0$ varies over I_j . Note that $|h| < r_\varepsilon$ for every $h \in [a_j - s_0, b_j - s_0]$, so γ_j is contained in the rectangle

$$[a_j - s_0 - \varepsilon\rho, b_j - s_0 + \varepsilon\rho] \times [-\varepsilon\rho, \varepsilon\rho].$$

Therefore, γ_j^δ is contained in the rectangle

$$[a_j - s_0 - \varepsilon\rho - \delta, b_j - s_0 + \varepsilon\rho + \delta] \times [-\varepsilon\rho - \delta, \varepsilon\rho + \delta],$$

so

$$m(\gamma_j^\delta) \leq (\rho + 2\varepsilon\rho + 2\delta)(2\varepsilon\rho + 2\delta) = 2\delta\rho + O(\varepsilon\rho^2 + \varepsilon\rho\delta + \delta^2).$$

For bad intervals, $|z(s) - z(s')| \leq |s - s'|$ for every $s, s' \in I_j$, so γ_j is contained in a ball of radius ρ , hence γ_j^δ is contained in a ball of radius $\rho + \delta$, so $m(\gamma_j^\delta) \leq \pi(\rho + \delta)^2 = O(\rho^2 + \rho\delta + \delta^2)$. Since there are at most $\frac{L}{\rho} + 1$ intervals, so at most $\frac{\varepsilon}{\rho} + 1$ bad intervals, we have

$$\begin{aligned} m(\gamma^\delta) &\leq \sum_{j=1}^N m(\gamma_j^\delta) \\ &\leq \left(\frac{L}{\rho} + 1\right)(2\delta\rho + O(\varepsilon\rho^2 + \varepsilon\rho\delta + \delta^2)) + \left(\frac{\varepsilon}{\rho} + 1\right)O(\rho^2 + \rho\delta + \delta^2) \\ &= 2\delta L + 2\delta\rho + O\left(\varepsilon\delta + \frac{\delta^2}{\rho} + \varepsilon\rho\right) + O\left(\left(\frac{\varepsilon}{\rho} + 1\right)(\rho^2 + \delta^2)\right). \end{aligned}$$

take $\rho = \frac{\delta}{\sqrt{\varepsilon}}$, then

$$\frac{m(\gamma^\delta)}{2\delta} \leq L + O\left(\frac{\delta}{\sqrt{\varepsilon}} + \varepsilon + \sqrt{\varepsilon} + \frac{\delta}{\varepsilon}\right).$$

Letting $\delta \rightarrow 0$ gives $\overline{M}(\gamma) \leq L$. ■

THEOREM 3.41. Suppose that Ω is a bounded open subset and its boundary $\overline{\Omega} \setminus \Omega$ is a rectifiable curve γ , then

$$L(\gamma)^2 \geq 4\pi m(\Omega).$$

Proof. For each $\delta > 0$, consider

$$\Omega_+(\delta) = \{x \in \mathbb{R}^2 : d(x, \overline{\Omega}) < \delta\}, \quad \Omega_-(\delta) = \{x \in \mathbb{R}^2 : d(x, \mathbb{R}^2 \setminus \Omega) \geq \delta\}.$$

Thus $\Omega_-(\delta) \subseteq \Omega \subseteq \Omega_+(\delta)$. Note that

$$\Omega_+(\delta) = \Omega_-(\delta) \cup \gamma^\delta$$

and the union is disjoint. We have

$$\Omega + B(0, \delta) \subseteq \Omega_+(\delta), \quad \Omega_-(\delta) + B(0, \delta) \subseteq \Omega.$$

By the Brunn-Minkowski inequality, we have

$$m(\Omega_+(\delta)) \geq \left(m(\Omega)^{\frac{1}{2}} + m(B(0, \delta))^{\frac{1}{2}}\right)^2 = \left(m(\Omega)^{\frac{1}{2}} + \sqrt{\pi}\delta\right)^2 \geq m(\Omega) + 2\sqrt{\pi}\delta\sqrt{m(\Omega)}.$$

Similarly, we have

$$m(\Omega) \geq m(\Omega_-(\delta)) + 2\sqrt{\pi}\delta\sqrt{m(\Omega_-(\delta))}.$$

Therefore,

$$m(\gamma^\delta) = m(\Omega_+(\delta)) - m(\Omega_-(\delta)) \geq 2\sqrt{\pi}\delta\left(\sqrt{m(\Omega)} + \sqrt{m(\Omega_-(\delta))}\right),$$

dividing both sides by 2δ and taking limsup as $\delta \rightarrow 0$ gives

$$\mathcal{M}(\gamma) \geq 2\sqrt{\pi}\sqrt{m(\Omega)}.$$

By the previous propositions, we have $L(\gamma) \geq \mathcal{M}(\gamma)$, so $L(\gamma) \geq 2\sqrt{\pi}\sqrt{m(\Omega)} \rightsquigarrow L(\gamma)^2 \geq 4\pi m(\Omega)$. The equality holds if and only if Ω is a disk, which can be seen from the equality case of the Brunn-Minkowski inequality. ■

Hilbert Spaces

CHAPTER 4

Let X be a vector space. An inner product is a map $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{C}$ such that

- (i) $\langle \alpha_1 x_1 + \alpha_2 x_2, y \rangle = \alpha_1 \langle x_1, y \rangle + \alpha_2 \langle x_2, y \rangle$ for every $x_1, x_2, y \in X$ and $\alpha_1, \alpha_2 \in \mathbb{C}$;
- (ii) $\langle y, x \rangle = \overline{\langle x, y \rangle}$ for every $x, y \in X$;
- (iii) $\langle x, x \rangle \geq 0$ for every $x \in X$, and $\langle x, x \rangle = 0$ if and only if $x = 0$.

In general, the inner product is a continuous map from $X \times X$ to $\mathbb{C}$. Two vectors x, y are orthogonal if $\langle x, y \rangle = 0$. The norm of a vector x is defined by $|x| = \sqrt{\langle x, x \rangle}$. A complete inner product space is called a Hilbert space.

Example 4.1. ℓ^2 is the space of square-summable sequences, i.e.

$$\ell^2 = \left\{ (x_n)_{n=1}^{\infty} : \sum_{n=1}^{\infty} |x_n|^2 < \infty \right\},$$

and the inner product is defined by

$$\langle (x_n), (y_n) \rangle = \sum_{n=1}^{\infty} x_n \overline{y_n}.$$

Example 4.2. Recall that $L^2(\mathbb{R}^d)$ is the space of square-integrable functions on $\mathbb{R}^d$, and it is a Hilbert space with inner product defined by

$$\langle f, g \rangle = \int_{\mathbb{R}^d} f(x) \overline{g(x)} dx.$$

If $|\cdot|$ is the norm induced by the inner product $\langle \cdot, \cdot \rangle$ and $|\cdot|'$ is the completion of $|\cdot|$, then there exists an inner product $\langle \cdot, \cdot \rangle'$ extending $\langle \cdot, \cdot \rangle$ such that $|\cdot|'$ is the norm induced by $\langle \cdot, \cdot \rangle'$.

PROPOSITION 4.1 (Parallelogram Law). *Let $|\cdot|$ be a norm on X induced by an inner product $\langle \cdot, \cdot \rangle$, then for any $x, y \in X$, we have*

$$|x + y|^2 + |x - y|^2 = 2|x|^2 + 2|y|^2.$$

Proof. By definition of the norm, we have

$$|x + y|^2 = \langle x + y, x + y \rangle = \langle x, x \rangle + \langle y, y \rangle + 2 \operatorname{Re} \langle x, y \rangle,$$

and

$$|x - y|^2 = \langle x - y, x - y \rangle = \langle x, x \rangle + \langle y, y \rangle - 2 \operatorname{Re} \langle x, y \rangle.$$

Adding these two equations gives the desired result. ■

Example 4.3. Consider ℓ^p spaces for $p \in [1, \infty)$ with norm defined by

$$|(x_n)|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{\frac{1}{p}}.$$

Suppose that ℓ^p is a Hilbert space. Take $x = (1, 1, 0, \dots)$ and $y = (1, -1, 0, \dots)$, then

$$|x + y|_p^2 + |x - y|_p^2 = 2 \cdot 2^2 = 8,$$

while

$$2 |x|_p^2 + 2 |y|_p^2 = 2 \cdot 2^{\frac{2}{p}} + 2 \cdot 2^{\frac{2}{p}} = 4 \cdot 2^{\frac{2}{p}},$$

so we must have $p = 2$. Therefore, ℓ^p is a Hilbert space if and only if $p = 2$.

PROPOSITION 4.2 (Polarization Identity). *Let $|\cdot|$ be a norm on X induced by an inner product $\langle \cdot, \cdot \rangle$, then for any $x, y \in X$, we have*

$$\operatorname{Re} \langle x, y \rangle = \frac{1}{4}(|x + y|^2 - |x - y|^2), \quad \operatorname{Im} \langle x, y \rangle = \frac{1}{4}(|x + iy|^2 - |x - iy|^2).$$

THEOREM 4.3. *Let K be a closed, convex, proper subset of a Hilbert space X , and let $x_0 \in X \setminus K$. Then there exists a unique $x^* \in K$ such that*

$$|x_0 - x^*| = \inf_{x \in K} |x_0 - x|.$$

Proof. Let $d = \inf_{x \in K} |x_0 - x|$. Let (x_n) be a sequence in K such that $|x_0 - x_n| \rightarrow \inf_{x \in K} |x_0 - x|$. We claim that (x_n) is a Cauchy sequence. Note that

$$\begin{aligned} |x_n - x_m|^2 &= |(x_n - x_0) - (x_m - x_0)|^2 \\ &= 2 |x_n - x_0|^2 + 2 |x_m - x_0|^2 - |(x_n - x_0) + (x_m - x_0)|^2 \\ &= 2 |x_n - x_0|^2 + 2 |x_m - x_0|^2 - 4 \left| x_n + \frac{x_m}{2} - x_0 \right|^2. \end{aligned}$$

Note that $x_n + \frac{x_m}{2} \in K$ by convexity of K , so $|x_n + \frac{x_m}{2} - x_0| \geq d$, hence

$$|x_n - x_m|^2 \leq 2 |x_n - x_0|^2 + 2 |x_m - x_0|^2 - 4d^2 \rightarrow 0$$

as $n, m \rightarrow \infty$, so (x_n) is a Cauchy sequence. Since X is complete, there exists $x^* \in X$ such that $x_n \rightarrow x^*$. Since K is closed, we have $x^* \in K$. By continuity of the norm, we have $|x_0 - x^*| = \lim_{n \rightarrow \infty} |x_0 - x_n| = d$, so x^* is a minimizer.

If $x' \in K$ is another minimizer, then

$$\begin{aligned} |x' - x^*|^2 &= 2 |x' - x_0|^2 + 2 |x^* - x_0|^2 - 4 \left| x' + \frac{x^*}{2} - x_0 \right|^2 \\ &\leq -4d^2 + 2d^2 + 2d^2 = 0, \end{aligned}$$

so $x' = x^*$, hence the minimizer is unique. ■

4.1. Orthogonal Decomposition

THEOREM 4.4. *Let Y be a closed proper subspace of a Hilbert space X , and let $x_0 \in X \setminus Y$. The point y_0 that minimizes the distance between x_0 and Y satisfies*

$$\langle x_0 - y_0, y \rangle = 0$$

for every $y \in Y$. Conversely, if $z \in Y$ satisfies $\langle x_0 - z, y \rangle = 0$ for every $y \in Y$, then $z = y_0$. In this case,

$$|x_0 - y_0|^2 + |y_0|^2 = |x_0|^2.$$

Proof. The existence and uniqueness of y_0 follows from the previous theorem.

For $y \in Y$ and $h \in \mathbb{R}$, define

$$\varphi(h) = |x_0 - y_0 + hy|^2,$$

then φ attains its minimum at $h = 0$. By differentiating φ at $h = 0$, we have

$$0 = \varphi'(0) = 2 \operatorname{Re} \langle x_0 - y_0, y \rangle.$$

Replacing y by iy gives $\operatorname{Im} \langle x_0 - y_0, y \rangle = 0$, so $\langle x_0 - y_0, y \rangle = 0$ for every $y \in Y$.

Conversely, if $z \in Y$ satisfies $\langle x_0 - z, y \rangle = 0$ for every $y \in Y$, then $\langle x_0 - z, z - y \rangle = 0$ for every $y \in Y$, so

$$|x_0 - y|^2 = |x_0 - z|^2 + |z - y|^2 \geq |x_0 - z|^2$$

for every $y \in Y$, hence z minimizes the distance between x_0 and Y , so $z = y_0$. In this case, we have

$$|x_0|^2 = |x_0 - y_0|^2 + |y_0|^2$$

since $\langle x_0 - y_0, y_0 \rangle = 0$. ■

DEFINITION 4.5. Let Y be a closed subspace of a Hilbert space X . The projection operator $\mathcal{P}$ of X onto Y is defined by

$$\mathcal{P}(x_0) = \begin{cases} y_0 x_0 \in X \setminus Y \\ x_0 x_0 \in Y \end{cases}$$

where y_0 is the unique point in Y that minimizes the distance between x_0 and Y .

It's easy to see that $\mathcal{P}$ is linear and continuous, and $\mathcal{P}^2 = \mathcal{P}$. Moreover, the operator norm of $\mathcal{P}$ equals

$$|\mathcal{P}| = \sup_{x \neq 0} \frac{|\mathcal{P}x|}{|x|} = 1.$$

4.1.1. Self-Duality

Hilbert spaces have self-duality. For each $z \in X$, we can associate a bounded linear functional Λ_z defined by $\Lambda_{z(x)} = \langle x, z \rangle$. The map

$$\Phi : X \rightarrow X^*, \quad z \mapsto \Lambda_z$$

is a sesquilinear map, i.e. $\Phi(\alpha z_1 + \beta z_2) = \bar{\alpha}\Phi(z_1) + \bar{\beta}\Phi(z_2)$ for every $z_1, z_2 \in X$ and $\alpha, \beta \in \mathbb{C}$.

THEOREM 4.6 (Riesz Representation Theorem). Let X be a Hilbert space. If Λ is a bounded linear functional on X , then there exists a unique $z \in X$ such that $\Lambda = \Lambda_z$. Moreover, $|\Lambda_z| = |z|$.

Proof. Let Λ be a bounded linear functional on X . Consider $Y = \{x \in X : \Lambda(x) = 0\}$, then Y is a closed subspace of X . If $Y = X$, then $\Lambda = 0$, so we can take $z = 0$. Otherwise, take any $x_0 \in X \setminus Y$, so

$$\langle x_0 - \mathcal{P}x_0, y \rangle = 0$$

for every $y \in Y$. Set $z_0 = x_0 - \mathcal{P}x_0$. Fix $x \in X$. Let $u = (\Lambda x)z_0 - (\Lambda z_0)x$, then $u \in Y$ since $\Lambda u = 0$. Therefore, we have

$$0 = \langle u, z_0 \rangle = \langle (\Lambda x)z_0 - (\Lambda z_0)x, z_0 \rangle = (\Lambda x)|z_0|^2 - (\Lambda z_0)\langle x, z_0 \rangle,$$

so

$$\Lambda x = \frac{\Lambda z_0}{|z_0|^2} \langle x, z_0 \rangle.$$

Set $z = \frac{\overline{\Lambda z_0}}{|z_0|^2} z_0$, then $\Lambda x = \langle x, z \rangle$ for every $x \in X$, so $\Lambda = \Lambda_z$. Moreover,

$$|\Lambda_z| = \sup_{x \neq 0} \frac{|\langle x, z \rangle|}{|x|} \leq |z|$$

by the Cauchy-Schwarz inequality, and $|\langle z, z \rangle| = |z|^2$, so $|\Lambda_z| = |z|$. ■

Remark. This shows that Φ is an isometric sesquilinear isomorphism between X and X^* .

4.1.2. Direct Sum Decomposition

We want to write $X = X_1 \oplus X_2$ for some closed subspaces X_1, X_2 of X , so that the projections onto X_1 and X_2 are continuous. This gives rise to the following question: gives a closed subspace X_1 of a Banach space X , does there exist a closed subspace X_2 such that $X = X_1 \oplus X_2$. The answer is negative for general Banach spaces, but it is positive for Hilbert spaces, as we can set $X_2 = X_1^\perp$. In fact, if a Banach space satisfies this property, then its norm must be equivalent to a norm induced by a complete inner product.

DEFINITION 4.7. Let X be a Hilbert space, and let X_1 be a closed subspace of X . The orthogonal complement of X_1 is defined by

$$X_1^\perp = \{x \in X : \langle x, x_1 \rangle = 0 \text{ for every } x_1 \in X_1\}.$$

PROPOSITION 4.8. $X_1^\perp$ is a closed subspace of X , and $X = X_1 \oplus X_1^\perp$.

Proof. It's easy to see that $X_1^\perp$ is a closed subspace of X since the inner product is continuous. For each $x \in X$, we can write

$$x = \mathcal{P}x + (x - \mathcal{P}x),$$

where $\mathcal{P}$ is the projection operator of X onto X_1 , so $\mathcal{P}x \in X_1$ and $x - \mathcal{P}x \in X_1^\perp$, hence $X = X_1 + X_1^\perp$. If $x_0 \in X_1 \cap X_1^\perp$, then $\langle x_0, x_0 \rangle = 0$, so $x_0 = 0$, hence $X_1 \cap X_1^\perp = \{0\}$, implying that $X = X_1 \oplus X_1^\perp$. ■

The bounded operators $\mathcal{P}$ and $I - \mathcal{P}$ are the projections of X onto X_1 and $X_1^\perp$, respectively.

THEOREM 4.9. For any closed subspace X_1 of a Hilbert space X , we have $X = X_1 \oplus X_1^\perp$. Moreover, if $\mathcal{P} : X \rightarrow X_1$ is the projection operator, then $\forall x \in X$, $\mathcal{P}x$ is the unique point in X_1 such that

$$|x - \mathcal{P}x| = \inf_{x_1 \in X_1} |x - x_1|,$$

and the projection $\mathcal{Q} : X \rightarrow X_1^\perp$ is given by $\mathcal{Q}x = x - \mathcal{P}x$.

4.1.3. Complete Orthonormal Sets

Given an orthonormal spanning set, we can compute the projection $\mathcal{P}x$ of x onto the closed subspace spanned by the set efficiently.

LEMMA 4.10 (Bessel's Inequality). *Let $S = \{x_\alpha : \alpha \in I\}$ be an orthonormal set in a Hilbert space X . Then for every $x \in X$, $\langle x, x_\alpha \rangle = 0$ for all but countably many $\alpha \in I$. Write B for the set of $\alpha \in I$ such that $\langle x, x_\alpha \rangle \neq 0$, then for any sequence (α_k) in B , we have*

$$\sum_k |\langle x, x_{\alpha_k} \rangle|^2 \leq |x|^2.$$

Proof. Assume that S is finite. Let $S = \{x_1, \dots, x_n\}$. Let $y = \sum_{k=1}^n \langle x, x_k \rangle x_k$, then $\langle x - y, x_k \rangle = 0$ for every k , so

$$|x|^2 = |x - y|^2 + |y|^2 \geq |y|^2 = \sum_{k=1}^n |\langle x, x_k \rangle|^2.$$

In general, let $x \in X$ and $l \in \mathbb{N}$. Consider

$$S_l = \left\{ x_\alpha \in S : |\langle x, x_\alpha \rangle| > \frac{1}{l} \right\}.$$

By the finite case, we have $\sum_{x_\alpha \in S_l} |\langle x, x_\alpha \rangle|^2 \leq |x|^2$, so S_l is finite for every l . Therefore,

$$S_x = \{x_\alpha \in S : \langle x, x_\alpha \rangle \neq 0\} = \bigcup_{l=1}^{\infty} S_l$$

is countable. The general case follows from the finite case by taking limit as $n \rightarrow \infty$. ■

THEOREM 4.11. *Let Y be a closed subspace of a Hilbert space X . Suppose that S is an orthonormal subset of Y such that $\text{span}(S)$, the set of finite linear combinations of elements in S , is dense in Y . Then for every $x \in X$, its orthogonal projection on Y is given by*

$$\mathcal{P}x = \sum_k \langle x, x_k \rangle x_k,$$

where $B = \{x_k : k \in \mathbb{N}\}$ is the set of elements in S that are not orthogonal to x .

Proof. We first verify that the sum is convergent. Let $z_n = \sum_{k=1}^n \langle x, x_k \rangle x_k$. It suffices to show that (z_n) is a Cauchy sequence. For $m < n$, we have

$$|z_m - z_n|^2 = \left| \sum_{k=m+1}^n \langle x, x_k \rangle x_k \right|^2 = \sum_{k=m+1}^n |\langle x, x_k \rangle|^2 \rightarrow 0$$

as $m, n \rightarrow \infty$ by Bessel's inequality. Next we verify that $\mathcal{P}x = \sum_k \langle x, x_k \rangle x_k$. If y is a finite linear combination of elements in S , then $\langle x - z_n, y \rangle = 0$ for sufficiently large n , so $\langle x - \sum_k \langle x, x_k \rangle x_k, y \rangle = 0$. Since $\text{span}(S)$ is dense in Y , we can take $n \rightarrow \infty$ and by continuity of the inner product, we have $\langle x - \sum_k \langle x, x_k \rangle x_k, y \rangle = 0$ for every $y \in Y$, so $\mathcal{P}x = \sum_k \langle x, x_k \rangle x_k$. ■

DEFINITION 4.12. *A subset B of a Hilbert space X is called a complete orthonormal set if*

- (i) B is an orthonormal set, and
- (ii) $\text{span}(B)$ is dense in X .

THEOREM 4.13. *Every nonzero Hilbert space has a complete orthonormal set.*

Proof. Let $\mathcal{O}$ be the collection of orthonormal sets in X , partially ordered by inclusion, which is nonempty since X is nonzero. Let $\mathcal{C}$ be a chain in $\mathcal{O}$, then $\bigcup_{O \in \mathcal{C}} O$ is an upper bound of $\mathcal{C}$, so by Zorn's lemma, there exists a maximal orthonormal set B . We claim that B is a complete orthonormal set. If not, then there exists $z \in X \setminus \overline{\text{span}(B)}$. Let $\mathcal{P}$ be the projection operator of X onto $\overline{\text{span}(B)}$ and

$$z' = z - \frac{\mathcal{P}z}{|z - \mathcal{P}z|},$$

then $B \cup \{z'\}$ is an orthonormal set, contradicting the maximality of B . Therefore, B is a complete orthonormal set. ■

THEOREM 4.14. *Let B be an orthonormal set in a Hilbert space X . The following are equivalent:*

- (i) B is a complete orthonormal set;
- (ii) $x = \sum_{x_\alpha \in B} \langle x, x_\alpha \rangle x_\alpha$ for every $x \in X$;
- (iii) $|x|^2 = \sum_{x_\alpha \in B} |\langle x, x_\alpha \rangle|^2$ for every $x \in X$ (Parseval's identity);
- (iv) if $\langle x, x_\alpha \rangle = 0$ for every $x_\alpha \in B$, then $x = 0$.

Proof.

- (i) $\Rightarrow$ (ii): Since $x \in X = \overline{\text{span}(B)}$, the orthogonal projection of x onto $\overline{\text{span}(B)}$ is x itself, so $x = \sum_{x_\alpha \in B} \langle x, x_\alpha \rangle x_\alpha$.
- (ii) $\Rightarrow$ (iii): We have

$$|x|^2 = \langle x, x \rangle = \left\langle \sum_{x_\alpha \in B} \langle x, x_\alpha \rangle x_\alpha, x \right\rangle = \sum_{x_\alpha \in B} |\langle x, x_\alpha \rangle|^2.$$

- (iii) $\Rightarrow$ (iv): Straightforward.
- (iv) $\Rightarrow$ (i): If B is not complete, then there exists $x \in X \setminus \overline{\text{span}(B)}$. Let $\mathcal{P}$ be the projection operator of X onto $\overline{\text{span}(B)}$, then $\langle x - \mathcal{P}x, x_\alpha \rangle = 0$ for every $x_\alpha \in B$, but $x - \mathcal{P}x \neq 0$, contradicting (iv). Therefore, B is a complete orthonormal set.

■

PROPOSITION 4.15. *A Hilbert space X has a countable complete orthonormal set if and only if X is separable.*

Proof. “ $\Rightarrow$ ”: Let B be a countable orthonormal set. Then rational linear combinations of elements in B form a countable dense subset of X , so X is separable. “ $\Leftarrow$ ”: Let $D = \{x_1, x_2, \dots\}$ be a countable dense subset of X . For each n , we throw away x_n if it is linearly dependent on $x_1, \dots, x_{n-1}$, obtaining a linearly independent subset $D' = \{y_1, y_2, \dots\}$ whose closure of span is still X . Applying the Gram-Schmidt process to D' gives a countable complete orthonormal set. ■

THEOREM 4.16. *Let X be an infinite dimensional separable Hilbert space. Then exists an inner-product preserving isomorphism $\Phi : X \rightarrow \ell^2$.*

Proof. Let (x_k) be a countable complete orthonormal set in X . Define

$$\Phi : X \rightarrow \ell^2, \quad x \mapsto (\langle x, x_k \rangle)_{k=1}^\infty.$$

By the Parseval's identity, we have $|\Phi x|_2^2 = \sum_{k=1}^{\infty} |\langle x, x_k \rangle|^2 = |x|^2$, so Φ is an isometry hence linear.

For $(b_k) \in \ell^2$, define

$$y_n = \sum_{k=1}^n b_k x_k, \quad y = \lim_{n \rightarrow \infty} y_n.$$

The limit exists since (y_n) is a Cauchy sequence by definition of ℓ^2 . By completeness of X , we have $y \in X$. Moreover, $\Phi y = (b_k)_{k=1}^{\infty}$, so Φ is surjective. Φ is injective since $\Phi x = 0$ implies $|x|^2 = \sum_{k=1}^{\infty} |\langle x, x_k \rangle|^2 = 0$, so $x = 0$. By Polarization Identity, Φ preserves the inner product. Therefore, Φ is an inner-product preserving isomorphism between X and ℓ^2 . ■

Fourier Analysis

CHAPTER 5

5.1. Fourier Series

For $f \in L^1([-\pi, \pi])$, we can define its n -th Fourier coefficient $n \in \mathbb{Z}$ by

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx.$$

$\hat{f}(n)$ is well-defined since the absolute value of the integrand is $|f(x)|$, which is integrable. Write

$$f(x) \sim \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{inx}$$

for the Fourier series of f . In general, we don't expect the Fourier series of f to converge to f pointwise. By Riemann-Lebesgue lemma, we have $\hat{f}(n) \rightarrow 0$ as $|n| \rightarrow \infty$. We want to know whether there is a certain rate at which $\hat{f}(n)$ goes to zero, and conversely, given a sequence (a_n) that goes to zero, can we find a function f such that $\hat{f}(n) = a_n$ for every n ?

Recall the Fejér kernel F_N defined by

$$F_N(x) = \frac{1}{N} \left(\frac{\sin(N\frac{x}{2})}{\sin(\frac{x}{2})} \right)^2.$$

We have $|F_N|_{L^1([-\pi, \pi])} = 2\pi$, and

$$\hat{F}_N(n) = \begin{cases} 1 - \frac{|n|}{N} & |n| < N \\ 0 & |n| \geq N \end{cases}$$

If g is continuous and 2π -periodic, and define convolution

$$(g * F_N)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(y) F_N(x-y) dy = \sum_{n=-(N-1)}^{N-1} \left(1 - \frac{|n|}{N}\right) \hat{g}(n) e^{inx}.$$

then $g * F_N \rightrightarrows g$ uniformly as $N \rightarrow \infty$.

PROPOSITION 5.1. *Let (a_n) be a sequence of nonnegative numbers such that $\lim_{|n| \rightarrow \infty} a_n = 0$. Suppose that $a_{-n} = a_n$ and $a_{n-1} + a_{n+1} \geq 2a_n$ for every $n \geq 1$, then there exists a nonnegative function $f \in L^1([-\pi, \pi])$ such that $\hat{f}(n) = a_n$ for every n .*

Proof. Note that $(a_n - a_{n+1})$ is a decreasing sequence. Also,

$$\sum_{n=0}^N (a_n - a_{n+1}) = a_0 - a_{N+1} \rightarrow a_0$$

as $N \rightarrow \infty$, so

$$\lim_{n \rightarrow \infty} n(a_n - a_{n+1}) = 0.$$

We have

$$\sum_{n=1}^N n(a_{n-1} + a_{n+1} - 2a_n) = a_0 - a_N - N(a_N - a_{N+1}) \rightarrow a_0$$

as $N \rightarrow \infty$. Define

$$f(x) = \sum_{n=1}^{\infty} n(a_{n-1} + a_{n+1} - 2a_n) F_{n(x)}.$$

Since $|F_n|_{L^1([-\pi, \pi])} = 2\pi$, the series converges in $L^1([-\pi, \pi])$. Also, f is nonnegative since F_n is nonnegative and $a_{n-1} + a_{n+1} \geq 2a_n$. By dominated convergence theorem, we have

$$\begin{aligned} \hat{f}(m) &= \sum_{n=1}^{\infty} n(a_{n-1} + a_{n+1} - 2a_n) \hat{F}_{n(m)} \\ &= \sum_{n=|m|+1}^{\infty} n(a_{n-1} + a_{n+1} - 2a_n) \left(1 - \frac{|m|}{n}\right) \\ &= \sum_{n=|m|+1}^{\infty} (a_{n-1} + a_{n+1} - 2a_n)(n - |m|) \\ &= \sum_{n=|m|+1}^{\infty} (a_{n-1} + a_{n+1} - 2a_n)n - |m| \sum_{n=|m|+1}^{\infty} (a_{n-1} + a_{n+1} - 2a_n) \\ &= a_{|m|} - \lim_{n \rightarrow \infty} a_n - |m| \cdot 0 = a_{|m|} = a_m. \end{aligned}$$

■

Remark. The condition $a_{n-1} + a_{n+1} \geq 2a_n$ roughly says that a_n is convex in n . If we can construct convex functions of arbitrarily slow decay, then we can construct functions whose Fourier coefficients decay arbitrarily slowly.

PROPOSITION 5.2. *Let $f \in L^1([-\pi, \pi])$ and assume that $\hat{f}(|n|) = -\hat{f}(-|n|)$ for every $n \in \mathbb{Z}$. Then*

$$\sum_{n \neq 0} \frac{|\hat{f}(n)|}{|n|} < \infty.$$

Proof. Note that $\hat{f}(0) = 0$, hence

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = 0,$$

so we can extend f to a 2π -periodic function on $\mathbb{R}$ and define

$$F(x) = \int_{-\pi}^x f(t) dt,$$

then F is also 2π -periodic and absolutely continuous. By integration by parts, we have

$$\hat{F}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(x) e^{-inx} dx = -\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \frac{e^{-inx}}{-in} dx = \frac{\hat{f}(n)}{in}.$$

Let F_N be the N -th Fejér kernel, then $F * F_N \rightrightarrows F$ uniformly as $N \rightarrow \infty$, so

$$F(0) = \lim_{N \rightarrow \infty} (F * F_N)(0)$$

$$\begin{aligned}
&= \lim_{N \rightarrow \infty} \sum_{n=-(N-1)}^{N-1} \left(1 - \frac{|n|}{N}\right) \hat{F}(n) \\
&= \lim_{N \rightarrow \infty} \sum_{n=-(N-1)}^{N-1} \left(1 - \frac{|n|}{N}\right) \frac{\hat{f}(n)}{in} \\
&= \lim_{N \rightarrow \infty} 2 \sum_{n=1}^{N-1} \left(1 - \frac{n}{N}\right) \frac{\hat{f}(n)}{in} + \hat{F}(0), \\
&\Rightarrow \lim_{N \rightarrow \infty} \sum_{n=1}^{N-1} \left(1 - \frac{n}{N}\right) \frac{\hat{f}(n)}{n} = \frac{i}{2} (F(0) - \hat{F}(0)) \\
&\Rightarrow \sum_{n=1}^{\infty} \frac{\hat{f}(n)}{n} - \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^{N-1} \hat{f}(n) = \frac{i}{2} (F(0) - \hat{F}(0)).
\end{aligned}$$

By Riemann-Lebesgue lemma, we have $\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^{N-1} \hat{f}(n) = 0$, so $\sum_{n=1}^{\infty} \frac{\hat{f}(n)}{n}$ converges. ■

Remark. If we choose (a_n) such that $a_n > 0$, $a_n \rightarrow 0$ as $|n| \rightarrow \infty$, and $\sum_n \frac{a_n}{|n|} = \infty$, then

$$\sum_{n=1}^{\infty} a_n \sin(nx)$$

is not the Fourier series of any function in $L^1([-\pi, \pi])$. In particular,

$$\sum_{n=2}^{\infty} \frac{\sin(nx)}{\log n}$$

is not the Fourier series of any function in $L^1([-\pi, \pi])$.

By Riemann-Lebesgue lemma, the map

$$\mathcal{F} : f \mapsto \left(\hat{f}(n) \right)_{n \in \mathbb{Z}}$$

defines a linear map from $L^1([-\pi, \pi])$ to $c_0(\mathbb{Z})$, where

$$c_0(\mathbb{Z}) = \left\{ (a_n)_{n \in \mathbb{Z}} : a_n \rightarrow 0 \text{ as } |n| \rightarrow \infty \right\}.$$

Since $|\hat{f}|_{\infty} \leq \frac{1}{2\pi} \|f\|_{L^1([-\pi, \pi])}$, $\mathcal{F}$ is a bounded linear operator from $L^1([-\pi, \pi])$ to $c_0(\mathbb{Z}) \subseteq \ell^{\infty}(\mathbb{Z})$. Also, $\mathcal{F}$ is not surjective.

5.1.1. Fourier Series for L^2 -Functions

$L^2([-\pi, \pi])$ can be equipped with the inner product

$$\langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx.$$

The set $B = \{e^{inx} : n \in \mathbb{Z}\}$ is a countable orthonormal set in $L^2([-\pi, \pi])$. If $f \in L^2([-\pi, \pi])$, then $f \in L^1([-\pi, \pi])$, so we can define its Fourier coefficients $\hat{f}(n)$ for every $n \in \mathbb{Z}$. In fact, the Fourier series $\sum_{n=-\infty}^{\infty} \hat{f}(n) e^{inx}$ can be viewed as the orthogonal projection of f onto the closed subspace $\text{span}(B)$. By the Weierstrass approximation theorem, $\text{span}(B)$ is dense in the space of all continuous, 2π -periodic functions under the sup-norm. Since

$$|f - g|_2 \leq \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x) - g(x)|^2 dx \right)^{\frac{1}{2}} \leq |f - g|_{\infty},$$

$\text{span}(B)$ is also dense in $L^2([-\pi, \pi])$, so B is a complete orthonormal set in $L^2([-\pi, \pi])$. For all $f \in L^2([-\pi, \pi])$, the series

$$\sum_{n=-\infty}^{\infty} \hat{f}(n) e^{inx}$$

converges to f in L^2 -norm, and

$$|f|_2^2 = \sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2$$

by Parseval's identity. Thus $\mathcal{F}$ is an isometric isomorphism between $L^2([-\pi, \pi])$ and $\ell^2(\mathbb{Z})$.

5.2. Fourier Transform

DEFINITION 5.3. If $f \in L^1(\mathbb{R})$, then we can define its Fourier transform $\hat{f}$ by

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx.$$

For $f, g \in L^1(\mathbb{R})$, we have

- (i) $|\hat{f}|_{L^{\infty}} \leq |f|_{L^1}$
- (ii) $\widehat{f + ag} = \hat{f} + a\hat{g}$ for every $a \in \mathbb{C}$
- (iii) $\widehat{f_{y(x)}} = \hat{f}(\xi) e^{-2\pi i \xi y}$ for every $y \in \mathbb{R}$, where $f_{y(x)} = f(x - y)$
- (iv) $\widehat{f_{\lambda(x)}} = \hat{f}\left(\frac{\xi}{\lambda}\right)$ for every $\lambda > 0$, where $f_{\lambda(x)} = \lambda f(\lambda x)$
- (v) $\widehat{f * g} = \hat{f} \cdot \hat{g}$,
- (vi) If $f' \in L^1(\mathbb{R})$, then $\hat{f}'(\xi) = 2\pi i \xi \hat{f}(\xi)$.

PROPOSITION 5.4. Let $f \in L^1(\mathbb{R})$, then $\hat{f}$ is uniformly continuous on $\mathbb{R}$. If further $xf(x) \in L^1(\mathbb{R})$, then $\hat{f}$ is differentiable and

$$\frac{d}{d\xi} \hat{f}(\xi) = -\widehat{2\pi i x f(x)}.$$

Proof. Let $\xi, \eta \in \mathbb{R}$. We have

$$\begin{aligned} \hat{f}(\xi + \eta) - \hat{f}(\xi) &= \int f(x) (e^{-2\pi i (\xi + \eta)x} - e^{-2\pi i \xi x}) dx \\ \Rightarrow |\hat{f}(\xi + \eta) - \hat{f}(\xi)| &\leq \int |f(x)| |e^{-2\pi i \eta x} - 1| dx \rightarrow 0 \end{aligned}$$

as $\eta \rightarrow 0$ by dominated convergence theorem, so $\hat{f}$ is uniformly continuous. If $xf(x) \in L^1(\mathbb{R})$, then

$$\begin{aligned} \frac{\hat{f}(\xi + \eta) - \hat{f}(\xi)}{\eta} &= \int f(x) e^{-2\pi i \xi x} e^{-2\pi i \eta x} - \frac{1}{\eta} dx \\ \Rightarrow \lim_{\eta \rightarrow 0} \frac{\hat{f}(\xi + \eta) - \hat{f}(\xi)}{\eta} &= \int f(x) e^{-2\pi i \xi x} \lim_{\eta \rightarrow 0} \frac{e^{-2\pi i \eta x} - 1}{\eta} dx = -\widehat{2\pi i x f(x)} \end{aligned}$$

by dominated convergence theorem, so $\hat{f}$ is differentiable and $\frac{d}{d\xi}\hat{f}(\xi) = -2\pi i x f(\xi)$. ■

THEOREM 5.5 (Riemann-Lebesgue lemma). *If $f \in L^1(\mathbb{R})$, then*

$$\lim_{|\xi| \rightarrow \infty} \hat{f}(\xi) = 0.$$

Proof. If f in C^1 with compact support, then $\hat{f}(\xi) = \frac{1}{2\pi i \xi} \hat{f}'(\xi)$, so $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$. In general, let $\varepsilon > 0$, then there exists $g \in C^1$ with compact support such that $\|f - g\|_{L^1} < \varepsilon$, so

$$\begin{aligned} |\hat{f}(\xi) - \hat{g}(\xi)| &\leq \|f - g\|_{L^1} < \varepsilon, \\ \Rightarrow \limsup_{|\xi| \rightarrow \infty} |\hat{f}(\xi)| &\leq \limsup_{|\xi| \rightarrow \infty} |\hat{f}(\xi) - \hat{g}(\xi)| + \limsup_{|\xi| \rightarrow \infty} |\hat{g}(\xi)| < \varepsilon. \end{aligned}$$

Since ε is arbitrary, we have $\lim_{|\xi| \rightarrow \infty} \hat{f}(\xi) = 0$. ■

The Fejér kernel on $\mathbb{R}$ is defined by

$$F_{\lambda(x)} = \lambda \left(\frac{\sin(\pi \lambda x)}{\pi \lambda x} \right)^2,$$

which comes from the Fourier transform of the function

$$f(x) = \begin{cases} 1 - |x| & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

F_λ is an approximate identity as $\lambda \rightarrow \infty$, so $F_\lambda * f \rightarrow f$ a.e. and in L^1 -norm for every $f \in L^1(\mathbb{R})$.

LEMMA 5.6. *If $f, g \in L^1(\mathbb{R})$ and*

$$g(x) = \int G(\xi) e^{2\pi i \xi x} d\xi$$

for some $G \in L^1(\mathbb{R})$, then

$$(f * g)(x) = \int \hat{f}(\xi) G(\xi) e^{2\pi i \xi x} d\xi.$$

Proof. By Fubini's theorem, we have

$$\begin{aligned} (f * g)(x) &= \int f(y) g(x - y) dy \\ &= \int f(y) \int G(\xi) e^{2\pi i \xi (x - y)} d\xi dy \\ &= \int G(\xi) e^{2\pi i \xi x} \int f(y) e^{-2\pi i \xi y} dy d\xi = \int \hat{f}(\xi) G(\xi) e^{2\pi i \xi x} d\xi. \end{aligned}$$

■

CLAIM 5.7.

$$(f * F_\lambda)(x) = \int_{-\lambda}^{\lambda} \left(1 - \frac{|\xi|}{\lambda} \right) \hat{f}(\xi) e^{2\pi i \xi x} d\xi.$$

Proof. Since

$$F_{\lambda(x)} = \int_{-\lambda}^{\lambda} \left(1 - \frac{|\xi|}{\lambda}\right) e^{2\pi i \xi x} d\xi,$$

the conclusion follows from the previous lemma. ■

CLAIM 5.8 (Uniqueness of Fourier transform). *If $f \in L^1(\mathbb{R})$ and $\hat{f}(\xi) = 0$ for every $\xi \in \mathbb{R}$, then $f = 0$ a.e.*

Proof. Let F_λ be the Fejér kernel on $\mathbb{R}$. We have

$$(f * F_\lambda)(x) = \int_{-\lambda}^{\lambda} \left(1 - \frac{|\xi|}{\lambda}\right) \hat{f}(\xi) e^{2\pi i \xi x} d\xi = 0$$

for every x and λ . Since $F_\lambda * f \rightarrow f$ a.e. as $\lambda \rightarrow \infty$, we have $f = 0$ a.e. ■

CLAIM 5.9 (Fourier inversion formula). *If $f \in L^1(\mathbb{R})$ and $\hat{f} \in L^1(\mathbb{R})$, then*

$$f(x) = \int \hat{f}(\xi) e^{2\pi i \xi x} d\xi$$

for a.e. x .

Proof. Let F_λ be the Fejér kernel on $\mathbb{R}$. We have

$$(f * F_\lambda)(x) = \int_{-\lambda}^{\lambda} \left(1 - \frac{|\xi|}{\lambda}\right) \hat{f}(\xi) e^{2\pi i \xi x} d\xi.$$

Since $F_\lambda * f \rightarrow f$ a.e. as $\lambda \rightarrow \infty$, we have

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi x} d\xi$$

for a.e. x . ■

CLAIM 5.10. *The functions whose Fourier transform are compactly supported are dense in $L^1(\mathbb{R})$.*

Proof. Let $f \in L^1(\mathbb{R})$ and $\varepsilon > 0$. Since $\hat{f}$ is uniformly continuous, there exists $\lambda > 0$ such that $|\hat{f}(\xi)| < \varepsilon$ for every $|\xi| > \lambda$. Define

$$g(x) = \int_{-\lambda}^{\lambda} \hat{f}(\xi) e^{2\pi i \xi x} d\xi,$$

then $\hat{g}$ is compactly supported and

$$\begin{aligned} |f - g|_{L^1} &= \left| \int_{-\infty}^{\infty} (\hat{f}(\xi) - \hat{g}(\xi)) e^{2\pi i \xi x} d\xi \right|_{L^1} \\ &\leq \int_{-\infty}^{\infty} |\hat{f}(\xi) - \hat{g}(\xi)| d\xi = \int_{|\xi| > \lambda} |\hat{f}(\xi)| d\xi < \varepsilon. \end{aligned}$$

■

We may also want to define the Fourier transform for functions in $L^2(\mathbb{R})$.

LEMMA 5.11. *If f is continuous with compact support on $\mathbb{R}$, then*

$$|f|_{L^2}^2 = |\hat{f}|_{L^2}^2.$$

Proof. Write $\tilde{f}(x) = \overline{f(-x)}$, and let $g(x) = f * \tilde{f}$, then

$$g(0) = \int f(x) \overline{f(x)} dx = \|f\|_{L^2}^2,$$

and

$$\hat{g}(\xi) = \int f(x) \overline{f(x-y)} dx = \int f(x) \overline{f(x-y)} e^{-2\pi i \xi y} dy = |\hat{f}(\xi)|^2,$$

so by continuity of f , g is also continuous, thus $g(0) = \lim_{\lambda \rightarrow \infty} (g * F_\lambda)(0)$. Therefore,

$$\begin{aligned} \|f\|_{L^2}^2 &= \lim_{\lambda \rightarrow \infty} (g * F_\lambda)(0) \\ &= \lim_{\lambda \rightarrow \infty} \int |\hat{f}(\xi)|^2 \left(1 - \frac{|\xi|}{\lambda}\right) e^{2\pi i \xi \cdot 0} d\xi = \int |\hat{f}(\xi)|^2 d\xi = \|\hat{f}\|_{L^2}^2. \end{aligned}$$

■

THEOREM 5.12 (Plancherel). *There exists a unique surjective bounded linear operator*

$$\mathcal{F} : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$$

such that $\mathcal{F}f = \hat{f}$ for every $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ and $\|\mathcal{F}f\|_{L^2} = \|f\|_{L^2}$ for every $f \in L^2(\mathbb{R})$.

Proof. Let $S = L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, then S is dense in $L^2(\mathbb{R})$ since $C_c(\mathbb{R}) \subseteq S$ is dense in $L^2(\mathbb{R})$. Hence, any bounded linear operator on $L^2(\mathbb{R})$ is determined by its values on S , so there is at most one operator $\mathcal{F}$ satisfying the conditions. By lemma, we have $\|\hat{f}\|_{L^2} = \|f\|_{L^2}$ for every $f \in S$, so by continuity, $\mathcal{F}$ defines an isometry from $L^2(\mathbb{R})$ to $L^2(\mathbb{R})$.

It remains to show that $\mathcal{F}$ is surjective. Let g be a C^2 -function with compact support. We claim that g equals $\mathcal{F}f$ for some $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Define

$$f(x) = \int g(\xi) e^{2\pi i \xi x} d\xi.$$

Since g has compact support, so does g'' . Define

$$h(x) = \int g''(\xi) e^{2\pi i \xi x} d\xi.$$

Integrating by parts, we have

$$h(x) = -4\pi^2 x^2 \int g(\xi) e^{2\pi i \xi x} d\xi = -4\pi^2 x^2 f(x),$$

so $|f(x)| \leq \frac{|h(x)|}{4\pi^2 x^2}$ for every $x \neq 0$. Since h is bounded and $\frac{1}{x^2}$ is integrable on $\mathbb{R}$, also f is bounded near 0, we have $f \in L^1(\mathbb{R})$. Therefore, the range of $\mathcal{F}$ is dense in $L^2(\mathbb{R})$ since $C_c^2(\mathbb{R})$ is dense in $L^2(\mathbb{R})$. Let $(\mathcal{F}f_n)$ be a Cauchy sequence in $\mathcal{F}(L^2(\mathbb{R}))$. Since $\mathcal{F}$ is an isometry, (f_n) is Cauchy in $L^2(\mathbb{R})$, so there exists $f \in L^2(\mathbb{R})$ such that $f_n \rightarrow f$ in L^2 -norm. By continuity of $\mathcal{F}$, we have $\mathcal{F}f_n \rightarrow \mathcal{F}f$ in L^2 -norm, so $\mathcal{F}(L^2(\mathbb{R}))$ is closed in $L^2(\mathbb{R})$. ■

Remark.

- (i) If $f \in L^2(\mathbb{R})$, we will simply write $\hat{f} = \mathcal{F}f$. Given $f \in L^2(\mathbb{R})$, we can define $\hat{f}$ to be the L^2 -limit of $\hat{f}_n$, where (f_n) is any sequence in $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ such that $f_n \rightarrow f$ in L^2 -norm. A natural choice is $f_n := f \cdot \chi_{[-n,n]}$. For this choice of (f_n) , we see that the sequence

$$\hat{f}_{n(\xi)} = \int_{-n}^n f(x) e^{-2\pi i \xi x} dx$$

converges to a function $\hat{f}$ in L^2 .

(i) Fourier transform on $L^2(\mathbb{R})$ is invertible. We can obtain the inverse map by

$$f(x) = \lim_{n \rightarrow \infty} \int_{-n}^n \hat{f}(\xi) e^{2\pi i \xi x} d\xi$$

for every $f \in L^2(\mathbb{R})$, where the limit is taken in L^2 -norm.

(i) By polarization identity, we have

$$\langle f, g \rangle = \langle \hat{f}, \hat{g} \rangle$$

for every $f, g \in L^2(\mathbb{R})$, so $\mathcal{F}$ preserves the inner product.

Abstract Measure

CHAPTER 6

6.1. Abstract Measure Spaces

DEFINITION 6.1. Let X be a nonempty set.

- (i) A σ -algebra $\mathcal{M}$ is a collection of subsets of X that contains $\emptyset$ and is closed under complements and countable unions. Elements of $\mathcal{M}$ are called measurable sets.
- (ii) A measure $\mu : \mathcal{M} \rightarrow [0, \infty]$ is a function such that $\mu(\emptyset) = 0$ and if $E_1, E_2, \dots$ is a countable sequence of pairwise disjoint sets in $\mathcal{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n).$$

- (iii) A measure space is a triple $(X, \mathcal{M}, \mu)$, where X is a nonempty set, $\mathcal{M}$ is a σ -algebra on X , and μ is a measure on $\mathcal{M}$.
- (iv) A measure space $(X, \mathcal{M}, \mu)$ is σ -finite if X can be written as a countable union of elements of $\mathcal{M}$ with finite measure.

Example 6.1.

- Lebesgue measure.
- Let X be a countable set, write $X = \{x_n : n \in \mathbb{N}\}$, and let $\mathcal{M} = 2^X$. Let (μ_n) be a sequence in $[0, \infty]$. For all $E \subseteq X$, define

$$\mu(E) = \sum_{x_n \in E} \mu_n,$$

then μ is a measure on $\mathcal{M}$. In particular, if $\mu_n = 1$ for every n , then μ is the counting measure on X .

- Let $X = \mathbb{R}^d$ and $\mathcal{M}$ be the collection of all Lebesgue measurable sets in $\mathbb{R}^d$. Define

$$\mu(E) = \int_E f(x) dx$$

for every $E \in \mathcal{M}$, where f is a nonnegative measurable function on $\mathbb{R}^d$. If $f \equiv 1$, then μ is the Lebesgue measure on $\mathbb{R}^d$.

6.1.1. Outer Measures

Recall that the Lebesgue measure on $\mathbb{R}$ is first defined as an outer measure, and then we restrict it to the σ -algebra of Lebesgue measurable sets.

DEFINITION 6.2. Let X be a nonempty set. An outer measure μ_* on X is a function from the power set of X to $[0, \infty]$ that satisfies

- (i) $\mu_*(\emptyset) = 0$;
- (ii) If $E_1 \subseteq E_2$, then $\mu_*(E_1) \leq \mu_*(E_2)$;
- (iii) If $E_1, E_2, \dots$ is a countable sequence of subsets of X , then

$$\mu_* \left(\bigcup_{n=1}^{\infty} E_n \right) \leq \sum_{n=1}^{\infty} \mu_*(E_n).$$

DEFINITION 6.3. Let μ_* be an outer measure on X . A set $E \subseteq X$ is Carathéodory measurable, or simply measurable, if

$$\mu_*(A) = \mu_*(A \cap E) + \mu_*(A \cap E^c)$$

for every $A \subseteq X$. Define $\mathcal{M}$ to be the collection of all measurable sets.

To prove that a set E is measurable, it suffices to show that $\mu_*(A) \geq \mu_*(A \cap E) + \mu_*(A \cap E^c)$ for every $A \subseteq X$, since the opposite inequality always holds by subadditivity of μ_* . In particular, every set E with $\mu_*(E) = 0$ is measurable, since $\mu_*(A) \geq \mu_*(A \cap E) + \mu_*(A \cap E^c)$ for every $A \subseteq X$.

LEMMA 6.4. If $E_1, E_2, \dots \in \mathcal{M}$ are pairwise disjoint, then for all $A \subseteq X$,

$$\mu_* \left(A \cap \bigcup_{n=1}^{\infty} E_n \right) = \sum_{n=1}^{\infty} \mu_*(A \cap E_n).$$

Proof. When there is only one set E_1 , there's nothing to prove. For two disjoint sets E_1 and E_2 , we have

$$\begin{aligned} \mu_*(A \cap (E_1 \cup E_2)) &= \mu_*(A \cap (E_1 \cup E_2) \cap E_2) + \mu_*(A \cap (E_1 \cup E_2) \cap E_2^c) \\ &= \mu_*(A \cap E_2) + \mu_*(A \cap E_1). \end{aligned}$$

By induction, the finite case follows. For the infinite case, we have

$$\begin{aligned} \mu_* \left(A \cap \bigcup_{n=1}^{\infty} E_n \right) &\leq \sum_{n=1}^{\infty} \mu_*(A \cap E_n), \\ \Rightarrow \mu_* \left(A \cap \bigcup_{n=1}^{\infty} E_n \right) &\geq \mu_* \left(A \cap \bigcup_{n=1}^N E_n \right) = \sum_{n=1}^N \mu_*(A \cap E_n) \end{aligned}$$

for every N , so $\mu_* \left(A \cap \bigcup_{n=1}^{\infty} E_n \right) \geq \sum_{n=1}^{\infty} \mu_*(A \cap E_n)$. ■

THEOREM 6.5. Given an outer measure μ_* on X , then $\mathcal{M}$ is a σ -algebra on X and the restriction of μ_* to $\mathcal{M}$ is a measure.

Proof. For all $A \subseteq X$, we have

$$\mu_*(A) = \mu_*(X \cap A) + \mu_*(\emptyset \cap A),$$

so $\emptyset \in \mathcal{M}$. If $E \in \mathcal{M}$, then since the definition of measurability is symmetric in E and E^c , we have $E^c \in \mathcal{M}$.

If $E, F \in \mathcal{M}$, then for all $A \subseteq X$, we have

$$\begin{aligned} \mu_*(A) &= \mu_*(A \cap E) + \mu_*(A \cap E^c) \\ &= \mu_*(A \cap E \cap F) + \mu_*(A \cap E \cap F^c) + \mu_*(A \cap E^c \cap F) + \mu_*(A \cap E^c \cap F^c) \\ &\geq \mu_*(A \cap (E \cup F)) + \mu_*(A \cap (E \cup F)^c), \end{aligned}$$

so $E \cup F \in \mathcal{M}$. By induction, we have $\bigcup_{n=1}^N E_n \in \mathcal{M}$ for every N . If $E_1, E_2, \dots$ are pairwise disjoint, then for each N , we have $\bigcup_{n=1}^N E_n \in \mathcal{M}$, so for all $A \subseteq X$, we have

$$\begin{aligned}\mu_*(A) &= \mu_*\left(A \cap \bigcup_{n=1}^N E_n\right) + \mu_*\left(A \cap \left(\bigcup_{n=1}^N E_n\right)^c\right) \\ &\geq \sum_{n=1}^N \mu_*(A \cap E_n) + \mu_*\left(A \cap \left(\bigcup_{n=1}^{\infty} E_n\right)^c\right).\end{aligned}$$

When $N \rightarrow \infty$, we have

$$\begin{aligned}\mu_*(A) &\geq \sum_{n=1}^{\infty} \mu_*(A \cap E_n) + \mu_*\left(A \cap \left(\bigcup_{n=1}^{\infty} E_n\right)^c\right) \\ &= \mu_*\left(A \cap \bigcup_{n=1}^{\infty} E_n\right) + \mu_*\left(A \cap \left(\bigcup_{n=1}^{\infty} E_n\right)^c\right),\end{aligned}$$

so $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$. If $E_1, E_2, \dots$ are not pairwise disjoint, then we can write

$$\begin{aligned}F_1 &= E_1, \\ F_2 &= E_2 \cap E_1^c, \\ F_3 &= E_3 \cap E_1^c \cap E_2^c, \\ &\vdots \\ F_n &= E_n \cap \bigcap_{k=1}^{n-1} E_k^c, \\ &\vdots\end{aligned}$$

so $F_1, F_2, \dots$ are pairwise disjoint and $\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} F_n$, thus $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$. ■

PROPOSITION 6.6. *In $\mathbb{R}^d$ with the Lebesgue outer measure, Lebesgue measurability is equivalent to Carathéodory measurability.*

PROPOSITION 6.7. *Suppose that $E_1, E_2, \dots$ are measurable sets.*

- (i) *If $E_n \uparrow E$, then $m(E_n) \rightarrow m(E)$.*
- (ii) *If $E_n \downarrow E$ and $m(E_1) < \infty$, then $m(E_n) \rightarrow m(E)$.*

Proof. Same as the proof of continuity of measure for Lebesgue measurable sets (See Chapter 1 Proposition 1.20). ■

6.1.2. Metric Outer Measures

Suppose that X is a metric space. The Borel σ -algebra $\mathcal{B}_X$ on X is the smallest σ -algebra on X that contains all open sets. We want to construct a measure defined on $\mathcal{B}_X$.

For $A, B \subseteq X$, define

$$d(A, B) = \inf\{d(x, y) : x \in A, y \in B\}.$$

DEFINITION 6.8. *An outer measure μ_* on X is a metric outer measure if*

$$\mu_*(A \cup B) = \mu_*(A) + \mu_*(B)$$

for every $A, B \subseteq X$ with $d(A, B) > 0$.

THEOREM 6.9. *If μ_* is a metric outer measure on X , then every Borel set is μ_* -measurable. In particular, the restriction of μ_* to $\mathcal{B}_X$ is a measure.*

Proof. It suffices to show that every closed set is μ_* -measurable. Let F be a closed set and $A \subseteq X$ with $\mu_*(A) < \infty$. For each n , define

$$A_n = \left\{ x \in A : d(x, F) > \frac{1}{n} \right\}.$$

Then $A_n \subseteq A_{n+1}$. Since F is closed, we have $F^c \cap A = \bigcup_{n=1}^{\infty} A_n$. Also, $d(F, A_n) > 0$ for every n , so

$$\begin{aligned} \mu_*(A) &= \mu_*((F \cap A) \cup (F^c \cap A)) \\ &\geq \mu_*((F \cap A) \cup A_n) = \mu_*(F \cap A) + \mu_*(A_n) \end{aligned}$$

for every n . Let $B_n = A_{n+1} \setminus A_n$. Note that

$$d(B_{n+1}, A_n) \geq \frac{1}{n} - \frac{1}{n+1} + 1 > 0,$$

so

$$\begin{aligned} \mu_*(A_{2n+1}) &\geq \mu_*(B_2 \cup B_4 \cup \cdots \cup B_{2n}) = \sum_{k=1}^n \mu_*(B_{2k}), \\ \mu_*(A_{2n}) &\leq \mu_*(B_1 \cup B_3 \cup \cdots \cup B_{2n-1}) = \sum_{k=1}^n \mu_*(B_{2k-1}). \end{aligned}$$

Since $\mu_*(A) < \infty$, both series $\sum_{k=1}^{\infty} \mu_*(B_{2k})$ and $\sum_{k=1}^{\infty} \mu_*(B_{2k-1})$ are convergent. By countable subadditivity, we have

$$\mu_*(A_n) \leq \mu_*(F^c \cap A) = \mu_*\left(\bigcup_{k=1}^{\infty} A_k\right) \leq \mu_*(A_n) + \sum_{k=n+1}^{\infty} \mu_*(B_{2k-1}),$$

so $\lim_{n \rightarrow \infty} \mu_*(A_n) = \mu_*(F^c \cap A)$. Therefore, $\mu_*(A) \geq \mu_*(F \cap A) + \mu_*(F^c \cap A)$, so F is μ_* -measurable.

If $\mu_*(A) = \infty$, then $\mu_*(A) \geq \mu_*(F \cap A) + \mu_*(F^c \cap A)$ holds trivially, so F is μ_* -measurable. ■

DEFINITION 6.10. A measure μ defined on the Borel sets of a metric space X is called a Borel measure.

If a Borel measure is finite on all balls (of finite radius), then it has nice regularity properties.

PROPOSITION 6.11. Suppose that the Borel measure μ is finite on all balls in X of finite radius. Then for every Borel set E and $\varepsilon > 0$, there exists an open set U and a closed set F such that $F \subseteq E \subseteq U$ and

$$\mu(G \setminus E) < \varepsilon, \quad \mu(E \setminus F) < \varepsilon.$$

Proof. We will first show that inner regularity holds for F_σ -sets. Let $F^* = \bigcup_{n=1}^{\infty} F_n$ be an F_σ -set, where F_n are closed sets. We may assume that $F_n \subseteq F_{n+1}$ for every n . Fix $x_0 \in X$ and let $B_n = \{x \in X : d(x, x_0) < n\}$ for every n , then

$$X = \bigcup_{n=1}^{\infty} B_n, \quad F^* = \bigcup_{n=1}^{\infty} (F^* \cap (\overline{B_n} \setminus B_{n-1})).$$

Note that $F^* \cap (\overline{B_n} \setminus B_{n-1})$ is the increasing limit of $F_k \cap (\overline{B_n} \setminus B_{n-1})$ as $k \rightarrow \infty$, hence

$$\mu(F^* \cap (\overline{B_n} \setminus B_{n-1})) = \lim_{k \rightarrow \infty} \mu(F_k \cap (\overline{B_n} \setminus B_{n-1})).$$

Since $\mu(\overline{B_n}) < \infty$, we can find an integer N_n such that

$$\mu((F^* \setminus F_{N_n}) \cap (\overline{B_n} \setminus B_{n-1})) < \frac{\varepsilon}{2^n}.$$

Let $F = \bigcup_{n=1}^{\infty} F_{N_n} \cap (\overline{B_n} \setminus B_{n-1})$, then $\mu(F^* \setminus F) < \varepsilon$. Since $F \cap \overline{B_n}$ is a finite union of closed sets, it is closed, hence F itself is closed.

Let $\mathcal{C}$ be the collection of sets that satisfy both outer and inner regularity (by closed sets). Clearly $\mathcal{C}$ is closed under complements. Suppose that $E_1, E_2, \dots$ are sets in $\mathcal{C}$. We want to prove that $E = \bigcup_{n=1}^{\infty} E_n \in \mathcal{C}$. We can find open sets U_n and closed sets F_n such that $F_n \subseteq E_n \subseteq U_n$ and

$$\mu(U_n \setminus E_n) < \frac{\varepsilon}{2^n}, \quad \mu(E_n \setminus F_n) < \frac{\varepsilon}{2^n}$$

for every n . Let $U = \bigcup_{n=1}^{\infty} U_n$, then U is open and $\mu(U \setminus E) \leq \sum_{n=1}^{\infty} \mu(U_n \setminus E_n) < \varepsilon$. Let $F^* = \bigcup_{n=1}^{\infty} F_n$, then F^* is a F_σ -set and $\mu(E \setminus F^*) < \varepsilon$. By the first step, we can find a closed set F such that $F \subseteq F^*$ and $\mu(F^* \setminus F) < \varepsilon$, hence $\mu(E \setminus F) < 2\varepsilon$. Therefore, $E \in \mathcal{C}$.

It remains to show that every open set is in $\mathcal{C}$. Let U be an open set, then outer regularity is trivial. For inner regularity, write

$$F_k = \left\{ x \in \overline{B_k} : d(x, U^c) \geq \frac{1}{k} \right\}.$$

Then F_k is closed and $U = \bigcup_{k=1}^{\infty} F_k$, so U is a F_σ -set and we can find a closed set F such that $F \subseteq U$ and $\mu(U \setminus F) < \varepsilon$. Therefore, $U \in \mathcal{C}$. ■

6.1.3. Premeasures

DEFINITION 6.12. Let X be a set. An algebra $\mathcal{A}$ on X is a collection of subsets of X that contains $\emptyset$ and is closed under complements and finite unions.

DEFINITION 6.13. Let $\mathcal{A}$ be an algebra on X . A premeasure $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ is a function such that

- (i) $\mu_0(\emptyset) = 0$, and
- (ii) if $E_1, E_2, \dots$ is a countable collection of pairwise disjoint sets in $\mathcal{A}$ such that $\bigcup_{n=1}^{\infty} E_n \in \mathcal{A}$, then

$$\mu_0\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu_0(E_n).$$

In particular, μ_0 is finitely additive on $\mathcal{A}$.

LEMMA 6.14. Let μ_0 be a premeasure on an algebra $\mathcal{A}$. For any $E \subseteq X$, define

$$\mu_*(E) = \inf \left\{ \sum_{n=1}^{\infty} \mu_0(E_n) : E_n \in \mathcal{A}, E \subseteq \bigcup_{n=1}^{\infty} E_n \right\}.$$

Then μ_* is an outer measure on X that satisfies

- (i) $\mu_*(E) = \mu_0(E)$ for every $E \in \mathcal{A}$;
- (ii) all sets in $\mathcal{A}$ are μ_* -measurable.

Proof. We will first show that μ_* is an outer measure. Since $\mu_0(\emptyset) = 0$, we have $\mu_*(\emptyset) = 0$. If $E_1 \subseteq E_2$, then every cover of E_2 is also a cover of E_1 , so $\mu_*(E_1) \leq \mu_*(E_2)$. If $E_1, E_2, \dots$ is a countable sequence of subsets of X , then for each n , we can find a sequence $(E_{n,k})_{k=1}^\infty$ in $\mathcal{A}$ such that $E_n \subseteq \bigcup_{k=1}^\infty E_{n,k}$ and

$$\sum_{k=1}^\infty \mu_0(E_{n,k}) < \mu_*(E_n) + \frac{\varepsilon}{2^n}.$$

Then $\bigcup_{n=1}^\infty E_n \subseteq \bigcup_{n=1}^\infty \bigcup_{k=1}^\infty E_{n,k}$ and

$$\sum_{n=1}^\infty \sum_{k=1}^\infty \mu_0(E_{n,k}) < \sum_{n=1}^\infty \left(\mu_*(E_n) + \frac{\varepsilon}{2^n} \right) = \sum_{n=1}^\infty \mu_*(E_n) + \varepsilon,$$

so $\mu_*\left(\bigcup_{n=1}^\infty E_n\right) \leq \sum_{n=1}^\infty \mu_*(E_n)$.

We will then show that $\mu_*(E) = \mu_0(E)$ for every $E \in \mathcal{A}$. Since E itself is a cover of E , we have $\mu_*(E) \leq \mu_0(E)$. Suppose that $E \subseteq \bigcup_{n=1}^\infty E_n$ for some sequence (E_n) in $\mathcal{A}$. Let

$$F_j = E \cap \left(E_j \setminus \bigcup_{k=1}^{j-1} E_k \right),$$

then $F_1, F_2, \dots$ are pairwise disjoint sets in $\mathcal{A}$. Also, $F_j \subseteq E_j$ and $\bigcup_{j=1}^\infty F_j = E$, so by definition of premeasure, we have

$$\mu_0(E) = \mu_0\left(\bigcup_{j=1}^\infty F_j\right) = \sum_{j=1}^\infty \mu_0(F_j) \leq \sum_{j=1}^\infty \mu_0(E_j).$$

Taking infimum over all covers of E , we have $\mu_0(E) \leq \mu_*(E)$, hence $\mu_*(E) = \mu_0(E)$.

Finally, we will show that all sets in $\mathcal{A}$ are μ_* -measurable. Let $E \in \mathcal{A}$ and $A \subseteq X$. Let $\varepsilon > 0$. There exists $A_1, A_2, \dots \in \mathcal{A}$ such that $A \subseteq \bigcup_{n=1}^\infty A_n$ and

$$\sum_{n=1}^\infty \mu_0(A_n) < \mu_*(A) + \varepsilon.$$

We have

$$\begin{aligned} E \cap A_j &\in \mathcal{A}, \quad \bigcup_{n=1}^\infty (E \cap A_n) \supseteq E \cap A, \\ E^c \cap A_j &\in \mathcal{A}, \quad \bigcup_{n=1}^\infty (E^c \cap A_n) \supseteq E^c \cap A, \\ \Rightarrow \sum_{n=1}^\infty \mu_0(E \cap A_n) &\geq \mu_*(E \cap A), \quad \sum_{n=1}^\infty \mu_0(E^c \cap A_n) \geq \mu_*(E^c \cap A), \\ \Rightarrow \mu_*(E \cap A) + \mu_*(E^c \cap A) &\leq \sum_{n=1}^\infty \mu_0(A_n) < \mu_*(A) + \varepsilon. \end{aligned}$$

Since ε is arbitrary, we have $\mu_*(E \cap A) + \mu_*(E^c \cap A) \leq \mu_*(A)$, so E is μ_* -measurable. ■

COROLLARY 6.15. *Suppose that $\mathcal{A}$ is an algebra on X and μ_0 is a premeasure on $\mathcal{A}$. Let $\mathcal{M}$ be the σ -algebra generated by $\mathcal{A}$. Then there exists a measure μ on $\mathcal{M}$ that extends μ_0 . Moreover, if μ is σ -finite, then such μ is unique.*

Proof. By the previous lemma, we can find an outer measure μ_* on X such that $\mu_*(E) = \mu_0(E)$ for every $E \in \mathcal{A}$ and all sets in $\mathcal{A}$ are μ_* -measurable.

Suppose that ν is another measure on $\mathcal{M}$ that coincides with μ_0 on $\mathcal{A}$. Suppose that $F \in \mathcal{M}$ with $\mu(F) < \infty$. Let $E_1, E_2, \dots$ be a sequence in $\mathcal{A}$ be a cover for F , then

$$\nu(F) \leq \sum_{n=1}^{\infty} \nu(E_n) = \sum_{n=1}^{\infty} \mu_0(E_n).$$

Taking infimum over all covers of F , we have $\nu(F) \leq \mu(F)$. Suppose $E = \bigcup_{n=1}^{\infty} E_n$ with $E_n \in \mathcal{A}$. Define $F_n = E_n \setminus \bigcup_{k=1}^{n-1} E_k$, then $F_1, F_2, \dots$ are pairwise disjoint sets in $\mathcal{A}$ and $\bigcup_{n=1}^{\infty} F_n = E$, so

$$\nu(E) = \nu\left(\bigcup_{n=1}^{\infty} F_n\right) = \lim_{N \rightarrow \infty} \nu\left(\bigcup_{n=1}^N F_n\right) = \lim_{N \rightarrow \infty} \mu\left(\bigcup_{n=1}^N F_n\right) = \mu(E).$$

We can choose (E_j) in $\mathcal{A}$ such that $F \subseteq \bigcup_{j=1}^{\infty} E_j =: E$ and $\mu(E) < \mu(F) + \varepsilon$. Since F has finite measure, we have $\mu(E \setminus F) < \varepsilon$, so

$$\begin{aligned} \mu(F) &\leq \mu(E) = \nu(E) = \nu(F) + \nu(E \setminus F) \\ &\leq \nu(F) + \mu(E \setminus F) < \nu(F) + \varepsilon. \end{aligned}$$

Since ε is arbitrary, we have $\mu(F) \leq \nu(F)$, hence $\mu(F) = \nu(F)$.

Since we can write $X = \bigcup_{n=1}^{\infty} E_n$ with $E_n \in \mathcal{A}$ and $\mu(E_n) < \infty$ for every n , for general $F \in \mathcal{M}$, we have

$$\mu(F) = \sum_{n=1}^{\infty} \mu(F \cap E_n) = \sum_{n=1}^{\infty} \nu(F \cap E_n) = \nu(F).$$

■

6.2. Integration

Let $(X, \mathcal{M}, \mu)$ be a measure space. We will assume that μ is σ -finite.

DEFINITION 6.16. A function $f : X \rightarrow [-\infty, \infty]$ is measurable if $f^{-1}([-\infty, a)) \in \mathcal{M}$ for every $a \in \mathbb{R}$.

DEFINITION 6.17. A simple function on X is a function of the form

$$f = \sum_{k=1}^N a_k \chi_{E_k},$$

where $a_1, \dots, a_N$ are real numbers and $E_1, \dots, E_N$ are measurable sets of finite measure.

As before, if f is measurable, we can find a sequence of simple functions that converges to f pointwise. We can then define the Lebesgue integral first for simple functions, and then for general measurable functions by taking limits. A measurable function f is integrable if

$$\int_X |f(x)| d\mu(x) < \infty.$$

Basic properties such as linearity, monotonicity still holds, as well as limit theorems such as monotone convergence theorem, Fatou's lemma, and dominated convergence theorem.

For $p \in [1, \infty]$, we can still define the L^p -norm of a measurable function f by

$$|f|_p = \left(\int_X |f(x)|^p d\mu(x) \right)^{\frac{1}{p}}$$

and the associated L^p -space by

$$L^p(X, \mu) = \{f : X \rightarrow \mathbb{C} : f \text{ is measurable and } |f|_p < \infty\}.$$

As usual, $L^p(X, \mu)$ is a Banach space under the L^p -norm, and $L^2(X, \mu)$ is a Hilbert space under the inner product

$$\langle f, g \rangle = \int_X f(x) \overline{g(x)} d\mu(x).$$

6.2.1. Product Measures and Fubini's Theorem

Suppose that $(X_1, \mathcal{M}_1, \mu_1)$ and $(X_2, \mathcal{M}_2, \mu_2)$ are two measure spaces. We want to construct a measure on the product space $X_1 \times X_2$.

Throughout this section, we'll assume that the measure spaces are complete and σ -finite.

DEFINITION 6.18. *A measurable rectangle in $X_1 \times X_2$ is a set of the form $A \times B$ with $A \in \mathcal{M}_1$ and $B \in \mathcal{M}_2$. Let $\mathcal{A}$ be the collection of all finite unions of disjoint measurable rectangles in $X_1 \times X_2$. Then $\mathcal{A}$ is an algebra on $X_1 \times X_2$.*

For a measurable rectangle $A \times B$, define

$$\mu_0(A \times B) = \mu_1(A) \cdot \mu_2(B).$$

We claim that μ_0 is a premeasure. Assume that $A \times B = \bigcup_{j=1}^{\infty} A_j \times B_j$, where the rectangles $A_j \times B_j$ are pairwise disjoint. If $x_1 \in A$, then for each $x_2 \in B$, the pair (x_1, x_2) belongs to exactly one $A_j \times B_j$, so

$$\sum_{j=1}^{\infty} \chi_{A_j}(x_1) \cdot \chi_{B_j}(x_2) = \chi_{A(x_1)} \cdot \chi_{B(x_2)}.$$

Integrating both sides with respect to μ_2 gives

$$\sum_{j=1}^{\infty} \chi_{A_j}(x_1) \cdot \mu_2(B_j) = \chi_{A(x_1)} \cdot \mu_2(B).$$

Integrating both sides with respect to μ_1 gives

$$\sum_{j=1}^{\infty} \mu_0(A_j \times B_j) = \sum_{j=1}^{\infty} \mu_1(A_j) \cdot \mu_2(B_j) = \mu_1(A) \cdot \mu_2(B) = \mu_0(A \times B).$$

Therefore, μ_0 is a premeasure. We can then extend μ_0 to a measure μ on the σ -algebra generated by $\mathcal{A}$. Thus we have defined the product measure space $(X_1 \times X_2, \mathcal{M}, \mu_1 \times \mu_2)$.

Given $E \in \mathcal{M}$, define the slices

$$E_{x_1} = \{x_2 \in X_2 : (x_1, x_2) \in E\}, \quad E^{x_2} = \{x_1 \in X_1 : (x_1, x_2) \in E\}.$$

Write $\mathcal{A}_\sigma$ to be the collection of sets that are countable unions of elements of $\mathcal{A}$, and $\mathcal{A}_{\sigma\delta}$ to be the collection of sets that are countable intersections of elements of $\mathcal{A}_\sigma$.

LEMMA 6.19. *If $E \in \mathcal{A}_{\sigma\delta}$, then E^{x_2} is $\mathcal{M}_1$ -measurable for every $x_2 \in X_2$, and $x_2 \mapsto \mu_1(E^{x_2})$ is an $\mathcal{M}_2$ -measurable function on X_2 . Moreover,*

$$\int_{X_2} \mu_1(E^{x_2}) d\mu_2(x_2) = (\mu_1 \times \mu_2)(E).$$

Proof. If $E = A \times B$ is a measurable rectangle, then

$$E^{x_2} = \begin{cases} A & \text{if } x_2 \in B \\ \emptyset & \text{otherwise} \end{cases}$$

Also, $x_2 \mapsto \mu_1(E^{x_2}) = \mu_1(A) \cdot \chi_{B(x_2)}$ is $\mathcal{M}_2$ -measurable and

$$\int_{X_2} \mu_1(E^{x_2}) d\mu_2(x_2) = \mu_1(A) \cdot \mu_2(B) = (\mu_1 \times \mu_2)(E).$$

Next, we suppose that $E \in \mathcal{A}_\sigma$. We can write E as a countable disjoint union of measurable rectangles $\bigcup_{j=1}^\infty E_j$, where $E_j \in \mathcal{A}$. Note that

$$E^{x_2} = \bigcup_{j=1}^\infty E_j^{x_2},$$

and since E_j are disjoint, by monotone convergence theorem, we have

$$\int_{X_2} \mu_1(E^{x_2}) d\mu_2(x_2) = \sum_{j=1}^\infty \int_{X_2} \mu_1(E_j^{x_2}) d\mu_2(x_2) = \sum_{j=1}^\infty (\mu_1 \times \mu_2)(E_j) = (\mu_1 \times \mu_2)(E).$$

Finally, we deal with general $E \in \mathcal{A}_{\sigma\delta}$. Assume $(\mu_1 \times \mu_2)(E) < \infty$. By definition of $\mathcal{A}_{\sigma\delta}$, there exists a sequence (E_j) in $\mathcal{A}_\sigma$ such that $E = \bigcap_{j=1}^\infty E_j$ and $E_j \supseteq E_{j+1}$ for every j . Further assume that $(\mu_1 \times \mu_2)(E_1) < \infty$ and each slice also has finite measure $\mu_1(E_j^{x_2}) < \infty$. Set

$$f_{j(x_2)} = \mu_1(E_j^{x_2}), \quad f(x_2) = \mu_1(E^{x_2}).$$

Since $E^{x_2} = \bigcap_{j=1}^\infty E_j^{x_2}$ is the countable intersection of $\mathcal{M}_1$ -measurable sets, it is $\mathcal{M}_1$ -measurable, so f is well-defined, so for every x_2 ,

$$f(x_2) = \mu_1(E^{x_2}) = \mu_1\left(\bigcap_{j=1}^\infty E_j^{x_2}\right) = \lim_{j \rightarrow \infty} \mu_1(E_j^{x_2})$$

is $\mathcal{M}_2$ -measurable. Since (f_j) is decreasing and converges to f , by monotone convergence theorem, we have

$$\begin{aligned} \int_{X_2} \mu_1(E^{x_2}) d\mu_2(x_2) &= \lim_{j \rightarrow \infty} \int_{X_2} \mu_1(E_j^{x_2}) d\mu_2(x_2) \\ &= \lim_{j \rightarrow \infty} (\mu_1 \times \mu_2)(E_j) = (\mu_1 \times \mu_2)(E). \end{aligned}$$

For general E , by σ -finiteness, there exists sequences of sets $F_1 \subseteq F_2 \subseteq \dots \subseteq X_1$ and $G_1 \subseteq G_2 \subseteq \dots \subseteq X_2$ such that

$$X_1 = \bigcup_{j=1}^\infty F_j, \quad X_2 = \bigcup_{j=1}^\infty G_j, \quad \mu_1(F_j) < \infty, \quad \mu_2(G_j) < \infty$$

for every j . Replace E, E_k by $E \cap (F_j \times G_j), E_k \cap (F_j \times G_j)$ and take limit as $j \rightarrow \infty$, we have

$$\int_{X_2} \mu_1(E^{x_2}) d\mu_2(x_2) = (\mu_1 \times \mu_2)(E).$$

■

PROPOSITION 6.20. *For all $E \in \mathcal{M}$, exists $F \in \mathcal{A}_{\sigma\delta}$ such that $E \subseteq F$ and*

$$(\mu_1 \times \mu_2)(E) = (\mu_1 \times \mu_2)(F).$$

PROPOSITION 6.21. *Lemma 6.19 holds for $E \in \mathcal{M}$, except that we now only have E^{x_2} is $\mathcal{M}_1$ -measurable for μ_2 -a.e. x_2 and $\mu_1(E^{x_2})$ is defined for only μ_2 -a.e. x_2 .*

Proof. Follows from the previous proposition and Lemma 6.19. ■

THEOREM 6.22. *Suppose that $f \in L^1(X_1 \times X_2, \mu_1 \times \mu_2)$. Then*

- (i) *For μ_2 -a.e. x_2 , the slice $f^{x_2}(x_1) = f(x_1, x_2)$ is in $L^1(X_1, \mu_1)$;*
- (ii) *$x_2 \mapsto \int_{X_1} f(x_1, x_2) d\mu_1(x_1)$ is integrable on X_2 ;*
- (iii)
$$\int_{X_2} \left(\int_{X_1} f(x_1, x_2) d\mu_1(x_1) \right) d\mu_2(x_2) = \int_{X_1 \times X_2} f(x_1, x_2) d(\mu_1 \times \mu_2)(x_1, x_2).$$

Proof. By the previous proposition the results holds for $f = \chi_E$ with $E \in \mathcal{M}$. By linearity, it also holds for simple functions. For nonnegative measurable functions, we can find an increasing sequence of simple functions that converges to f pointwise, so the result follows from monotone convergence theorem. For general integrable functions, we can write $f = f^+ - f^-$, where $f^+, f^- \geq 0$ are integrable, so the result follows from the nonnegative case. ■

6.2.2. Stieltjes Integrals

Stieltjes integrals are a generalization of Riemann integrals, where the increments dx are replaced by increments $dF(x)$ for some increasing function F .

Since F is increasing, it has at most countably many discontinuities. At a discontinuity x_0 , the left and right limits of F exist, and we denote them by $F(x_0^-)$ and $F(x_0^+)$ respectively. We modify F at these discontinuities by defining $F(x_0) = F(x_0^+)$, so that F is right-continuous.

THEOREM 6.23. *Let F be an increasing right-continuous function on $\mathbb{R}$. Then there exists a unique Borel measure μ on $\mathbb{R}$ such that*

$$\mu((a, b]) = F(b) - F(a)$$

for every $a < b$. Conversely, if μ is a Borel measure on $\mathbb{R}$ that is finite on all bounded intervals, then F defined by

$$F(x) = \begin{cases} \mu((0, x]) & x \geq 0 \\ 0 & x = 0 \\ -\mu((x, 0]) & x < 0 \end{cases}$$

is an increasing right-continuous function on $\mathbb{R}$.

Proof. The proof is similar to the construction of Lebesgue measure.

Define

$$\mu_*(E) = \inf \left\{ \sum_{j=1}^{\infty} (F(b_j) - F(a_j)) : E \subseteq \bigcup_{j=1}^{\infty} (a_j, b_j] \right\}.$$

Then μ_* is an outer measure on $\mathbb{R}$. We will show that $\mu_*((a, b]) = F(b) - F(a)$ for every $a < b$. Since $(a, b]$ itself is a cover of $(a, b]$, we have $\mu_*((a, b]) \leq F(b) - F(a)$. Let $\varepsilon > 0$ and suppose that $\bigcup_{j=1}^{\infty} (a_j, b_j]$ is a cover of $(a, b]$, then it also covers $[a', b]$ for

all $a' \in (a, b)$. By right continuity of F , $\exists b_{j'} > b_j$ such that $F(b_{j'}) \leq F(b_j) + \frac{\varepsilon}{2^j}$. Then $\bigcup_{j=1}^{\infty} (a_j, b_{j'}]$ is a cover of $[a', b]$, so

$$\begin{aligned} F(b) - F(a') &\leq \sum_{j=1}^N (F(b_{j'}) - F(a_j)) \\ &\leq \sum_{j=1}^N \left(F(b_j) - F(a_j) + \frac{\varepsilon}{2^j} \right) \\ &\leq \sum_{j=1}^N (F(b_j) - F(a_j)) + \varepsilon. \end{aligned}$$

Taking limit as $N \rightarrow \infty$ and then $a' \rightarrow a^+$, we have $F(b) - F(a) \leq \sum_{j=1}^{\infty} (F(b_j) - F(a_j)) + \varepsilon$. Since ε is arbitrary, we have $\mu_*((a, b]) = F(b) - F(a)$.

As before, we can show that μ_* is a metric outer measure. This implies the existence of a Borel measure μ on $\mathbb{R}$ such that $\mu((a, b]) = \mu_*((a, b]) = F(b) - F(a)$ for every $a < b$.

Let ν be another Borel measure on $\mathbb{R}$ such that $\nu((a, b]) = F(b) - F(a)$ for every $a < b$. We can write an open interval (a, b) as a countable union of $(a_j, b_j]$, hence by continuity of measures, we have $\nu((a, b)) = F(b) - F(a) = \mu((a, b))$. Then by outer regularity, they agree on all Borel sets.

For the converse, increasingness of F is trivial. For right continuity, let $x > 0$ and define $E_n = (0, x + \frac{1}{n}]$, then E_n decreases to $(0, x] =: E$. Since $\mu(E_n) < \infty$, we have

$$\lim_{n \rightarrow \infty} F\left(x + \frac{1}{n}\right) = \lim_{n \rightarrow \infty} \mu(E_n) = \mu(E) = F(x),$$

so F is right-continuous at x . The case $x < 0$ is similar. ■

Remark. Alternatively, we can show that $\mu_0((a, b]) = F(b) - F(a)$ is a premeasure on the algebra of finite unions of disjoint intervals of the form $(a, b]$, and then extend μ_0 to a measure μ on the σ -algebra generated by these intervals, which is the Borel σ -algebra.

If F is an increasing right-continuous function on $\mathbb{R}$, from the previous theorem, we obtain the corresponding Borel measure μ on $\mathbb{R}$. We can then define the Stieltjes integral of a measurable function f with respect to F by

$$\int_a^b f(x) dF(x) = \int_a^b f(x) d\mu(x).$$

Remark.

- (i) We can extend the results to F that is of bounded variation, by decomposing into real and imaginary parts, then as differences of two increasing functions.
- (ii) If F is absolutely continuous, then F is differentiable a.e. and

$$\int_a^b f(x) dF(x) = \int_a^b f(x) F'(x) dx$$

for all functions f Borel measurable and integrable with respect to μ .

- (iii) Suppose that F is a right-continuous, pure jump function on $[a, b]$ with jumps α_n at points x_n , then whenever f is continuous and has compact support, we have

$$\int_a^b f(x) dF(x) = \sum_{n=1}^{\infty} \alpha_n f(x_n).$$

This corresponds to the measure μ such that $\mu(\{x_n\}) = \alpha_n$ and $\mu(E) = 0$ for any E that does not contain any x_n . A special case is the Heaviside function $H = \chi_{[0, \infty)}$, which has a jump of 1 at 0, so

$$\int_{-\infty}^{\infty} f(x) dH(x) = f(0).$$

6.3. Absolute Continuity of Measures

6.3.1. Signed Measures

A signed measure possesses all the properties of a measure except that it can take negative values.

DEFINITION 6.24. A signed measure on a σ -algebra $\mathcal{M}$ is a function $\nu: \mathcal{M} \rightarrow (-\infty, \infty]$ such that $\nu(\emptyset) = 0$ and if (E_n) is a sequence of pairwise disjoint sets in $\mathcal{M}$, then

$$\nu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \nu(E_n).$$

From the definition, for the equality to hold, the sum on the right hand side must be independent of rearrangement. In particular, if the left hand side is finite, then the sum must converge absolutely.

If f is a real-valued measurable function with $\int f^- d\mu < \infty$, then

$$\nu(E) = \int_E f d\mu$$

defines a signed measure on $\mathcal{M}$. We can also define complex measures in a similar way.

DEFINITION 6.25. Let ν be a signed measure on a σ -algebra $\mathcal{M}$. The total variation $|\nu|$ of ν is defined by

$$|\nu|(E) = \sup \sum_{j=1}^{\infty} |\nu(E_j)|,$$

where the supremum is taken over all countable partitions $E = \bigcup_{j=1}^{\infty} E_j$ of E into disjoint measurable sets.

PROPOSITION 6.26. Let ν be a signed measure on a σ -algebra $\mathcal{M}$. Then $|\nu|$ is a measure on $\mathcal{M}$, and for every $E \in \mathcal{M}$, we have

$$|\nu(E)| \leq |\nu|(E).$$

Proof. Clearly $|\nu|(\emptyset) = 0$. To show countable additivity, let $E_1, E_2, \dots$ be a sequence of disjoint sets in $\mathcal{M}$ and $E = \bigcup_{n=1}^{\infty} E_n$, then it suffices to show that

$$\sum_{n=1}^{\infty} |\nu|(E_n) \leq |\nu|(E) \quad \text{and} \quad |\nu|(E) \leq \sum_{n=1}^{\infty} |\nu|(E_n).$$

For the first inequality, let α_j be a real number such that $\alpha_j < |\nu|(E_j)$ or $\alpha_j = 0$ if $|\nu|(E_j) = 0$. Then by definition of total variation, we can find a partition $E_j = \bigcup_{k=1}^{\infty} E_{j,k}$ such that

$$\sum_{k=1}^{\infty} |\nu(E_{j,k})| \geq \alpha_j$$

for every j . Since $E = \bigcup_{j=1}^{\infty} \bigcup_{k=1}^{\infty} E_{j,k}$, we have

$$|\nu|(E) \geq \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} |\nu(E_{j,k})| \geq \sum_{j=1}^{\infty} \alpha_j.$$

Taking supremum over all α_j , we have $|\nu|(E) \geq \sum_{n=1}^{\infty} |\nu|(E_n)$.

For the second inequality, let (F_k) be a partition of E , then $F_k = \bigcup_{j=1}^{\infty} (F_k \cap E_j)$, so

$$\begin{aligned} \sum_{k=1}^{\infty} |\nu(F_k)| &= \sum_{k=1}^{\infty} \left| \nu \left(\bigcup_{j=1}^{\infty} (F_k \cap E_j) \right) \right| \\ &\leq \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} |\nu(F_k \cap E_j)| \\ &= \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} |\nu(F_k \cap E_j)| \leq \sum_{j=1}^{\infty} |\nu|(E_j). \end{aligned}$$

Taking supremum over all partitions (F_k) of E gives the desired inequality. ■

DEFINITION 6.27. Let ν be a signed measure on a σ -algebra $\mathcal{M}$. The positive variation and negative variation are

$$\nu^+(E) = \frac{|\nu|(E) + \nu(E)}{2}, \quad \nu^-(E) = \frac{|\nu|(E) - \nu(E)}{2}.$$

If $\nu(E) = |\nu|(E) = \infty$, we define $\nu^-(E) = 0$.

Since $|\nu|$ is a measure, ν^+ and ν^- are also measures. Moreover, we have $\nu = \nu^+ - \nu^-$ and $|\nu| = \nu^+ + \nu^-$.

DEFINITION 6.28. A signed measure ν is σ -finite if $|\nu|$ is σ -finite

Since $\nu \leq |\nu|$ and $|\nu| = \nu^+ + \nu^-$, we have

$$-\nu \leq \nu \leq |\nu|.$$

If ν is σ -finite, then so are ν^+ and ν^- .

6.3.2. Mutual Singularity and Absolute Continuity

DEFINITION 6.29.

- (i) A signed measure ν is supported on a set $A \in \mathcal{M}$ if $\nu(E) = \nu(E \cap A)$ for all $E \in \mathcal{M}$.
- (ii) Two signed measures ν and μ on $\mathcal{M}$ are mutually singular, denoted by $\nu \perp \mu$, if there exists disjoint sets $A, B \in \mathcal{M}$ such that ν is supported on A and μ is supported on B .
- (iii) If ν is a signed measure and μ is a (positive) measure, we say that ν is absolutely continuous with respect to μ , denoted by $\nu \ll \mu$, if $\mu(E) = 0$ implies $\nu(E) = 0$ for all $E \in \mathcal{M}$.

Example 6.2.

- Let m be the Lebesgue measure on $\mathbb{R}$ and δ_0 be the Dirac measure at 0, i.e. $\delta_0(E) = 1$ if $0 \in E$ and $\delta_0(E) = 0$ otherwise. Then m and δ_0 are mutually singular, since δ_0 is supported on $\{0\}$ and m is supported on $\mathbb{R} \setminus \{0\}$.
- Let f be a μ -measurable function with $\int f^- d\mu < \infty$, then the signed measure defined by

$$\nu(E) = \int_E f d\mu$$

is absolutely continuous with respect to μ , since if $\mu(E) = 0$, then $\nu(E) = \int_E f d\mu = 0$.

PROPOSITION 6.30. *If for each $\varepsilon > 0$ there exists $\delta > 0$ such that $|\nu(E)| < \varepsilon$ whenever $\mu(E) < \delta$, then $\nu \ll \mu$. The converse holds when $|\nu|$ is a finite measure, i.e. $|\nu|(X) < \infty$.*

Proof. If $\mu(E) = 0$, then for every $\varepsilon > 0$, we have $|\nu(E)| < \varepsilon$, so $\nu(E) = 0$.

Conversely, suppose that $\nu \ll \mu$ and $|\nu|(X) < \infty$. By replacing ν with $|\nu|$, we may assume that ν is a positive measure. If the conclusion does not hold, then there exists $\varepsilon > 0$ such that for every $\delta > 0$, we can find $E \in \mathcal{M}$ such that $\mu(E) < \delta$ and $\nu(E) \geq \varepsilon$. We can then find a sequence of sets $E_1, E_2, \dots$ such that $\mu(E_n) < 2^{-n}$ and $\nu(E_n) \geq \varepsilon$ for every n . Let

$$E = \limsup_{n \rightarrow \infty} E_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k,$$

then for each n , $\mu(\bigcup_{k=n}^{\infty} E_k) < 2^{-n}$, so $\mu(E) = 0$. On the other hand, since $\nu(E_n) \geq \varepsilon$ for every n , we have $\nu(\bigcup_{k=n}^{\infty} E_k) \geq \varepsilon$ for every n , so $\nu(E) \geq \varepsilon$. This contradicts the assumption that $\nu \ll \mu$. ■

THEOREM 6.31 (Radon-Nikodym Theorem). *Suppose that μ is a σ -finite positive measure and ν is a σ -finite signed measure on $(X, \mathcal{M})$. Then there exists a unique decomposition $\nu = \nu_a + \nu_s$ into signed measures ν_a and ν_s such that $\nu_a \ll \mu$ and $\nu_s \perp \mu$. Moreover, $d\nu_a = f d\mu$, i.e.*

$$\nu_a(E) = \int_E f d\mu$$

for some μ -measurable function f with $\int f^- d\mu < \infty$.

Proof. First consider the case that both ν and μ are positive and finite. Let $\lambda = \nu + \mu$. Define the linear functional $\ell : L^2(X, \lambda) \rightarrow \mathbb{C}$ by

$$\ell(\psi) = \int_X \psi d\nu.$$

Note that

$$\begin{aligned} \ell(\psi) &\leq \int_X |\psi| d\nu \leq \int_X |\psi| d\lambda \\ &\leq \left(\int_X |\psi|^2 d\lambda \right)^{\frac{1}{2}} \left(\int_X 1^2 d\lambda \right)^{\frac{1}{2}} = |\psi|_2 (\lambda(X))^{\frac{1}{2}}, \end{aligned}$$

so ℓ is a bounded linear functional on $L^2(X, \lambda)$. By Riesz representation theorem, there exists $g \in L^2(X, \lambda)$ such that

$$\ell(\psi) = \int_X \psi d\nu = \int_X \psi g d\lambda$$

for every $\psi \in L^2(X, \lambda)$.

Let $E \in \mathcal{M}$ with $\lambda(E) > 0$, then $0 \leq \frac{\mu(E)}{\lambda(E)} \leq 1$. Consider $\psi = \chi_E$, then

$$\ell(E) = \mu(E) = \int_E g d\lambda \leq \frac{1}{\lambda(E)} \int_E g d\lambda \leq 1.$$

Since E is arbitrary, we have $0 \leq g \leq 1$ λ -a.e. Note that

$$\int_X \psi d\nu = \int_X \psi g d\lambda = \int_X \psi g d\nu + \int_X \psi g d\mu,$$

so

$$\int_X \psi(1-g) d\nu = \int_X \psi g d\mu. \quad (1)$$

Consider the sets

$$A = \{x \in X : 0 \leq g(x) < 1\}, \quad B = \{x \in X : g(x) = 1\}.$$

Define ν_a and ν_s on $\mathcal{M}$ by

$$\nu_{a(E)} = \nu(E \cap A), \quad \nu_{s(E)} = \nu(E \cap B).$$

CLAIM 6.32. $\nu_s \perp \mu$

Proof. Substituting $\psi = \chi_B$ into (1), we have

$$0 = \int_X \chi_B(1-g) d\nu = \int_X \chi_B g d\mu = \mu(B),$$

so ν_s is supported on B and μ is supported on $X \setminus B$, hence $\nu_s \perp \mu$. ■

CLAIM 6.33. $\nu_a \ll \mu$ and $\nu_{a(E)} = \int_E f d\mu$ for some μ -integrable function f .

Proof. Substituting $\psi = \chi_E(1+g+\dots+g^n)$ into (1), we have

$$\int_E (1-g^{n+1}) d\nu = \int_E g(1+g+\dots+g^n) d\mu.$$

By dominated convergence theorem the left hand side converges to $\int_E \chi_A d\nu = \nu_{a(E)}$, and since $1+g+\dots+g^n$ converges to $\frac{1}{1-g}$, by monotone convergence theorem, we have

$$\nu_{a(E)} = \int_E \frac{g}{1-g} d\mu.$$

Let $f = \frac{g}{1-g}$, then f is μ -integrable since $\nu_{a(X)} \leq \nu(X) < \infty$. ■

If μ and ν are σ -finite and positive, we can write $X = \bigcup_{j=1}^{\infty} E_j$ with $E_j \in \mathcal{M}$ and $\nu(E_j), \mu(E_j) < \infty$ for every j . Define positive finite measures on $\mathcal{M}$ by

$$\nu_{j(E)} = \nu(E \cap E_j), \quad \mu_{j(E)} = \mu(E \cap E_j).$$

By the previous case, we can find decompositions $\nu_j = \nu_{j,a} + \nu_{j,s}$ such that $\nu_{j,s} \perp \mu_j$ and $\nu_{j,a} = f_j d\mu_j$ for some μ_j -integrable function f_j . It suffices to set

$$\nu_a = \sum_{j=1}^{\infty} \nu_{j,a}, \quad \nu_s = \sum_{j=1}^{\infty} \nu_{j,s} \quad \text{and} \quad f = \sum_{j=1}^{\infty} f_j.$$

If ν is signed, we can apply the above argument separately to ν^+ and ν^- .

For uniqueness, suppose $\nu = \nu_{a'} + \nu_{s'}$ is another decomposition with the same properties, then

$$\nu_a - \nu_{a'} = \nu_{s'} - \nu_s.$$

The left hand side is absolutely continuous with respect to μ , while the right hand side is singular with respect to μ , so both sides must be 0, hence $\nu_a = \nu_{a'}$ and $\nu_s = \nu_{s'}$. ■

Remark. The above f is unique up to a μ -null set. f is also called the Radon-Nikodym derivative of ν_a with respect to μ and is denoted by $\frac{d\nu_a}{d\mu}$.

Fractal Geometry

CHAPTER 7

7.1. Self-similarity

Informally, a set is self-similar if it is the union of several scaled down copies of itself.

Example 7.1.

- The Cantor set is self-similar, since it is the union of two copies of itself scaled down by a factor of $\frac{1}{3}$.
- The Sierpinski triangle is self-similar, since it consists of three copies of itself scaled down by a factor of $\frac{1}{2}$.
- the unit interval $[0, 1]$ is self-similar, since it consists of two copies of itself scaled down by a factor of $\frac{1}{2}$.

DEFINITION 7.1. Let (X, d) be a metric space. A function $\varphi : X \rightarrow X$ is a *similitude* if there exists $r \in (0, 1)$ such that

$$d(\varphi(x), \varphi(y)) = rd(x, y)$$

for every $x, y \in X$.

Note that a similitude is a contraction. It's straightforward to verify that $\text{diam}(\varphi(A)) = r \text{diam}(A)$ for every $A \subseteq X$.

DEFINITION 7.2. A family $(\varphi_1, \dots, \varphi_n)$ of finitely many contractions is called an *iterated function system (IFS)*. A compact set $K \subseteq X$ is called an *attractor* of the IFS $(\varphi_1, \dots, \varphi_n)$ if

$$K = \bigcup_{k=1}^n \varphi_k(K).$$

A set is called *self-similar in the wider sense* in X if it is the attractor of some IFS on X . A set is called *self-similar (in the strict sense)* if it is the attractor of some IFS in which each φ_k is a similitude.

If $c_1, \dots, c_n$ are the contraction ratios of $\varphi_1, \dots, \varphi_n$ respectively, then we say that the IFS has contraction ratio $c = \max\{c_1, \dots, c_n\}$.

Example 7.2.

- $[0, 1]$ is self-similar with similitudes

$$\varphi_1(x) = \frac{x}{2}, \quad \varphi_2(x) = \frac{x}{2} + \frac{1}{2}.$$

- The Cantor set is self-similar with similitudes

$$\varphi_1(x) = \frac{x}{3}, \quad \varphi_2(x) = \frac{x}{3} + \frac{2}{3}.$$

- Every finite set is self-similar in the wider sense, since we can define $\varphi_{k(x)} = x_k$ for every k , where $x_1, \dots, x_n$ are the points in the set. However, a finite set is self-similar in the strict sense if and only if it is a singleton.

7.1.1. Coding Theorem

Recall that for the middle-thirds Cantor set, there is a function $\pi : \{0, 2\}^{\mathbb{N}} \rightarrow C$ defined by

$$\pi((x_k)) = \sum_{k=1}^{\infty} \frac{x_k}{3^k},$$

which is a homeomorphism. The following definition generalizes this result to arbitrary attractors.

DEFINITION 7.3. Let $\sum_n = \{1, \dots, n\}^{\mathbb{N}}$ be the set of all sequences with values in $\{1, \dots, n\}$, called the code space in n symbols. For any two distinct sequences $(x_k), (y_k) \in \sum_n$, define

$$d_n((x_k), (y_k)) = 2^{-m}, \quad m = \min\{k : x_k \neq y_k\}.$$

Then d_n is a metric on $\sum_n$ and makes $\sum_n$ a compact metric space.

THEOREM 7.4 (Coding Theorem). Let K be the attractor of an IFS $(\varphi_1, \dots, \varphi_n)$ on a metric space (X, d) . Then there exists a continuous surjective function $\pi : \sum_n \rightarrow K$ defined by

$$\pi((y_k)) = \lim_{k \rightarrow \infty} \varphi_{y_1} \circ \varphi_{y_2} \circ \dots \circ \varphi_{y_k}(x)$$

for any $x \in K$, such that for all $(y_1, \dots, y_k) \in \{1, \dots, n\}^k$,

$$\pi\left(\left\{(x_k) \in \sum_n : x_1 = y_1, \dots, x_k = y_k\right\}\right) = \varphi_{y_1} \circ \varphi_{y_2} \circ \dots \circ \varphi_{y_k}(K).$$

If additionally, all the φ_k are injective and $\varphi_{i(K)} \cap \varphi_{j(K)} = \emptyset$ for all $i \neq j$, then π is a homeomorphism.